

Reconditioning Dharavi: A toolkit of strategies for incremental development

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Acknowledgment

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Abstract

The 2003 Global Report on Human Settlements (Un-Habitat, 2003) defines a “slum” as a densely populated metropolitan area that is distinguished by a variety of low-income settlements, subpar housing, and squalor. Dharavi, on the other hand, is far more than a “slum.” In the heart of Mumbai, Dharavi is an economically prosperous and socially active informal town. Mumbai is a thriving metropolis with many different realities and patterns, even though it appears to be a slum filled with squatters. However, the region has recently become a hub for informal settlements and urban problems associated with poor hygiene in developing countries. People’s misconceptions about Dharavi stem from a failure to recognize its social capital and economic power: the area encompasses a variety of economic networks, production types, income levels, land tenure arrangements, and religious activities and festivities. Dharavi is made up of 85 separate groups with a strong feeling of belonging and high expectations for stability and improved economic position and living standards. It is also clear that these folks are capable of building and enhancing their shelter if they have the resources to do so. To develop all these qualities, Dharavi’s Social Capital must be recognized and promoted as an asset to the city of Mumbai. A community such as Dharavi requires ‘urban acupuncture’; where mediation of the littlest kind will have the greatest effect. Dharavi, like any other “Informal” city, requires rigorous examination to be fully comprehended. It is a unique location where a large flood of migrants has managed to build jobs and their city. My underlying attitude to this location is a conflicting desire to save and replace it. The desire to save is linked to the aesthetic of informality as well as the intense sociality, diversity, and production of the streets and lanes - a fascinating and diversified urban ensemble. The desire to eliminate stems from hopeless states of sterilization, ventilation, light, open space, and congested areas. As a result, a reliable strategy for combining the two methodologies and locating a functioning arrangement should be developed. The government has been trying to redevelop this area for the past 50 years but hasn’t been successful in doing so. In contrast to the existing redevelopment plan, which promotes uniform top-down development, my concept anticipates techniques for progressive self-development, including “bottom-up” finance models and architectural approaches. After identifying various patterns and carefully examining behavior patterns, production systems, and existing community facilities, a toolkit of methods can be built that can be used in various places and “outboxes.” The simple homogeneity of solutions for Dharavi’s changing conditions has been avoided. Dharavi’s current identity and “mixed-use” paradigm have been respected, with Home recognized as an instrument of production. The proposed design has been tested for various environmental factors using different tools for natural lighting and ventilation. The outdoor areas are also analyzed for thermal comfort since a lot of social activities take place in these areas. Communal areas have been designed to accommodate micro infrastructure systems while also increasing productivity. As a result, a system of self-development triggers has been created that can improve present conditions while also supporting the community’s need for stability. Simultaneously, by focusing on property ownership as an economic driver, the proposed approach can provide a type of “social mobility” for Dharavi’s residents.

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INTRODUCTION

“The magic of squatter cities is that they are improved steadily and gradually by their residents” - Steward Brand

Urbanization has emerged as a key problem for global governance, prosperity, and security. According to experts, more than 75 percent of the world’s population will reside in cities by 2030. Millions of people could benefit from the increased living standards and possibilities that this rapid urbanization can bring, but it could also have a negative impact: cities that are unable to meet the demands of their expanding populations would see an increase in crime, violence, and poverty. Urban fragility has consequently become a major concern for national, municipal, and local security and development players. Megacities, urban sprawl, unplanned, massive metropolitan area growth, and rising economic, social, and spatial inequalities mark the dawn of a new era. Africa and Asia have the fastest rates of urbanization in the world, and by 2050, their respective urban populations will quadruple and double. Latin American cities will likewise continue to face issues imposed on by fragility and rapid growth. This global context has significant effects on urban vulnerabilities and hazards, from persistent crime and violence to potential urban growth plans and the growing significance of local administrations.

Towards Collective Urban Resilience

At. Urban surroundings are changing quickly today in many regions. Numerous cities’ densities, unpredictable nature, and vulnerability call for novel answers. There are numerous instances of how resilient and resourceful particular people and communities may be, some of which are documented in the case studies that follow. But whether the poor and marginalized have access to the essential components of resilient systems—safety, economic opportunity, spatial integration, and mobility—is the ultimate stress test for today’s cities. Policies that are holistic and sustainable can assist communities manage the challenges of growing urban complexity by removing both physical and social impediments to social cohesiveness. Collective resilience is a requirement for individual resilience as urban populations become more integrated and interdependent.

Mumbai: Plurality and Complexity of a Mega-city

Mumbai, the Maharashtra state’s capital, has a population of 12 million. Greater Mumbai is a 438 sq. km. area with an extraordinarily high population density (27,348 per sq.km). In addition, Mumbai Metropolitan Region, home to 18 million people, is the fifth-most populous metropolitan area in the world. Mumbai is the financial, entertainment, and slum center of India, as well as the city with the most slums. It provides 60% of the customs duty and 40% of the national income tax. According to purchasing power parity (PPP), Mumbai’s GDP is believed to be worth US\$143 billion. The average American’s income is \$12,070. The main concerns Mumbai is facing are slums, significant housing issues, flooding, and traffic congestion as well as the loss of wetlands and loss of wetlands. According to certain predictions, Mumbai may surpass Tokyo as the biggest metropolis in the world by 2050. (Correa, 1984) The national, state, metropolitan, and local levels are contrasted in the diagram below. Each layer in India begins at the national level and zooms in four times more to examine the urbanization setting in the city of Mumbai.

Scale can be advantageous in large clusters of various industries, marketplaces, and businesses. The city is rich and offers investment opportunities, contributing 40% of India’s tax revenues. There is a sizable workforce that is highly educated, competent, and unskilled. It’s the most cosmopolitan Indian city, with less caste distinctions, easier employment opportunities, and a vibrant cultural scene. More so than in most other Indian cities, the train and bus public transit systems, infrastructure, and telecommunications all function well here. But as Architect Charles Correa once famously said, “Bombay’s a great city...but a terrible place!”

Most ‘Mumbaikars’ (term used to address a resident of Mumbai), except for a small elite class, would concur that: Adequate housing is not affordable. The length of the commute and the discomfort are nearly unpleasant. Transport by road and by rail is under extreme strain. It is nearly tough to find good schools in the area. Playgrounds and open spaces are disappearing (0.03 acres per thousand compared to the international norm of 4 acres). Only a small percentage of slum dwellers benefit from slum relocation programs, which typically result in inappropriate (high-rise) and pricey to maintain homes. Only a small portion of residences where it is supplied by standard standpipes have access to piped water and sanitary facilities. (Correa, 1984) Only a few hours a day are filled with supplies. The city’s tap water is unfit for consumption. Civic services are not provided voluntarily, effectively, or without corruption. According to the above-mentioned criteria (affordable housing, proximity to employment, accessibility to clean water, a lack of significant open space, and supportive municipal services), the quality of life in the city is subpar and is only getting worse.

MUMBAI



INDIA					
Area (Sq.km)	Population (millions)	Density (pers/sq.km)	Rel.GDP (% higher level)	Number of states	No of elected officials
3,287,590	1,131.04	344	8.9%	29	545/250

STATE OF MAHARASHTRA					
Area (Sq.km)	Population (millions)	Density (pers/sq.km)	Rel.GDP (% higher level)	Number of districts	No of elected officials
307,713	105.51	343	13%	35	289/78

MUMBAI METROPOLITAN REGION					
Area (Sq.km)	Population (millions)	Density (pers/sq.km)	Rel.GDP (% higher level)	Number of districts	No of elected officials
4,355	17.76	4,080	39%	N/A	N/A

MUMBAI METROPOLITAN PEOPLE

17,768,993 people

MUMBAI: 24 WARDS

11,978,450 people

GREATER MUMBAI					
Area (Sq.km)	Population (millions)	Density (pers/sq.km)	Rel.GDP (% higher level)	Number of wards	No of elected officials
438	11.98	27,348	26%	24	277

Fig 1.1. Statistical data of India with focus on Maharashtra State
Source: Urban Age- LSE Cities- Redrawn by author

Understanding the maximum city

Governance

The government of Mumbai makes decisions at the federal, state, and local levels. Several strong ministries within the federal government offer the city resources and services. There is a strong level of state government, led by a Chief Minister, that manages numerous services within the city, including policing, housing, roads, education, and health care. A mayor who is elected has some authority over the city administration. The Municipal Commissioner and the Secretary for Special Projects, both civil employees chosen directly by the Maharashtra State government, hold the real executive power. A Metropolitan Planning Committee for the Metropolitan Area is about to be established by the state government in accordance with the Jawaharlal Nehru National Urban Renewal Mission (JNNURM). At the state and local levels, there is a lot of overlap in duties. As mandated by the JNNURM, the city administration is often less powerful than the state.

Density and Land-use

The island city's territorial restrictions have led to extraordinarily high urban density. When just considering the built-up region, the average density within the city limits exceeds 27,000 persons per square kilometer and rises to well over 50,000, exceedingly even the densest points in Manhattan, a borough of New York City. Furthermore, it is not unusual for the densest areas of Mumbai, like Dharavi, to house up to 100,000 people per square kilometer. A region the size of Berlin that is home to 12 million people has less than half of Greater Mumbai covered in infrastructure and built-up territory, with vast parts taken up by a national park, open spaces, coastal wetlands, mangroves, agricultural land, and beaches.

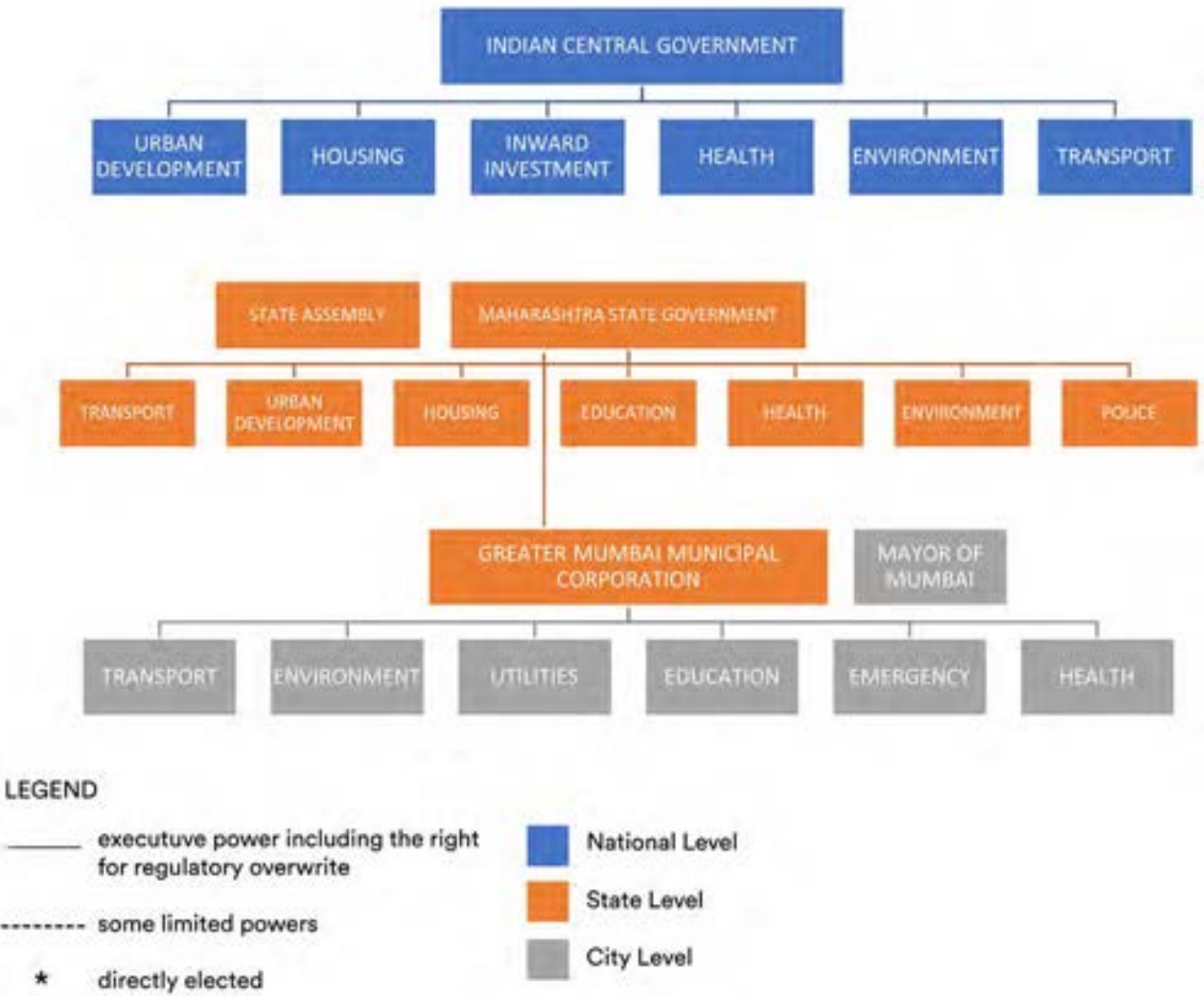


Fig 1.2. Governance Structure of Mumbai
Source: Urban Age- LSE Cities- Redrawn by author

Understanding the maximum city

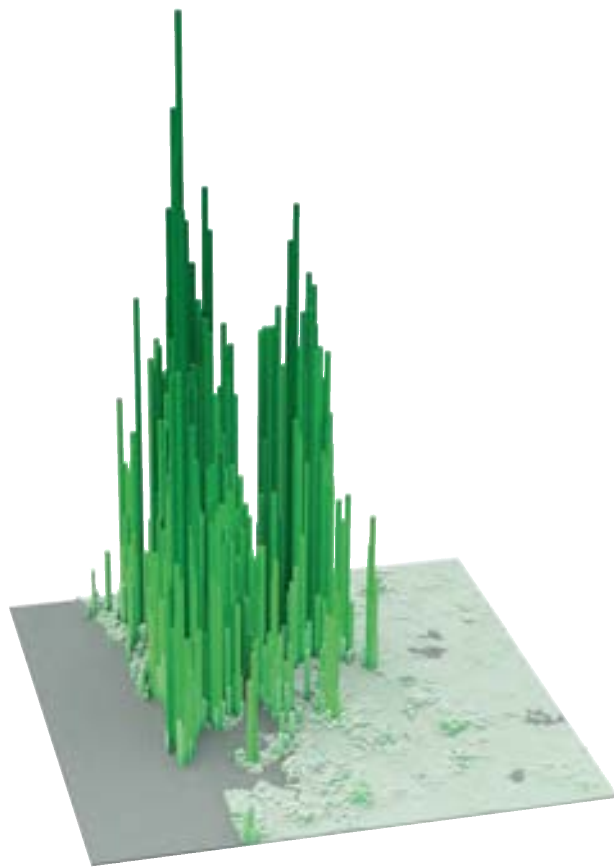


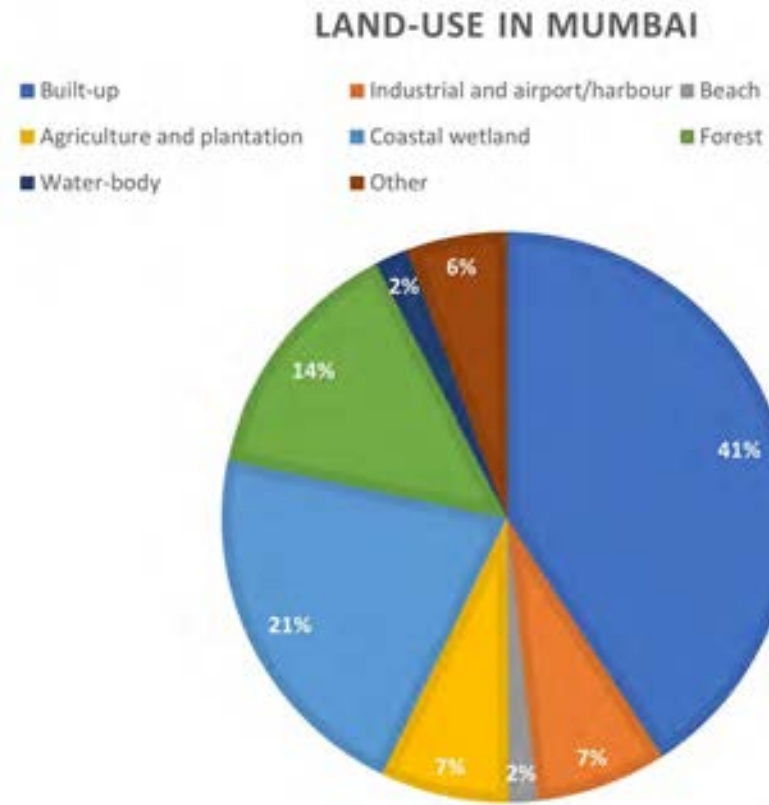
Fig 1.3. Density of Mumbai
Source: <https://urbanage.lsecities.net/>

Urban Morphology

Cities must adapt to the demographic and economic challenges driving the world’s brisk urbanization. The livability and adaptability of urban environments that are undergoing strong processes of change are significantly influenced by the design of city streets, buildings, and spaces—their spatial DNA. The wide “white” spaces that make up the public realm of the city are contained and compressed by an intense and close-knit arrangement of buildings and structures in Indian cities. This figure-ground is a useful tool for understanding how buildings and their surroundings succeed or fail in creating a continuous and integrated urban whole and for seeing the micro-scale of urban neighborhoods. Additionally, it offers a street-level view of the constructed shapes and volumetric configurations that influence urban daily life. It measures 1 square kilometer and depicts open landscapes in white and houses in black. (Correa, 1984)

Moving in the city

The largest suburban rail network on the subcontinent is in Mumbai. It is one of the busiest train systems in the world, carrying more than 6 million passengers daily. The enormous amount of rail infrastructure is highlighted by the projected GIS numbers for the length of regional rail networks in each city, with significant levels in Mumbai. Understanding how people move in cities can be improved by taking a closer look at various modes of transportation (modal divides). Due to the high proportions of people who commute by public transportation or walk to work, which is a direct result of the proximity of residential buildings (often slums) and offices in these high-density, mixed-use urban environments where commutes to work are typically less than 2 kilometers, Indian cities are more compact than other cities. In Mumbai, walking accounts for a staggering 55% of all modes of transportation, with vehicles accounting for only 5%. With 29 automobiles per 1000 persons, Mumbai has the lowest rate of car ownership.



City	Metropolitan Region	Administrative City	Inner City (10 km radius)	Peak Density
Mumbai	4,080	27,348	34,289	101,066

Fig 1.4. Land Use Statistics for Mumbai
Source: Urban Age- LSE Cities- Redrawn by author



Fig 1.5. Figure ground map of Mumbai City
Source: <https://urbanage.lsecities.net/>



Fig 1.6. Transport Infrastructure
Source: <https://urbanage.lsecities.net/>

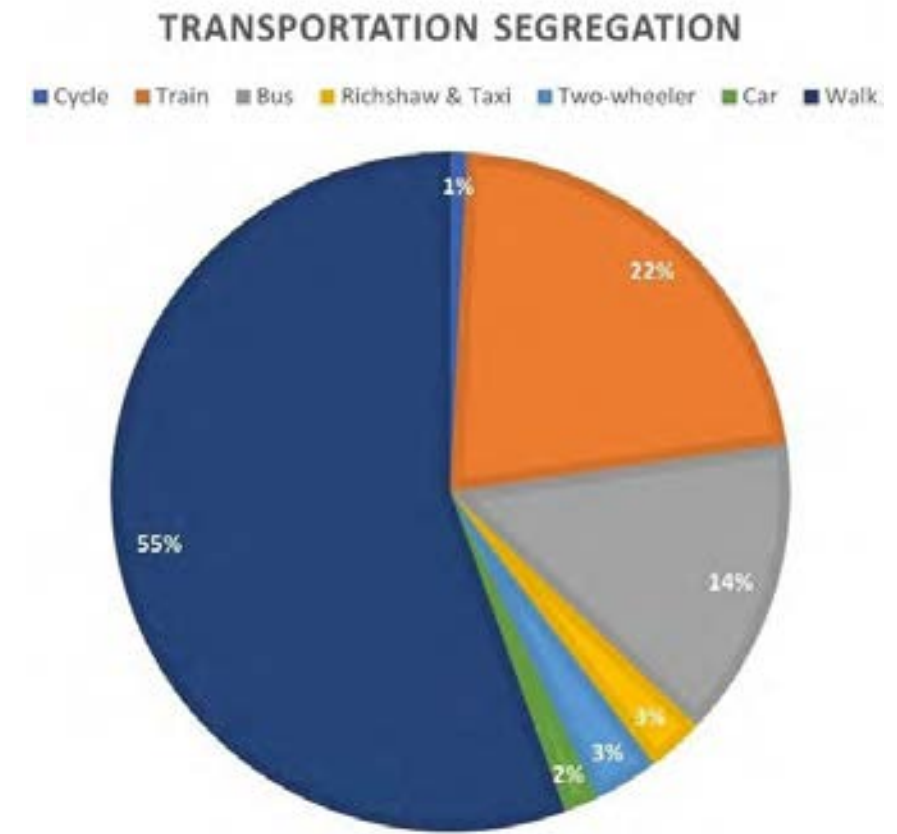
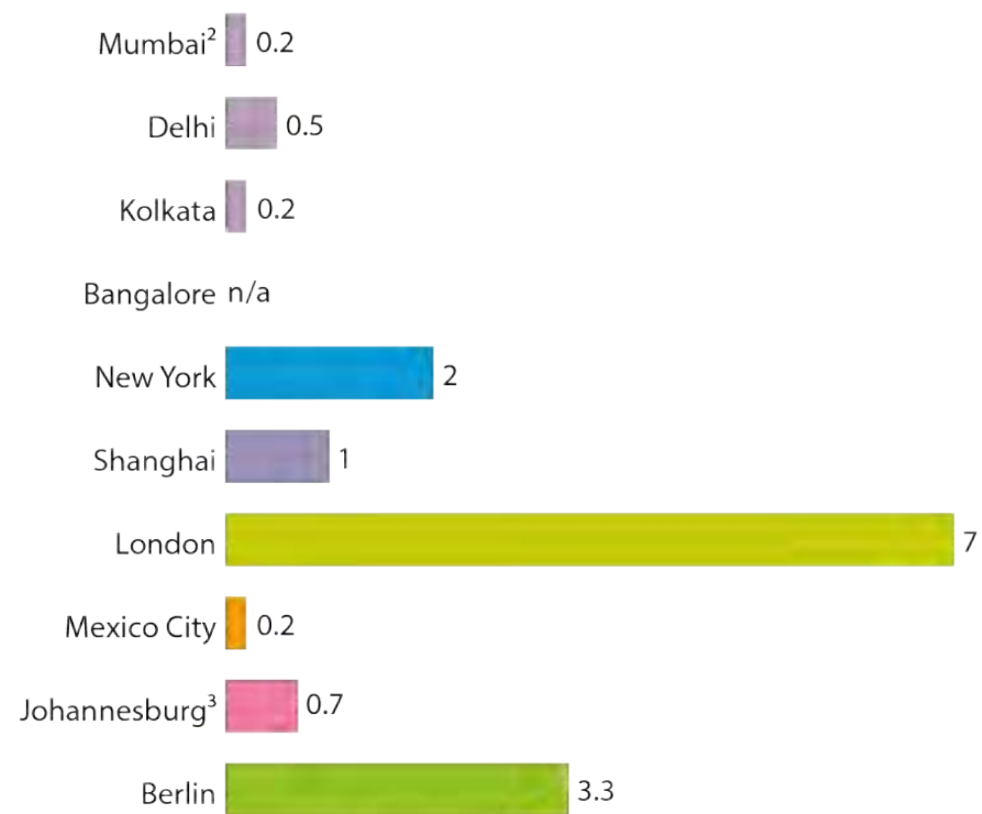


Fig 1.8. Transport Segregation
Source: Urban Age- LSE Cities- Redrawn by author

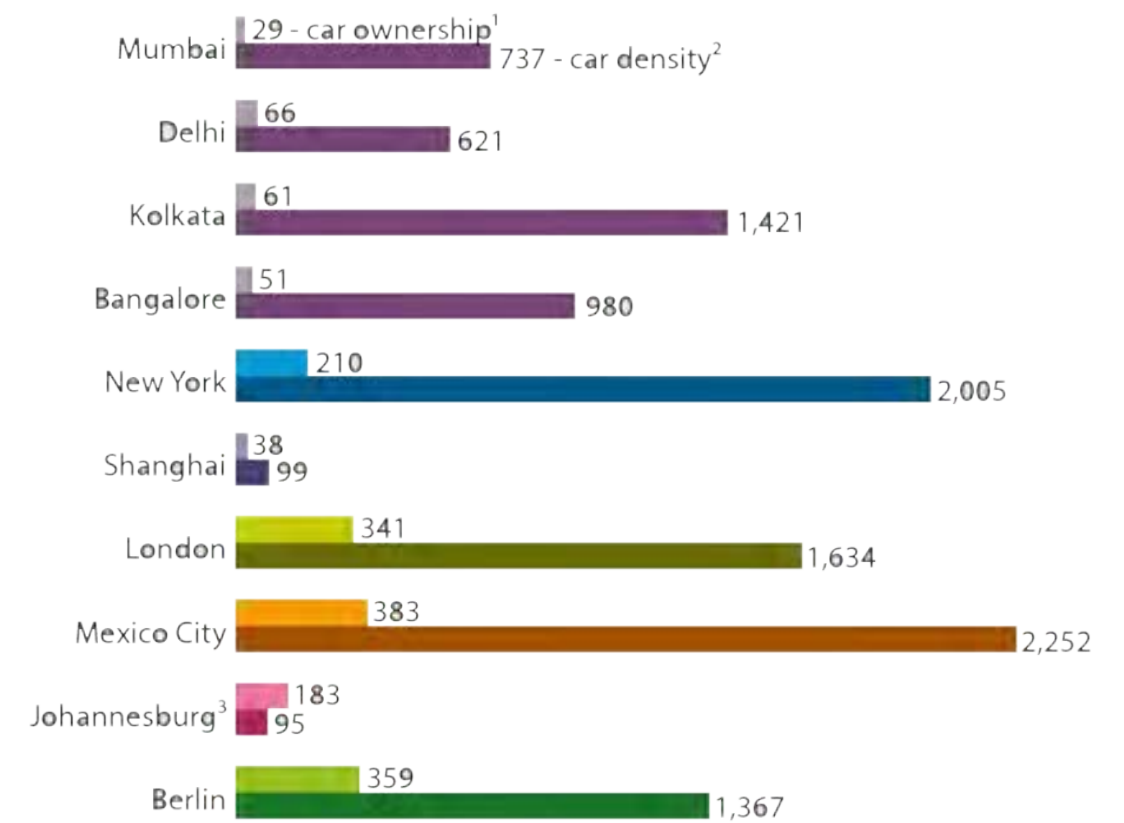


¹ price for similar single tickets

² Mumbai rail ticket for general class

³ Johannesburg rail

Fig 1.7. Relative costs for Transport Infrastructure
Source: <https://urbanage.lsecities.net/>



¹ car ownership in number of cars per 1,000 residents

² car density in cars per km²

³ Gauteng Province

Fig 1.9. Mumbai car ownership and densities
Source: <https://urbanage.lsecities.net/>

Real Estate, Housing and Slums

Mumbai’s housing crisis and high cost make renting a difficult situation. Demand for both affordable and opulent homes has surged due to the economy’s rapid expansion, and suburban areas are now seeing a spike in demand for commercial real estate. Due to the city’s constrained land area, prices have increased dramatically. As a result, a connection between builders and the mafia has developed wherein old homes in desirable downtown areas are bought, demolished, and rebuilt. Mumbai’s average flat costs 30 million rupees (\$440,000), forcing most residents to pay rising prices or live in neighborhoods with lower rents. However, even in areas where rentals are inexpensive, landlords demand sizable deposits to cover any damage or tenant nonpayment of bills. Many cases of renters who have refused to leave their homes are still pending in court.

A Mumbai court recently decided in favor of the landlord following a 42-year legal struggle in a case where the tenants wanted to stay in their home and argued that they were unable to locate alternative housing while the landlord was wealthy enough to do so. The reason that 41.3 percent of Mumbai’s population lived in slums in 2011—a decline from 57.6 percent in 2006—and the high cost of housing are due to these factors. All wards, except for Ward C, have some level of slum presence. In Ward S, slums account for about 86 percent of the population, while Dharavi, one of the largest slums in the world, covers an area of more than 200 hectares. Most of the 100 to 300 families who are thought to relocate to Mumbai everyday wind up living in slum colonies or shanties. (Batra, 2021)

Slums have a tremendous political impact, and governments have been elected based on their commitment to redeveloping them. A goal set by the Shiv Sena government to provide free accommodation for 400,000 Mumbai slum residents was not achieved. It remains to be seen whether the Bhartiya Janata Party administration, which took office in 2014, quickens the pace of the Slum Rehabilitation Authority’s projects, 117 of which are currently on hold at the BMC. Slum rehabilitation is a difficult task. Certain slums are situated on extremely costly real estate. Given the slum dwellers’ right to vote, no government would risk removing the slums and relocating the population because they would not be accepted by them.

To keep the voting bases in these neighborhoods, governments have refrained from razing illegal shanties or slums. Additionally, people who had new houses constructed for them often sell, rent, or abandon them and return to the slums. (S Tsenkova, 2009)

Slums also continue to exist because of the informal economic activity they produce, which provides financial incentives for it. However, most Mumbai’s slums continue to have horrendous living conditions. Slums are disaster-prone and environmentally dangerous. The most challenging and unhealthy component of living in a slum is sanitation. In Mumbai, public restrooms are necessary in three out of four slums. The state of these restrooms is shocking; they are rarely cleaned, are without electricity or running water, are overrun by dogs or antisocial characters, and pose serious risks to personal safety. As a result, almost 6% (420,000 people) of slum dwellers defecate in public.

With the help of the World Bank, the Slum Sanitation Program has attempted to offer the residents of Mumbai’s slums high-quality sanitation services. Some toilets were replaced by toilets constructed as part of this program, which are required to have electricity, running water, and to be cleaned three times each day. However, for many people, these latrines are far from their colonies, and the conditions are still appalling. By mid-afternoon, the latrines stop working, leaving many people little choice but to urinate in the open. Nongovernmental organizations (NGOs) now provide paid restrooms in many slum areas, where the sanitation, water, and electricity infrastructure are considerably better. These toilets cost a nominal charge, and slum inhabitants choose to use them. Additionally, they ease the strain on already-built government-built restroom facilities. The various wards are unequally distributed in terms of toilets. With an average of 81 individuals per toilet in the slums, the number of people per toilet ranges from 399 to 56. According to the BMC, the city needs to build 64,157 new toilet seats. However, just 6,050 are currently being built. (Philip, 2022)

Locating Dharavi: Slum of Enterprise

Mumbai’s city, known as Slumbai or Slumbay, is home to what is perhaps the highest concentration of slum inhabitants in the world (over six million). The largest mega-slum in Asia is Dharavi, which is an informal township within the city and one of the world’s 30 largest. The city’s 525 acres are home to a very vibrant mosaic of tens of thousands of small businesses and hundreds of thousands of people who are interdependent with one another and the city on a social, cultural, and economic level. These people come from various religions, castes, languages, provinces, and ethnicities. Its entrepreneurial locals also operate a thriving - and distinctive - recycling company in addition to producing clothing, leather items, cuisine, and pottery. (Dharavi, 2022).

Dharavi, which was formerly on the outskirts of the peninsula city of Bombay, is now in the center of the metropolitan network of traffic flows, well-served by infrastructure and services related to transportation. It is a triangular piece of land in the center of the city that is bordered by the Mahim Creek and its mangroves on three sides and railway lines on two of its sides. Three corners are marked by the Mahim, Matunga, and Sion train stations, and the Western Express Highway runs along the northern edge. Dharavi, which began as a fishing town on the northern tip of Parel island, was inexorably pushed into the heart of the city as Mumbai expanded through time and northward into the suburbs to accommodate the continually rising population. The location, which has accessible connections to three stations, is at the intersection of two main railway lines that connect central Bombay to the greater metropolitan area. Along both sides are busy thoroughfares. Dharavi is close to the Bandra-Kurla Complex (BKC), a significant new business zone that serves as a counterbalance to the ancient Central Business District (CBD) in south Mumbai. In many ways, the BKC is easier to get to than the CBD because it is located just south of the airport. Dharavi is of great interest to real estate promoters and developers due to its proximity to BKC and superb mass transit service. With its extremely high commercial real estate value and proximity to the upscale Bandra-Kurla complex, Dharavi has become a highly “hot property.” On one of Mumbai’s seven islands’ backwaters, Dharavi used to be a small fishing community.

It would rank among the most valuable real estate in the world if cleaned up and renovated! (Dharavi, 2022). Even on a holiday, the atmosphere in Dharavi is like being on a treadmill. Everyone is working; not many people loiter. There are hawkers lining the streets offering everything from bananas and bags to safety pins. There is a diverse selection of stores throughout the entire region like Satkar Jewelers, Ration Shop, Bhupendra Steel, Husain Hotel, Swastik Electric & Hardware, Shreenath Jewelers and Mumbai Polyclinic, just to name a few. Dharavi’s daily economic output can only be estimated because few residents will really admit to their income for fear of being visited by an official. A large portion of this region’s production is not registered with any authorities. There is little doubt, however, that it is in the millions of rupees. Residents of Dharavi made a quick computation that came to between Rs 1,500 crore and Rs 2,000 crore every year, or at least Rs 5 crore per day! Without even considering real estate costs, Dharavi is a “gold mine”!

The National Slum Dwellers’ Federation (NSDF) found 1,044 manufacturing facilities of different sizes in Dharavi during a study they conducted in 1986. The Society for Human and Environmental Development (SHED) later conducted a survey and found 1,700 units. Since many smaller operations, which operate out of homes and lofts, would have been excluded from the surveys, it is likely that the true number is higher. According to the NSDF survey, there are 244 small-scale manufacturers, each of which employs between 5 and 10 people. The 43 significant industries identified by the survey likely only represented medium-scale production units. According to the survey, there were 50 printing presses, 111 restaurants, 722 scrap and recycling units, 85 units working only for exports, 25 bakeries, and 152 units producing a range of culinary products like chikki, papad, and chana dal. Success tales abound in Dharavi’s lanes, including those of former tea boys who now export ready-mades to US malls and a former low-level worker at a coal industry who has advanced significantly in life by moving into a high-rise apartment! That a 12-year-old boy working a 12-hour shift at a tailoring shop dared to dream, “When I grow up, maybe I’ll also own a factory!” is therefore not surprising.

Dharavi: A New Urban Paradigm

Dharavi, a thriving neighborhood of around 700,000 people, is situated in the center of Mumbai, the financial and commercial center of India. Despite being a poor neighborhood, it is not impoverished because of a lack of grit, initiative, optimism, or social capital. Instead, it results from a lack of access to the facilities, social services, and other possibilities that are available to Mumbai’s society’s wealthier sectors. The people of Dharavi, who live on a landfill that is prone to flooding, make the most of their difficult circumstances. Both the young and the old are active and productive, making a substantial contribution to the city’s total economy. Dharavi is where over 75 percent of recycling in Mumbai is done. Small workshops produce leather and embroidery goods for customers in far-off locations like New York or Paris. Potters fire their kilns, tiny manufacturing businesses buzz, women toil in food processing plants, markets are bustling, and most kids go to school with the goal of continuing to college or getting a professional degree. Dharavi’s small businesses and

services bring in between US\$500 and US\$65 million for Mumbai’s economy. Religious holidays and social gatherings are joyfully observed. On streets loaded with food, gold jewelry, clothing, mosques, temples, and churches, women wearing lovely saris greet one another. Despite this seeming regularity, the residents of Dharavi live in constant fear of eviction because most of them are unauthorized occupiers of public land that the government is eager to renovate for higher-income housing and offices. Due to the lack of sufficient land to support the fast economic expansion that Mumbai has been experiencing over the past 20 years, Dharavi is under tremendous development pressure. (Chen, 2019)

Recently, Dharavi has also become a symbol of urban problems including informal communities in developing countries. While Dharavi undoubtedly raises unresolved issues related to land tenure, subpar construction quality, and inadequate physical infrastructure, it also possesses several positive qualities that should be acknowledged as potential answers to the problem of fast urbanization. These include the tenacious businesspeople and their social capital in Dharavi, the urban fabric’s sense of community, and the live-work model that was developed there after the industrial revolution. The severe housing shortage for all economic strata of Mumbai as well as the rest of the country is intimately related to issues in Dharavi. At the beginning of India’s 11th Five Year Plan (2007-2012), the official shortage of urban housing was expected to be 25 million units, with the lowest income groups accounting for most of it. The actual shortfall is probably far greater because it is difficult to find accurate data on urban migrants. Because of this, those who live in great poverty have little alternative but to build their own shelters wherever there is free land, including public lands, rail tracks, sidewalks, and ecologically sensitive regions like flood plains and wetlands. (Federighi, 2018)

With clouds of various development plans floating overhead, Dharavi was lately in the eye of a storm. Dharavi is seen as a special entryway to Mumbai’s change by local real estate developers, international capital investment firms, and the state government. Will these clouds break up, we wonder?

Current redevelopment plans appear to ignore the existing, deeply ingrained environment in Dharavi and instead treat it as a blank canvas where new buildings and thoroughfares can be built. Many locals’ lives and businesses could be at danger if these measures are not opposed. Now, suddenly, outsiders are creating plans without consulting the people of Dharavi, plans that seem to minimize what the locals have accomplished and are not guided by their long-term investments or best interests. The very substantial profits that developers and the State Government are looking to make are, in fact, what are driving efforts to redevelop Dharavi. The global South has largely ignored the needs of the poor in urban areas, and the gap between the planned, formal city and its shadow, the informal city, is widening. Ironically, official cities rely significantly on the labor and vigor of people who live in slums or shantytowns, but they hardly ever provide support for these employees. For the State, Dharavi represents a long-standing development nightmare for which it hasn’t been able to create any effective strategies for upgrading and improving on a wide scale, much like Kibera in Nairobi, one of Africa’s largest slums. It is, to put it succinctly, a gold mine for international financial institutions. (Gupta, 2012)

However, neither are there sufficient funds to replace these self-built shelters nor are new policies from municipal planners anticipated to lessen the dangers associated with lending to the poor. Global investors lack the expertise and motivation to handle decentralized, incremental investments, which Dharavi communities have mastered. But even the clouds have a silver lining. The Maharashtra government, which was first in denial about the locals' unhappiness, now recognizes the necessity of listening to their concerns. Of course, Dharavi residents and dissidents aren't trying to stop investment or growth; they just want to make sure that it will advance their lives as well as the overall city. The way that informal settlements respond to the needs of small companies and housing is something that Dharavi can teach us a lot about. The truth of how the poor not only survive, but prosper without handouts or charity, is flexible work schedules, home-based vocations, and enterprises of various scales connected with houses.

Objectives and Methodology:

Having studied the demographics and the growth of the region, it is evident that for the populations it serves, Dharavi's natural urban structure serves as a significant archive of memory and history. The area has developed incrementally over several decades and it continues to do so even today. To understand this in depth, I studied how the cities are shaped in the Urban Age, which can be found in the second chapter. I looked at some of the studies done by the London School of Economics (LSE) as well as by Council on Tall Buildings and Urban Habitat (CTBUH) to understand the psycho analysis of residents in a high-rise structure in Mumbai, since that was the strategy adopted by the architect in the latest redevelopment proposal. I was thus able to derive the problems faced by the residents in such a set-up and eventually derive my strategies to avoid those fallouts.

All parties involved, including the current residents, are aware that Dharavi needs to be redeveloped, looking at the current infrastructure conditions and innumerable problems concerning space and sanitation, but there are some key questions that prevent consensus on a solution. Is the tabula rasa strategy that the Slum Rehabilitation Authority is presently considering the only course of action, or are there other possibilities that need to be considered? Can development take place gradually to conserve and improve the social capital of Dharavi's communities and prevent the loss of a significant urban asset for Mumbai? Can the Dharavi urban fabric's scale and historical context make it a significant asset for the redevelopment? (Dighavkar, 2021)

To get answers to all these questions, I studied the redevelopment schemes right from 1972 to understand why they never took off. The third chapter 'Makeover or Takeover?' talks about the same in detail and highlights the key aspects and reasons for the failure of the latest redevelopment scheme which was proposed. From that I was able to gather the starting point for my thesis. Initial reactions to this location were frequently contradictory wishes to preserve and replace it. The aesthetic of informality, the intense sociality, diversity, and productivity of the streets and lanes—it is a wonderfully fascinating and varied urban assemblage—are all connected to the need to conserve at the same time. Poor sanitary, ventilation, lighting, open-space, and overcrowded circumstances in some regions are to blame for the desire to erase.

Before I could dive into the site analysis, selection, and design concepts, I wanted to understand how such informal communities operate over the world as well as in the Indian context. The 5 precedents analyzed in Chapter four helped me to compare the tools used in such communities, for designing. I was thus able to compare the results for all the case studies which helped me to derive concepts for my thesis.

Post that, I did a comprehensive analysis of all the existing zones in Dharavi to understand how the multiple industries work in such a diverse and complicated neighborhood. I was then able to select a critical site out of 85 different communities where I can carry out my design interventions and strategies, which is outlined in chapter five. The sixth and the final chapter talks about my design interventions and toolkit of strategies for progressive self-development of the residents.

SHAPING CITIES IN AN URBAN AGE

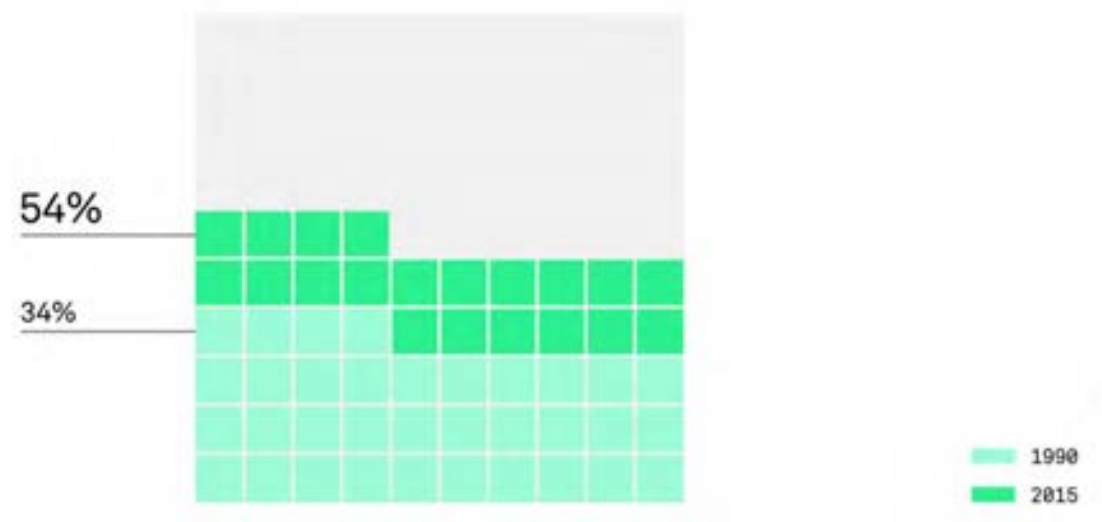


Fig 2.1. Comparison of people living in the slums in 1990 and 2015
Source: <https://urbanage.lsecities.net/data/living-in-slums-mumbai>

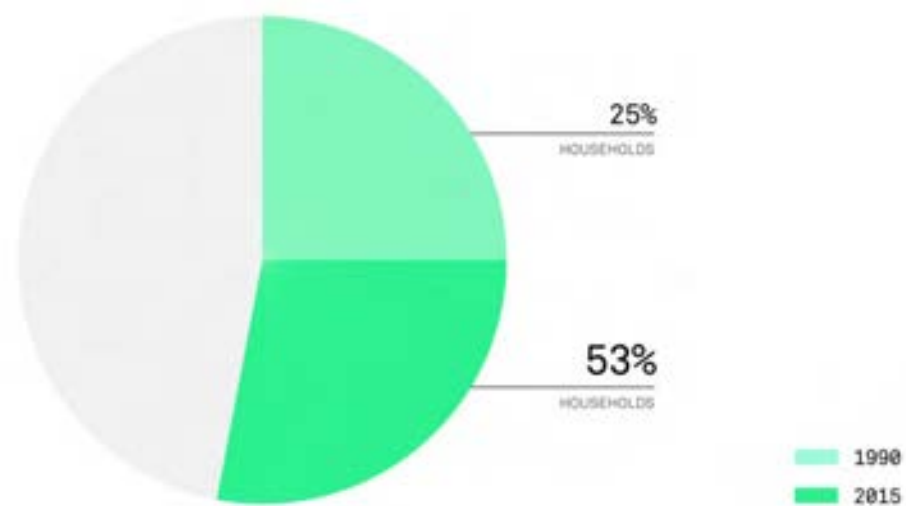


Fig 2.2. Comparison of people having households without toilets in 1990 and 2015
Source: <https://urbanage.lsecities.net/data/basic-facilities-mumbai>

The Urban Age

90% of the world's urban population expansion, or 2.5 billion additional people, will reside in Asia and Africa by 2050. Cities are complex entities. Urbanization is a complicated, drawn-out process that will never be finished. Urbanization is a naturally open, iterative, and unfinished process. One method to comprehend how these global trends affect cities to be more sustainable, more integrated, and more equal in the future is probably by understanding different levels of economic, social, and environmental complexity. Cities have evolved into vital areas for human existence and creativity of both good and evil things. To understand how the cities are shaped in an urban age, I did some study based on the research done by the London School of Economics and Council on Tall Buildings and Urban Habitat (CTBUH)

The London School of Economics (LSE)

With over 40 million inhabitants, Mumbai will be the greatest metropolis on Earth by 2050 and serve as the economic engine of the soon-to-be most populous country on the planet. A city with a limited land supply and increasing traffic is under pressure to accommodate the pace of growth as more than 50 new inhabitants move in every hour. Mumbai has grown substantially, although its densities have hardly changed. Mumbai amplifies the tensions of modern urbanization because more than half of its residents live in slums and the bulk of daily trips are taken by foot, bicycle, or public transportation. In Mumbai, where land is in short supply, informal settlements can be found all around the city, frequently in prime spots near open parks or the water. In a city where over 90% of the population does not rely on cars for their livelihoods, Mumbai's compact built form supports one of the most effective urban systems in the world, sustaining jobs and transactions associated with the informal economy.



Fig 2.3. Comparison of area per person living in 1990 and 2015
Source: <https://urbanage.lsecities.net/data/room-for-living-mumbai>

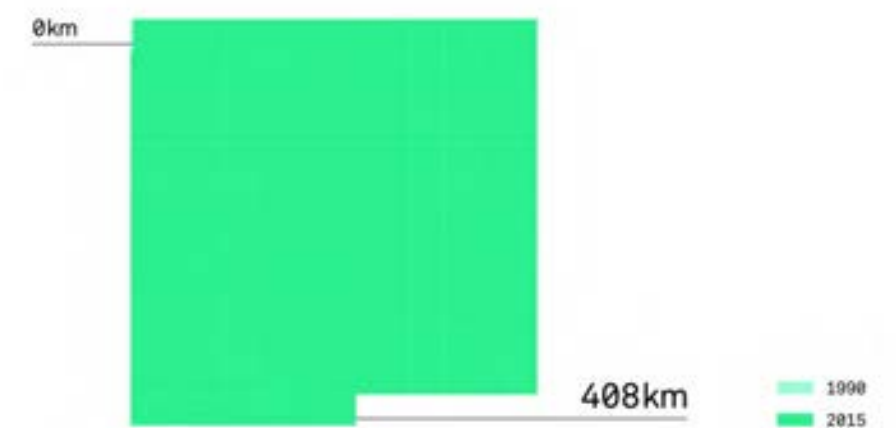


Fig 2.4. Comparison of investments in road network in 1990 and 2015
Source: <https://urbanage.lsecities.net/data/investing-in-roads-mumbai>

The London School of Economics (LSE)

As we can see from the figures, there is a substantial increase in the number of people living in the slums from 1990 to 2015. The density of the people living in an area is increasing day by day, thus deteriorating the living conditions, depriving people of their basic needs and amenities. (Burdett, 2015)

We can also see that the household started lacking basic facilities, such as toilets in the household, as the population started to increase. This was more evident in informal settlements.

While the density kept increasing over the decades, the room for living per person did not increase by much since a lot more people started cramping up on a smaller piece of land.

While development started happening gradually, the road network was developed in the major city areas. Narrow streets and lanes started to form in these dense settlements. Thus, we can say that among the most significant issues facing cities around the world and particularly in India, are inadequate housing and inadequate infrastructure. A priority for action in the modern urban period is these circumstances. The difficulties of raising urban quality of life in densely populated places experiencing rapid growth are best shown by the slums of India. Although the issues with precarious urbanization are not unique to the nation, India's cities are unmatched places to think about ways to better meet the expanding population of urban dwellers and their variety of requirements. (Burdett, 20195)

Council on Tall Buildings and Urban Habitat (CTBUH)

One might think that creating high-rise buildings can be one of the solutions to address the problem of densities in a smaller area. Hence, I started doing some research into why that scheme would or wouldn't work in the area I am dealing with. There was a lot of resistance to the process of redevelopment of Dharavi in general and the latest schemes where the architect wanted to house the people in tall towers (detailed description in Chapter 3). So why did it never work? To answer this question, I studied the research done by CTBUH on Psychoanalysis on Tall-buildings habitants in Mumbai.

A sizeable portion of Mumbai's population is moving from low-rise to high-rise structures, whether they are homes, workplaces, hotels, or slum restoration initiatives. A person living on the third floor finds himself having to acclimatize to the excitement of the thirty-first floor. Some people have been sought out and others have been compelled to move from a low-rise to a high-rise structure. Mumbai is therefore the perfect location for polling those who have recently undergone this drastic transformation. People who live in high-end luxury high-rises as well as low-income housing for those who have been rehabilitated from slums were surveyed. A questionnaire asking about many elements and problems of living in a tall building was produced and sent through a network of professional connections, families, and friends. Online surveys were completed anonymously by respondents. Personal interviews were done with residents of a Mumbai SRA construction project. Results for objective questions have been displayed as graphs with percentage values.

Concerns over the construction industry's negative environmental effects and the deterioration of the environment appeared to be widespread. Being actively "green" makes a building appealing as a place to live in, which increases its social sustainability. Living in a structure that absolves them of any direct or indirect environmental responsibility makes people feel more at ease. Therefore, a building that meets certain sustainability standards and makes use of clean technology may sell more quickly than one that does not. Figure 2.6 shows the most common concerns of such people in a high-rise structure. Since they are used to living in the low-rise setup, the idea of dependency on the elevator for going higher floors is something that worries them. The maintenance costs are also increased thus making it unaffordable to some of the residents (Figure 2.6)

Thus, people coming from such communities prefer earthy materials and finishes more since it might be economically viable for them (Figure 2.7). They also tend to focus more on natural light and ventilation rather than increasing their costs by dependency on artificial lighting systems. Thus, moving forward in my design, these were some of the concepts which I thought of picking up since my design was dealing with people from such communities

Low-income housing in groups is a complex issue. Numerous cottage enterprises that meet everyday necessities may be found in Mumbai's slums and shantytowns, including those that produce small foods, clothing, accessories, genuine leather goods, tanneries, clay pots, and recycled plastics. When people move to a little 269 square foot apartment, they lose the workspace and social connection that slums afford. Vendors, shop owners, potter's community and fishermen must live in ground-level residences where they can store their boats and nets, or pots and closely monitor their stores to accomplish their jobs. Moving to vertical developments upsets their livelihoods (Figure 2.8)

Slums tend to group together because of shared religious preferences, which allows political leaders to arouse emotions during campaigns. However, surrounding clusters of various religious groups peacefully coexist. When slums are moved into buildings, this coexistence may be lost. Vested interests, such as the dissolution of vote-banks for certain political leaders, are strongly opposed to the mixing of clusters. Thus, their livelihoods depend on the sprawling layout and the spill out spaces in front of their houses, they do not appreciate the high-rise set-up.

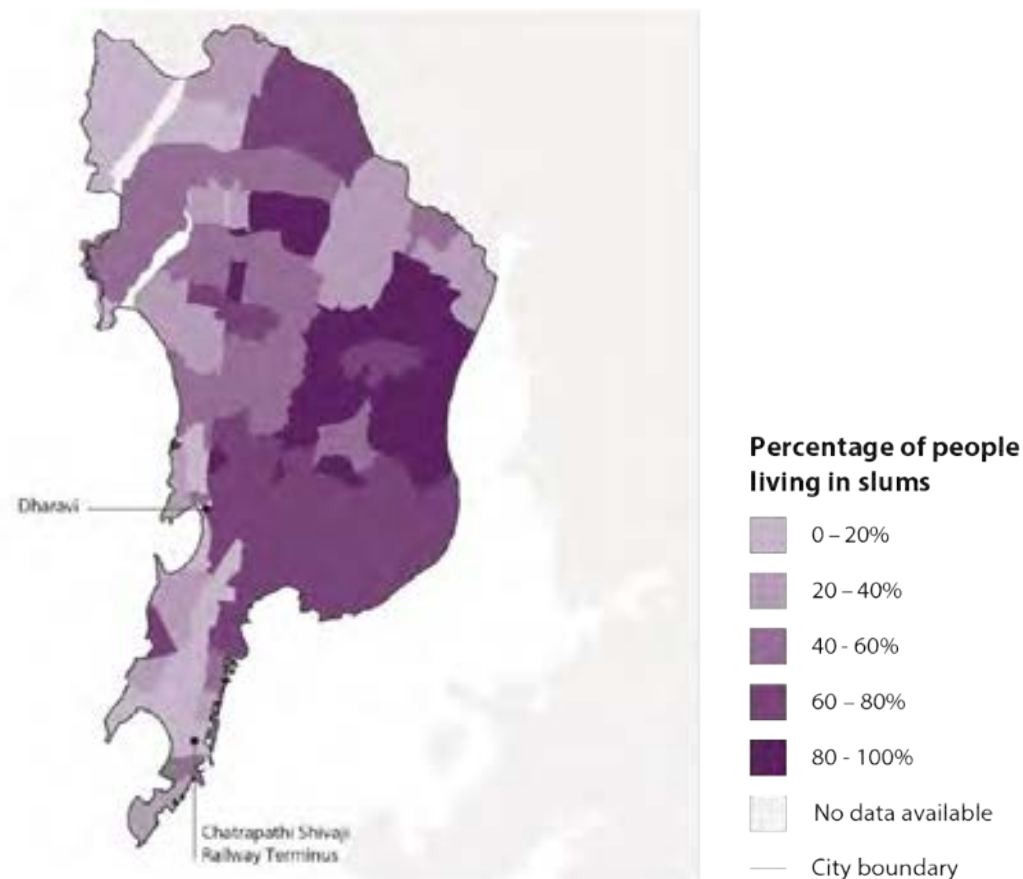


Fig 2.5. Housing India's urban poor, case of Mumbai city
Source: <https://urbanage.lsecities.net/data/housing-india-s-urban-poor-mumbai-2007>

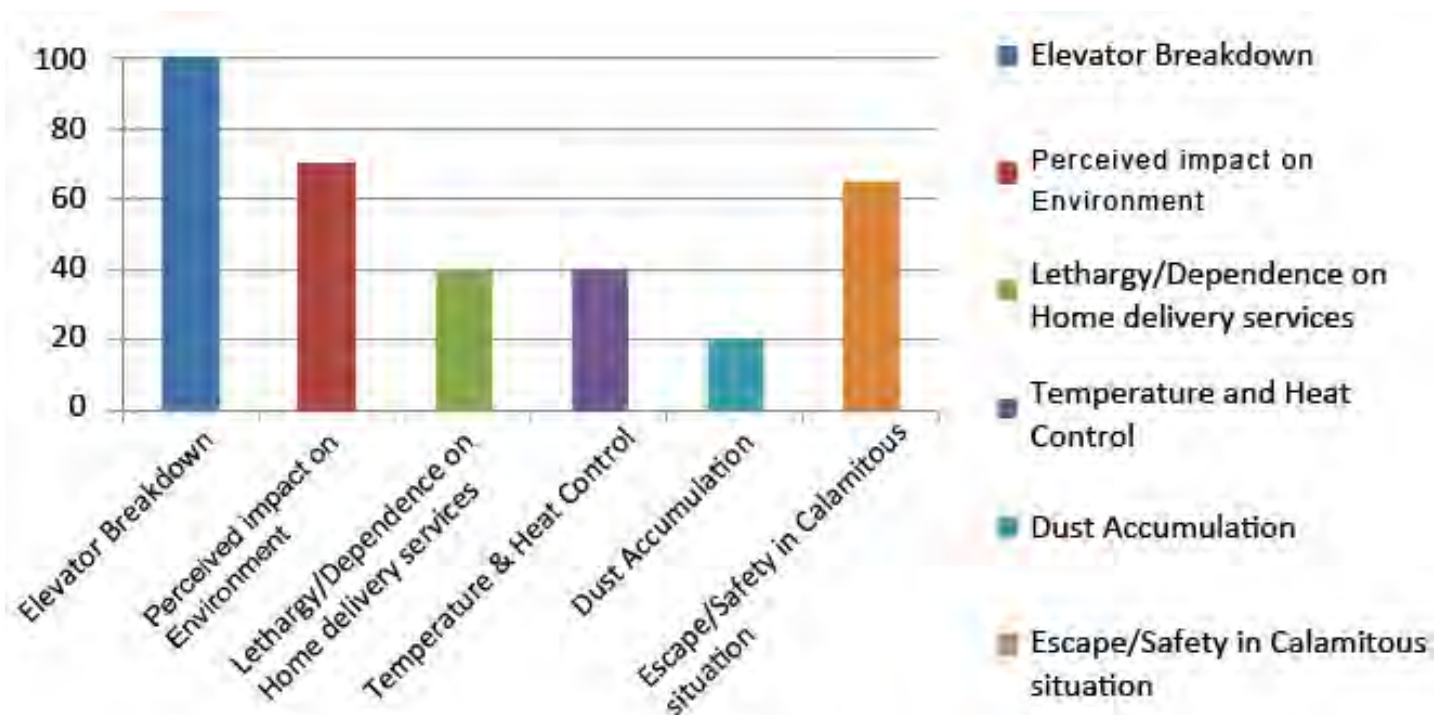


Fig 2.6. Most common concerns in a high-rise residence
Source: CTBUH Research, Redrawn by Author

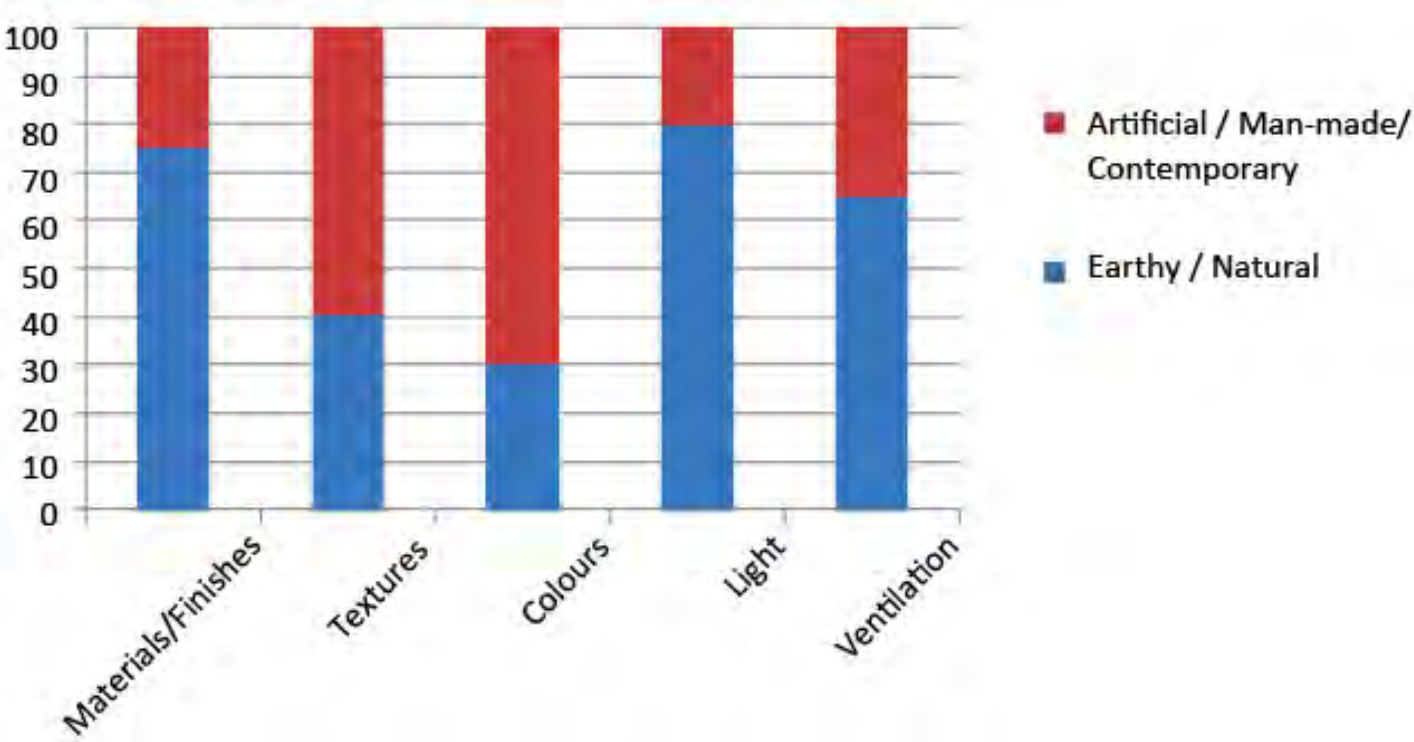


Fig 2.7. Consumer preferences
Source: CTBUH Research, Redrawn by Author

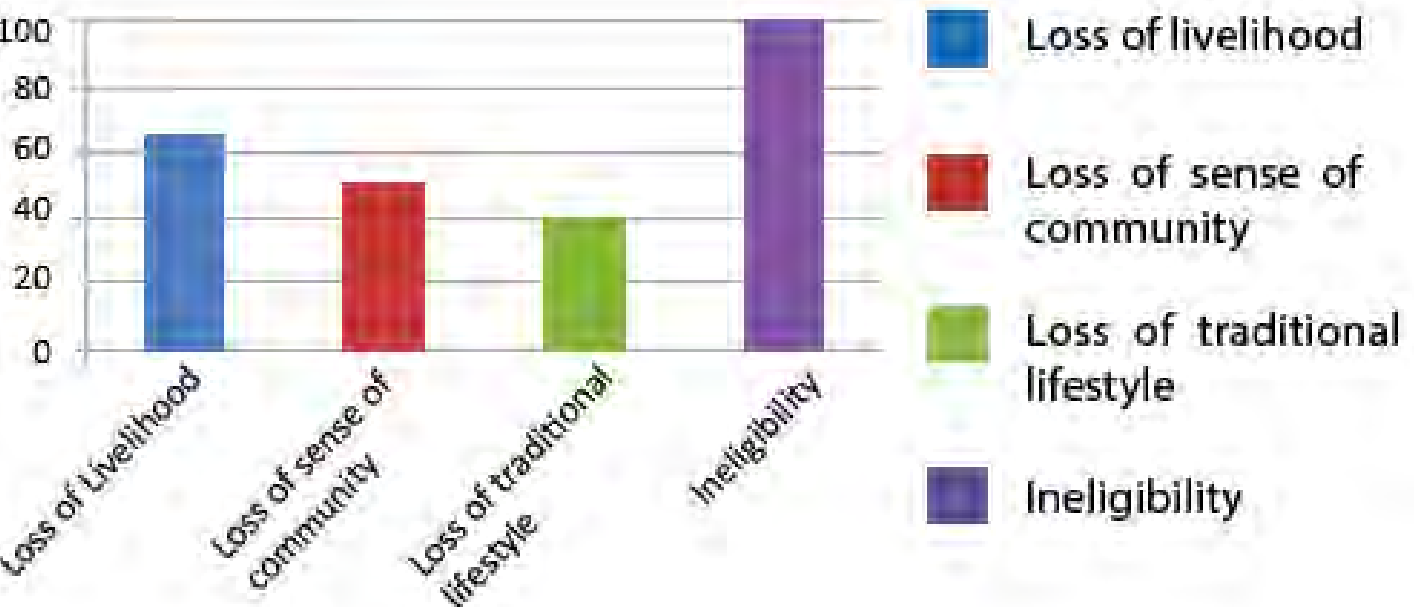


Fig 2.8. Primary reasons for not wanting to move
Source: CTBUH Research, Redrawn by Author

Council on Tall Buildings and Urban Habitat (CTBUH)

If a private developer builds a housing complex for slum rehabilitation on the same plot of land, he is permitted to use the space made available by the relocation of the slums for commercial endeavors at discounted rates. Due to the involvement of private developers competing for less expensive land, a terrible pattern of poor buildings being clumsily built for the rehabilitated slum people nestled in the shadow of a luxurious high-rise tower has emerged. These apartment buildings are cramped spaces without basic comforts like lighting, air, privacy, and sanitation (Figure 2.9).

The residents of the renovated buildings now incur significantly higher costs for housing maintenance than they did in the past due to licensed water and electricity connections and the creation of a co-operative housing society. This deters those who eventually leave these apartments and frequently relocate to another slum complex where water and electricity are provided free of charge due to unlawful usage. So essentially, they end up forming one more slum and we do not solve the core problem at hand.

In addition to architectural regulations, structural safety is a concern because little care has gone into their design. It becomes riskier to ensure the safety of such shabbily monitored structures as they get bigger.

Such low-income housing projects should be turned into business models for the developers to prevent them from becoming vertical slums. This would motivate designers to better plan and design them while also taking care of their maintenance. Keeping these factors in mind, I decided to explore the approach of having low-rise model for my project.

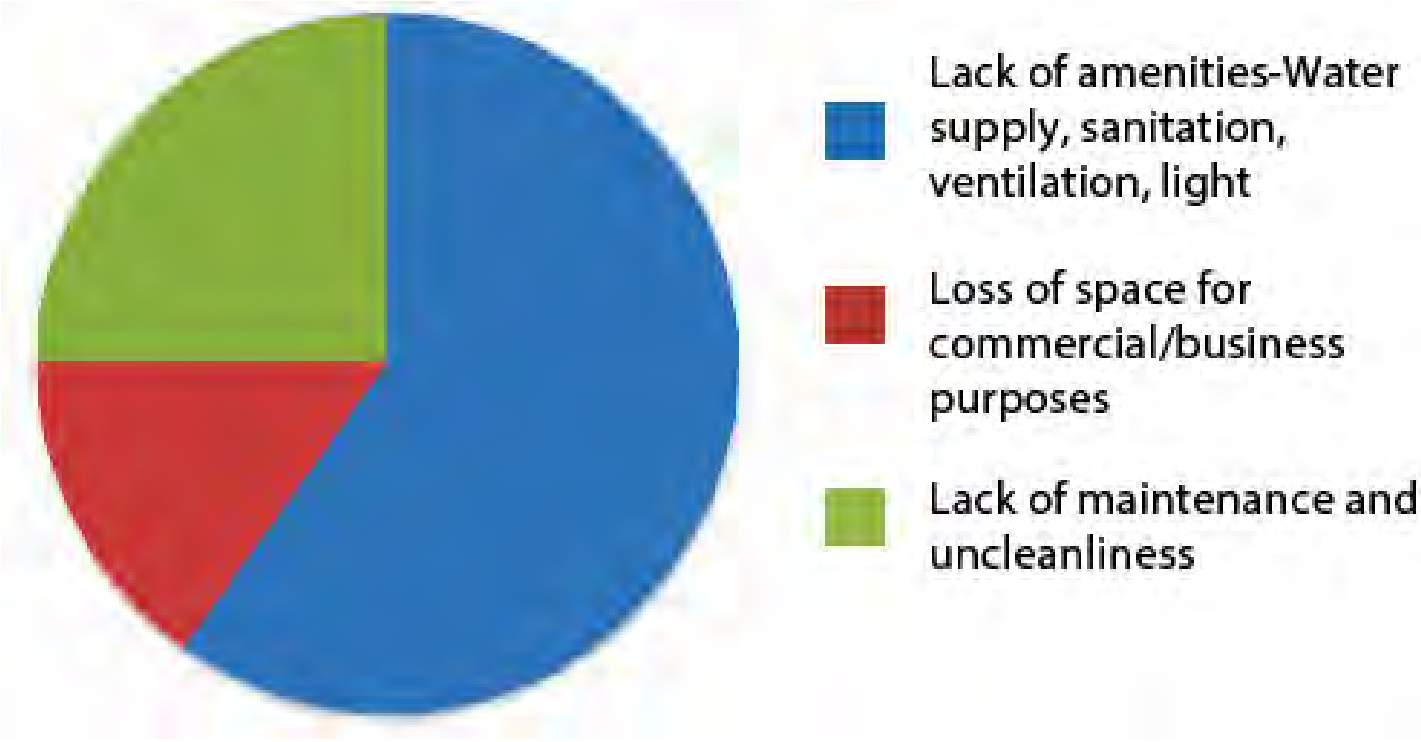


Fig 2.9. Grievances after moving in
Source: CTBUH Research, Redrawn by Author

Urban Analysis

We looked at how the cities are slowly shaped in an urban age and particularly how the conditions in Mumbai developed over time along with the problems faced by the residents in the high-rise model. To further strengthen my knowledge about it, I also studied why the urban vibe is important in any setup.

Urban fabric should be built on theory, and ideally this theory will result in a diversified and engaging metropolis. However, physical design frequently lacks fundamental philosophy and is motivated more by money and beauty than by functionality. To build new ideas in the future, I was able to learn information and concepts about the urban fabric through studying theories of urban life. The goal should always be to combine the effective aesthetic fabric with the dynamic urban fabric, where life may happen

For creating an analytical instrument that can be used to study the current urban fabric and extract its effective elements, the works of six urban theorists and architects who concentrate on urban living and cover a wide range of theories have been chosen. Theorists were picked for their contributions to urban planning and design as well as their original and precise theories since the area I am dealing with requires certain urban acupuncture. This tool is split into 6 sections: one for each urban thinker..

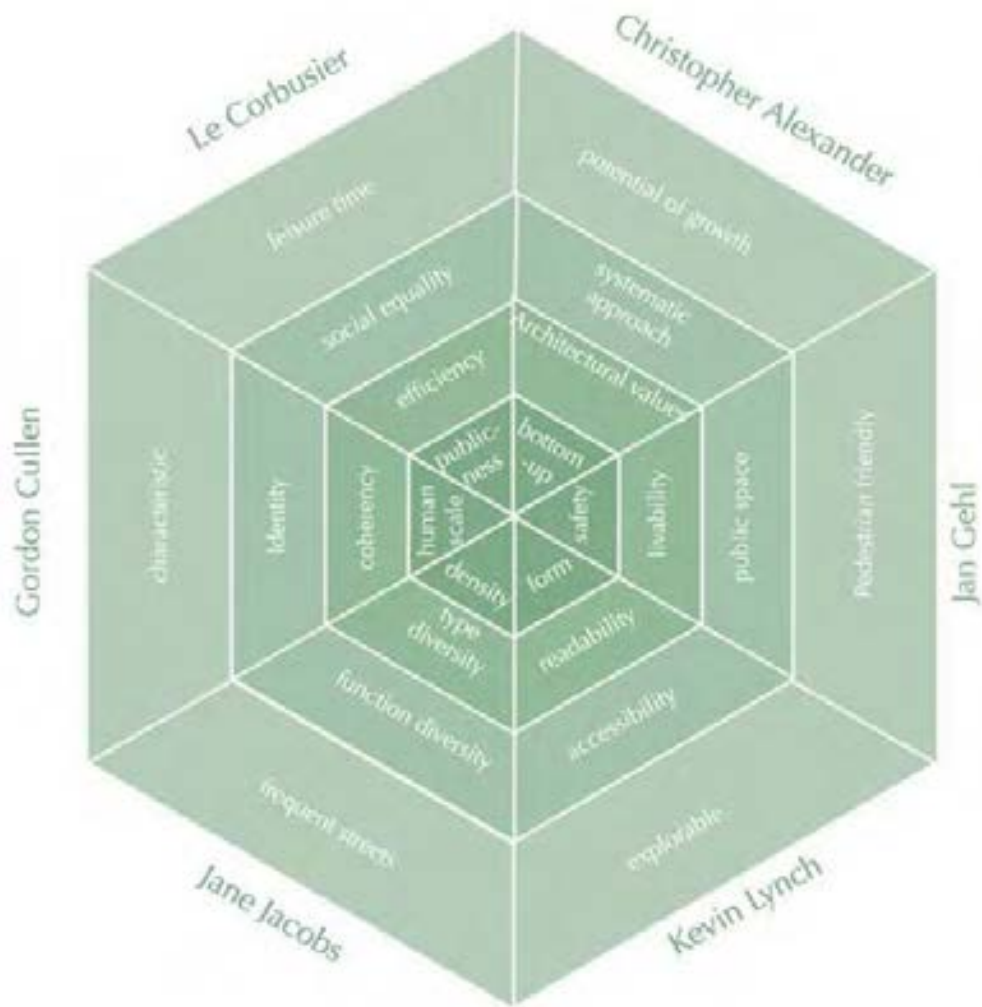


Fig 2.10. Theory of Urbanism as explained by six architects and theorists
Source: CTBUH Research, Redrawn by Author

Le Corbusier

The urban core of Le Corbusier's Radiant City is made up of identical prefabricated high-rise buildings with lots of open space in between. To create a sprawling, park-like city, each block is set onto pillars, leaving more space for nature. The apartment buildings feature a catering department that handles daily tasks, so the residents have more free time. The time saved gives the person more time to write, ponder, play sports, and engage in other leisure activities. The goal of Le Corbusier's architecture is to enhance human life. The concept was missing some of the most fundamental concepts of urban life, though. The idea of street life, which gives the city its vitality and excitement, is one of these. This theory should not be considered for its physical form, but the idea of lifestyle (Corbusier, 1924).



Fig 2.11. Le Corbusier's radiant city of 1935
Source: <https://www.archdaily.com/411878/ad-classics-ville-radieuse-le-corbusier>

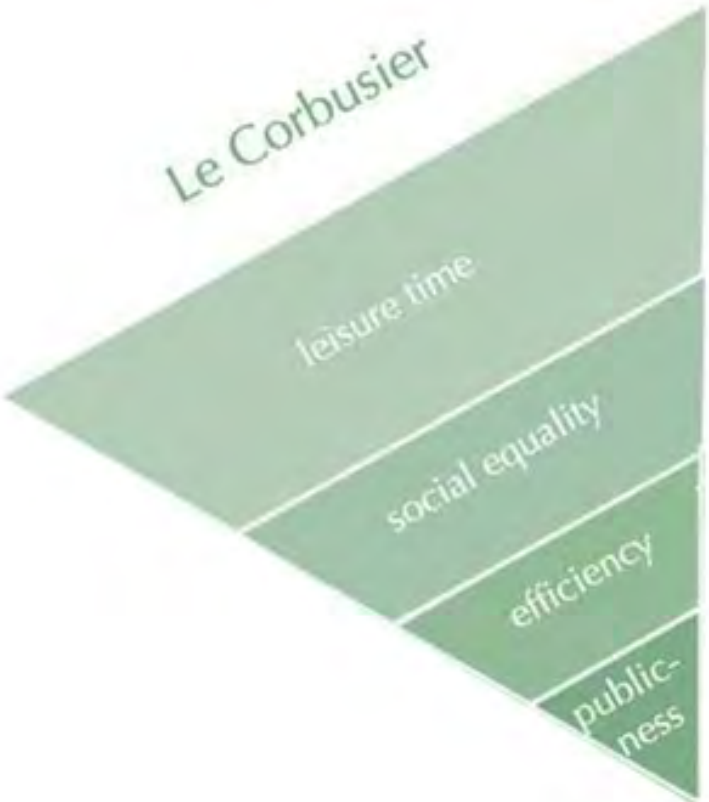


Fig 2.12. Theory of Urbanism as explained by Le Corbusier
Source: CTBUH Research, Redrawn by Author

Kevin Lynch
Kevin Lynch examines how people view the city based on its physical shape in his book, *The Image of the City* (Lynch, 1964). He concludes that humans comprehend a city by creating a mental map made up of five different types of information: pathways, edges, districts, nodes, and landmarks. Each of these components gives the city its shape and helps its inhabitants to understand it. Lynch suggests subheadings for each aspect that explain what makes it work or not. (Banerjee, 2002)

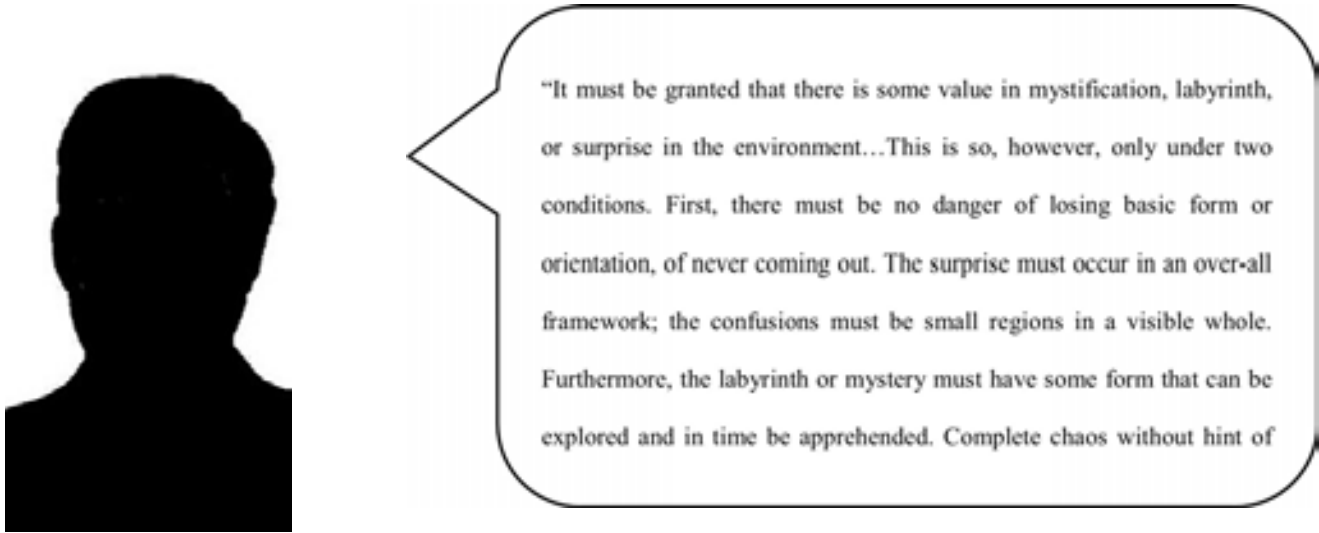


Fig 2.13. Kevin Lynch: *The image of the city*
Source: Silhouette image created by Author in Photoshop

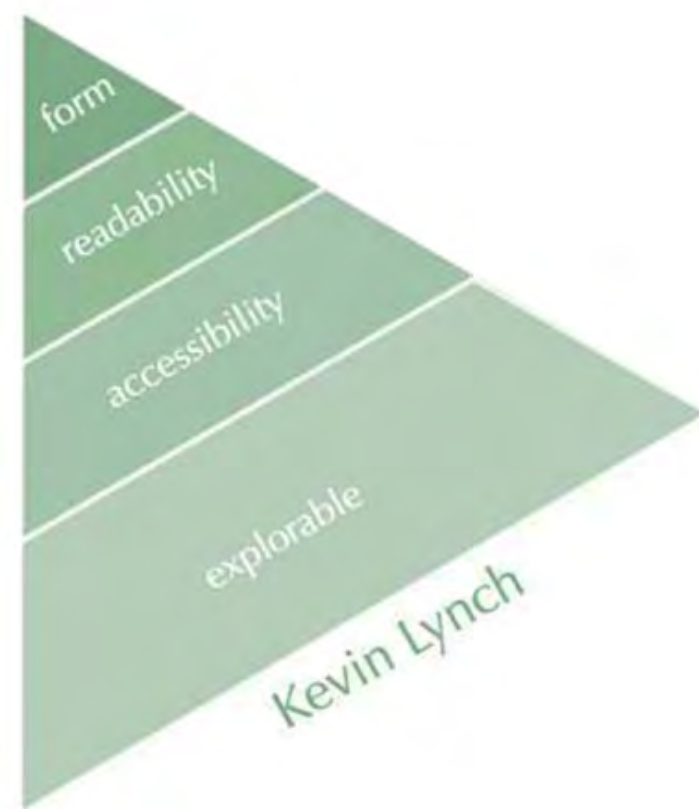


Fig 2.14. Theory of Urbanism as explained by Kevin Lynch
Source: CTBUH Research, Redrawn by Author

Jane Jacobs
Jane Jacobs, an influential urban theorist, frequently emphasizes the value of the urban form and how it connects to urban street life. She is a major supporter of the small-scale urban form, which allows for numerous, overlapping paths and uses throughout the city thanks to compact blocks. She thinks that to strengthen urban fabric, we should look at and take lessons from the city’s successful pathways. (Hughes, 2018)

She is well-known for coining the phrase “eyes on the street,” commonly referred to as the “natural surveillance technique,” (Jacobs, 1969) which expresses the notion that the most effective surveillance is that which occurs organically. People who use or merely observe the outside space keep an eye on what is going on and so naturally create a sense of security. Significant levels of observance also correlate with high levels of usage diversity. To generate “eyes on the street,” front porches and windows should be constructed in a way that encourages pedestrian activity. The street corner was cited by Jane Jacobs as a crucial urban component for the success of the urban fabric. As the location that receives the highest foot traffic on the street, retail is usually located here.

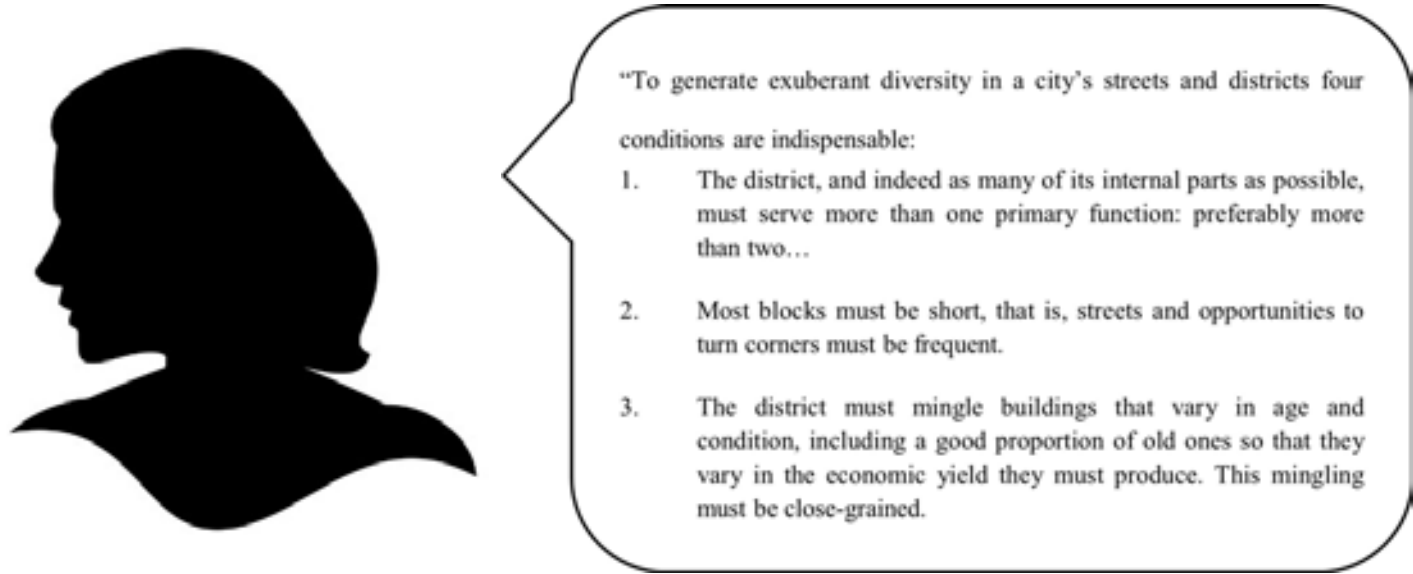


Fig 2.15. Jane Jacobs: *The death and life of Great American cities*
Source: Silhouette image created by Author in Photoshop

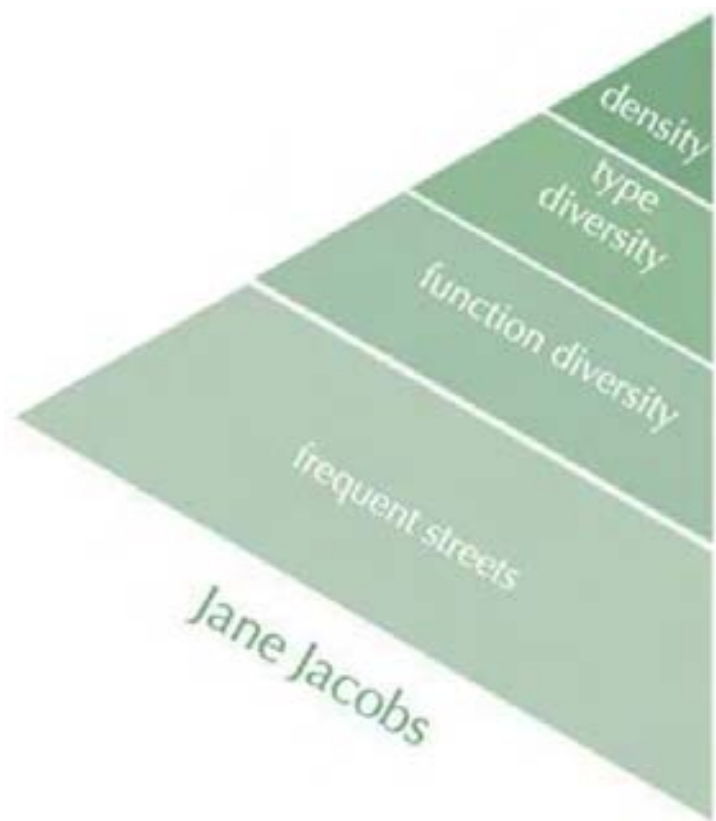


Fig 2.16. Theory of Urbanism as explained by Jane Jacobs
Source: CTBUH Research, Redrawn by Author

Gordon Cullen

Gordon Cullen wrote about the benefits of people congregating in urban areas because they produce a sum greater than the individuals who live there. Communities are preferable to living in isolation, and people are brought together to create a collective surplus of enjoyment. The aggregate of all individuals generates a collective amenity. Like this, grouping buildings together produces a pleasing visual effect that is impossible to achieve when they are constructed separately. (Gibbons, 2015)

- A. Regarding optics: How serial views bring the drama of juxtaposition to life in the city. When the developing view is unknown, a series of events are brought about by the existing and emerging views. This is accomplished by designing the city with various levels, corners, volumes, etc.
- B. Place-related reactions and feelings: These depend on where our bodies are in relation to their surroundings. Understanding how we react to the space we are in and how we use words like HERE and THERE should help us make better use of this in city planning. (Cullen, 2015)
- C. Content-related: The final point discusses the character of the city. We are affected by color, texture, scale, style, personality, and originality. Additionally, how the city should be differentiated to make it fascinating and livable.



A long straight road has little impact because the initial view is soon digested and becomes monotonous. Several things begin to happen in the group (of buildings), which would be impossible for the isolated building. We may walk through and past the buildings. And as a corner is turned, an unsuspected building is suddenly revealed...the space created between the buildings is seen to have a life of its own over and above the

Fig 2.17. Abstract from the book: The Concise Landscape
Source: Silhouette image created by Author in Photoshop

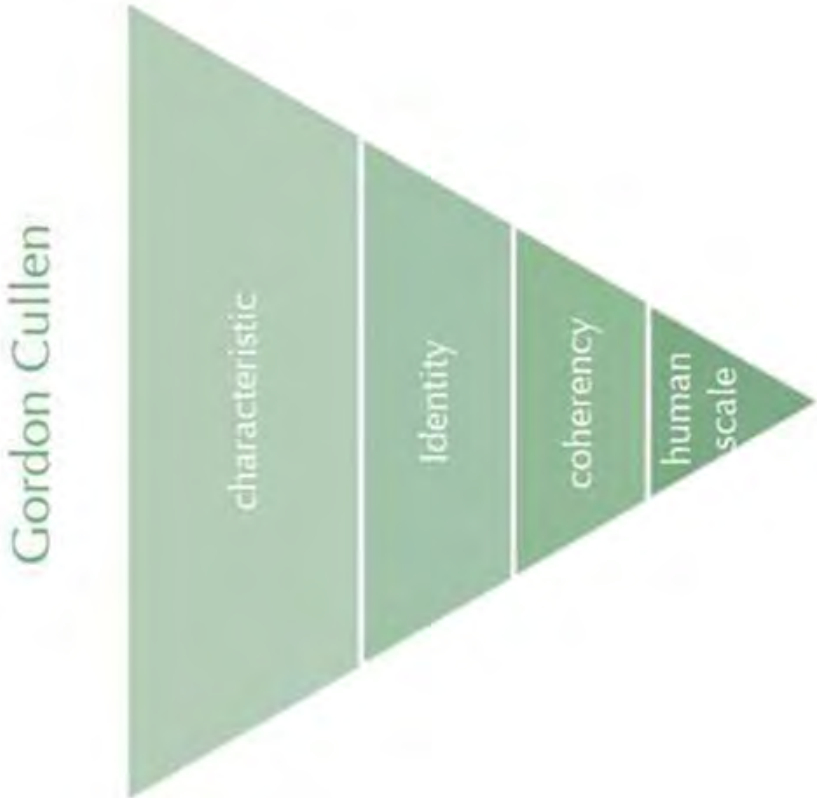


Fig 2.18.Theory of Urbanism as explained by Gordon Cullen
Source: CTBUH Research, Redrawn by Author

Christopher Alexander

According to Christopher Alexander, an effective design can be created by combining several systems and rules. All the techniques follow established patterns that have been proven effective in the actual world. He believes that the city should be created by individual actors and regional laws in respect to its neighbors and not by the concept of a single blueprint (Alexander, 1985)



...As far as I can see, many of the worst characteristics of western modernist cities are being replicated in the planning of fast-growing Chinese cities. The kind of 'New Towns, the Chinese are building look very much like the housing areas we now see being demolished in many places in both western and eastern Europe. Some of the worst models are being used in China to deal with the rapid growth of cities...This, I have to say, is a recipe for disaster, and I certainly see this as a major challenge for the sustainability of cities in the world.

Fig 2.19. Christopher A: A Pattern Language: Towns, Buildings, Construction
Source: Silhouette image created by Author in Photoshop

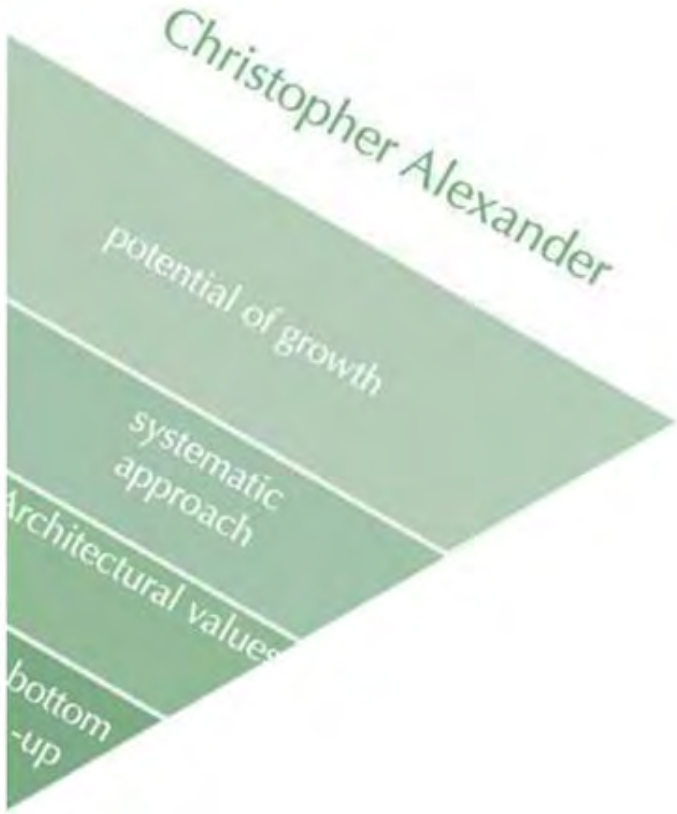


Fig 2.20. Theory of Urbanism as explained by Christopher Alexander
Source: CTBUH Research, Redrawn by Author

Jan Gehl

The concepts of Jan Gehl are supported by study into the urban form. He investigates how various factors, including squares and streets, sunlight and shade, and street width, affect individuals. Jan Gehl emphasizes the necessity for a small, people-friendly city that prioritizes public areas to aid in pedestrian circulation. According to him, creating vibrant, safe cities that people want to use, stroll through, and live in is the key to sustainability. In a nutshell: People must utilize public space if they want safe cities.



This is a fundamental view of the world. It says that when you build a thing you cannot merely build that thing in isolation, but must repair the world around it, and within it, so that the larger world at that one place becomes more coherent, and more whole; and the thing which you make takes its place in the web of nature, as you make it. But in practice master plans fail-because they create totalitarian order, not organic order. They are too rigid; they cannot easily adapt to the natural and unpredictable

Fig 2.21. Jan Gehl's interview from 2008
Source: Silhouette image created by Author in Photoshop

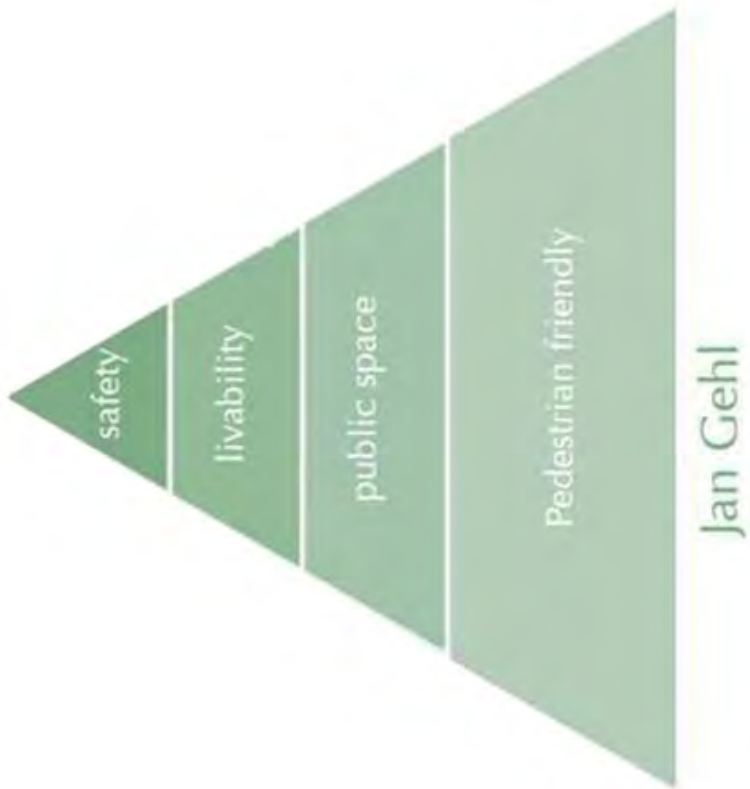


Fig 2.22.Theory of Urbanism as explained by Jan Gehl
Source: CTBUH Research, Redrawn by Author

Thus, after studying the theories chalked out by all these architects, I was able to derive certain principles of good development in an urban area which I could potentially use in my project.

A district should have more than two main functions - functions which bring people to a specific place (offices, dwellings, factories etc.)

The road structure should be maze like -but always with a sense of how to "get out" of it if feeling lost or in a hurry.

Smaller open spaces (horizontal + vertical) creates possibility for extended city life.

Closed off open spaces can act as safe areas for children (schools, nursery ect.) Important: it is also a "city life stopper"

Irregularities in the city scape creates possibility for informal situations to happen.

Certain functions y requires x amount of open space. Either on ground or plateau level. (A school requires a school-yard)

Landmarks attracts people (local+tourist) and bring life into the area.

Larger open spaces should not divide - but strengthen the neighbourhood. It should primarily be placed where the need is - in already dense areas.

Horizontal human scale is 100 meters. Minimum every 100 meters there needs to be a break in the block.

A understanding of the different roads is important for the understanding of city life.
- Primary road should be kept to a minimum, but is needed for transport
- Secondary roads are a mix of functions
- Tertiary roads are only for light traffic

Mono use residential buildings must, within the human scale, be a connection to ground, (that be plateaus or real ground) to ensure city life and a sense of community (belonging) of the residents.

Balconies bring life - the be building and make it more interesting/human to look at. But it can also kill street life by keeping the people .

Fig 2.24. Sketches for principles of good development- Part 02
Source: Drawn by author

These principles were the starting point for the development of concepts and strategies for my thesis.

Dharavi: Emergence in an Urban Context

Dharavi, which is currently recognized as Asia’s largest informal neighborhood and has about 1 million residents in a globalizing Mumbai used to be thought of as a town within the metropolis. When Mumbai was undergoing different changes, it was a fishing community called Koliwada prior to the colonial era. Mumbai (formerly known as Bombay) was divided into two main areas up until the end of the 19th century, with the native area to the north of the colonial area and the colonial half to the south. Dharavi was located on the northernmost edge of the original territory, close to the sea, giving it its reputation as a fishing community. The first tannery opened in Dharavi in 1887, bringing Muslim tanners from Tamil Nadu with it. For the first time since its inception, Dharavi was inhabited by new social classes and professions, elevating its economic standing through further business ventures. Dharavi began to take on her current form at this point. Geographically, Dharavi is in the prime zone of Mumbai city. It’s one of the major economic hubs for Mumbai, being in the Central business district.

Over the following few decades, Dharavi underwent a significant demographic and geographic area change. Immigrants began forming communities and creating communal buildings like temples and schools. As additional immigrants from various countries joined the initial settlers, the number of immigrants grew steadily over time. As shown in the picture below, the city began reconstructing its center and expanding towards the north to handle the massive influx of people. Dharavi, which was formerly part of the city’s northern perimeter, is now located in the center. (Baweja, 2015).

Dharavi is 519 acres of informal settlement with variegated nature of land use. Dharavi has an enormous spread almost the size of a small township or around half the size of Central Park in Manhattan. What is even more intimidating than its size is the density of population, an estimated of 18,000 persons per acre. In this densely packed area one can find 27 temples, 11 mosques and 6 churches. Most of the structures are declared as hut by the Government, due to their construction quality and structurally unsound nature. Most of the Commercial space and major industries are lined along the main roads and main streets whereas the housing is in the inner clusters. In some communities, there is no clear distinction between the living, commercial and social spaces. Major industries are in the western most part whereas smaller industries are scattered in other parts of the settlement hiding in between housing or sometimes on upper floors of housing. (Allaway, 2020). The common characteristic feature between all neighborhoods are houses connected to vast labyrinth of narrow streets, public toilets shared by the neighborhood, lack of basic infrastructure such as toilets, lack of light and ventilation as the houses are built close to one another without setbacks and more people living in a small space. The settlements converge and part in mysterious ways, where one Nagar ends, and another begins only the residents can tell us.



Fig 2.25. Dharavi's geographical location
Source: <https://urbanage.lsecities.net/data/housing-india-s-urban-poor-mumbai-2007>



Fig 2.26. Dharavi's land-use map
Source: <https://urbanage.lsecities.net/data/housing-india-s-urban-poor-mumbai-2007>



Fig 2.27. Unequal scenes by Johnny Miller
Source: <https://unequalscenes.com/mumbai>

Fig 2.28. Slums and sprawl by Johnny Miller
Source: <https://unequalscenes.com/mumbai>



Fig 2.29. "Life in Dharavi" by Swaminathan is licensed under CC by 2.0.

Source: https://commons.wikimedia.org/wiki/File:Life_in_Dharavi.jpg, License: <https://creativecommons.org/licenses/by/2.0/deed.en>



Fig 2.31. "Unclean streets of Dharavi" by Kounosu is licensed under CC by SA 3.0.

Source: https://commons.wikimedia.org/wiki/File:Dharavi_Slum_in_Mumbai.jpg, License: <https://creativecommons.org/licenses/by-sa/3.0/deed.en>



Fig 2.30. "Inside the Dharavi slum" by Mark Hillary is licensed under CC by 2.0. Source: https://commons.wikimedia.org/wiki/File:Dharavi_slum_in_Mumbai_India_August_2009_4.jpg, License: <https://creativecommons.org/licenses/by/2.0/deed.en>



Fig 2.32. "Sanitation issues in Dharavi highlight inhabitable conditions" by Kristian Bertel is licensed under CC by SA 4.0. Source: https://commons.wikimedia.org/wiki/File:Slum_in_Dharavi_-_Kristian_Bertel.jpg, License: <https://creativecommons.org/licenses/by-sa/4.0/deed.en>

“MAKEOVER OR TAKEOVER”

Even during the colonial era, Dharavi was of concern to authorities, particularly during the outbreak of the bubonic plague in 1896 because of the filthy conditions in the neighborhood. The government then proposed a redevelopment plan after deciding to give the neglected Dharavi neighborhood more attention and better facilities after the epidemic was eliminated.

Development for whom?

“The Fortune at the Bottom of the Pyramid” is a book by the late management guru C.K. Prahalad. Because it highlighted how the impoverished represent a big and mostly untapped market for industry, the book became a best-seller. It recommended businesses to modify their plans considering the needs of the underprivileged. For instance, the idea that poor people can only spend small sums of money at a time led to the popularity of selling tiny sachets of tea, sugar, or shampoo. It wouldn’t be overstating things to suggest that the initial goal of the redevelopment of Dharavi was to earn a fortune by taking advantage of the land’s worth, which served as the foundation for the lives of the poor.

Pre Independence-Redevelopment proposal

On May 31st, 1944, the Municipal Redevelopment Plan for Dharavi was made public. Despite the lack of papers for the Dharavi masterplan, a masterplan for the Koliwada (fishermen community) was discovered, in which it was suggested to split the village in two, with one-third set aside for the fish market and the other half for Koli (fishermen) houses. These intentions were never carried out, however, because India attained independence not long after.

Post Independence-Redevelopment proposal

Various attempts have been made in the past, beginning with the post-independence period, to improve the situation in Dharavi before the Redevelopment proposal of 2004. However, none of them could be fully implemented because of a lack of funding, a gap in the survey, and a lack of knowledge. This resulted in an even more rapid incremental development during those years, which helped to create several clusters as well as business facilities and other unofficial sectors

The initial discussions about redevelopment date back to the 1950s, when Dharavi had already been encircled by the city and had developed into the center of the informal sector. At the time, the Maharashtra government suggested plans for rehabilitation, with the goal of demolishing illegal structures and relocating residents into long-term residences in a separate area of the city. And while a few evictions occurred because of this strategy, the government lacked the resources to erect long-term structures, leaving many of the occupants homeless. The failure of the redevelopment plan was caused by this factor.

While this was going on, there were more and more immigrants arriving every day, and by the 1960s, there were about 1.5 million. (Saglio-Yatzimirsky, 2013) When it was first conceptualized a few years ago, the Dharavi Redevelopment Project (DRP) plan called for the separation of Dharavi into 5 sections. Consortia of national and international developers were asked to submit bids to build free housing and infrastructure for the people of Dharavi. The government required the developers to pay a premium in exchange for the Floor Space Index (FSI), which would allow them to construct more commercial buildings and other structures to resell on the open market. The free housing and infrastructure were cross subsidized with a portion of the profit. Developers would have made windfall profits because of the inflated land values in the area, and the government would have made a ton of money. However, a common query is whether land should be viewed primarily as a source of income for governments and developers.

The initial suggested method for redeveloping Dharavi has significant drawbacks. For lack of room, we shall concentrate on a select selection of them. The fact that the entire plan was developed without any community input and is a classic example of top-down planning is arguably the most significant criticism. Even worse, the plan openly sought to eliminate the public’s consent to the type of growth that would occur. Earlier slum restoration programs required the permission of at least 70% of slum residents, and even if this requirement was not adequately executed, it provided a democratic check on the developers’ plans. The question of whose interests such redevelopment serves arises if the public is not consulted at all during the process. Unsurprisingly, global money and its local affiliates are the answer. (Babar, 2022)

The topic of democratic participation or community involvement is central to the concept of development and is not merely a theoretical or scholarly matter. It can be incredibly destructive to ignore other people’s involvement. For instance, the types of urban structure and design the plan had in mind would have meant the annihilation of the Dharavi population’s means of subsistence. Since around 80% of its residents both live and work there, Dharavi has a unique quality that is the intimate relationship between home and workplace. The multitude of small firms and organizations would not be able to survive in buildings of 30 or 50 stories. People wouldn’t stay in the new Dharavi if their means of subsistence were destroyed; instead, they would go to a slum where they could carry on with their daily lives. This freshly constructed housing would have changed hands quickly, and the neighborhood would have undergone gentrification due to the acute housing crisis in Mumbai. Dharavi would have grown into a sizable residential and commercial complex without any accommodation for the underprivileged

The fact that there was no baseline survey, no transport study, no environmental assessment, and no mechanism in place to coordinate the proposed infrastructures in various sectors with one another and with the rest of the city’s infrastructure at the time the original plans were proposed must be emphasized.

Was there any attempt to determine whether these products and services would be offered in the new Dharavi in an environment where water and power are in low supply? It was unknown how many households would need to be relocated because there had never been a survey. Again, since most huts in Dharavi have one or two mezzanine floors, there had not been an attempt to count them and consider whether those individuals and families living and working there were eligible for rehabilitation. On the administrative side, no processes were established for handling grievances or resolving conflicts of interest. Another remarkable feature is the lack of a Development Plan for Dharavi, which was mandated by law. (Hoffmann-Brandt, 2017)

It appears that the original concept has been abandoned, at least temporarily, for reasons that are not obvious. Most of the international partners have withdrawn from the fight because of the global financial crisis and the numerous uncertainties that surround the project. It's also possible that Dharavi's citizens' opposition influenced the result.

Since many years ago, the Dharavi Vikas Samiti (Dharavi Development Committee), a local affiliate of the National Slum Dwellers Federation, has maintained a presence in Dharavi. Residents of Dharavi have formed the Dharavi Bachao Andolan or Save Dharavi Campaign during the past few years by banding together in a diverse coalition of political parties, NGOs, various social groupings, and individuals. This grassroots organization challenged the current plans and began working with a team of experts, including retired officials, architects, planners, and NGO representatives to develop alternatives. This group later became known as the Committee of Experts (CoE). The Institute of Kamala Raheja Vidyaniidhi KRVIA architects offered their expertise and began to examine Dharavi by use of its citizens' perspectives. Consequently, for instance, instead of drawing random lines throughout Dharavi to distinguish between various neighborhoods, an effort was made to group cooperative housing organizations, chawls, and those living in them all proposed within accepted social limits. (Adimulam, 2022). One industry has completed this work and further research and work are required to develop a complete replacement. (Scott, 2020)

It must be acknowledged that many of the difficulties stated above were resolved because of the 2008 hiring of a compassionate and understanding administrator for the Dharavi Redevelopment Project. At the State Government level, there were a few more senior bureaucrats who cared about the needs of the underprivileged. The CoE* was also established at this time by the government to support and counsel the authorities.

To create a Master Plan that is sensitive to the needs and circumstances of the poor, it will be important to consult the residents of Dharavi. It is necessary to create small groups and subclusters that can connect to the larger plan as and when they are ready to do so. State authorities must make a strong case for themselves as protectors of the underprivileged, mediating conflicts and enforcing the agreements made by the various parties concerned. The challenge we face is clear: how do we fulfill the hopes of the populace while enlisting their enthusiastic involvement in the planning and execution of the reconstruction project? (Nandgaonkar, 2022)



Fig 3.1. Newspaper headline for Dharavi redevelopment project
Source: IndiaTimes, 2007

Process of resistance by Dharavi residents

The Dharavi community started opposing the government plans since it was not sensitive to their needs and their livelihood depended on the sprawling layout. Newspaper articles from Times of India (TOI), Mumbai Mirror, Hundustan Times and DNA Mumbai were extensively studied to understand the process and impacts.

February 16, 2007- All Dharavi does not qualify as slum

Former Chief Secretary and afterwards CoE member D. M. Sukthankar raises objections to the Development Control Rules' adjustments made to meet and sends the first of several letters to the then-proposed redevelopment Director of Housing and Area Development for Maharashtra I.S. Authority (MHADA) Chahal. He claims in his letter: "...For slum redevelopment zones, the Authority has been granted the status of SPA. This, however, was insufficient in the instance of Dharavi because the entire region is not a 'slum'..."

March 14, 2007- Lowrise, high-density more suitable to Dharavi

The group of experts agreed, according to D.M. Sukthankar, in a letter sent in March 2007 to Swadheen Kshatriya, Principal Secretary of the Housing Department of the Government of Maharashtra: "A low-rise, high-density model is more appropriately suited to the existing lifestyles in Dharavi than a high-rise high-density model."

May 9, 2007- People of Dharavi have no info on govt.rehanb plans

A letter was sent to the Maharashtra's Chief minister in compliance with earlier, many concerns were highlighted in letters: ...The "sector"-based strategy fully disregards the preexisting divisions while enforcing new ones both inside and between communities. The planning procedure must be centered around the current Nagar limits... ...Dharavi residents essentially know nothing about the DRP (Dharavi Redevelopment Project), other than the fact that it is a sector plan. They don't know who qualifies for rehabilitation, what their rights are, where the transit tenements are, or where they will live permanently. They are unsure of the precautions to take to safeguard their livelihoods and the kinds of homes that will be offered. The 225 sq. ft. space is insufficient for their needs because other residents have larger families. Has the government thought about making more space available to them, either as a means of profit-sharing with the developers or as extra real estate that can be purchased? Like this, shouldn't private developers' residential developments (as a "free-sale" component) include a required portion of lower- and middle-class housing?...

June 18, 2007- Redevelopment plan ignored living/working conditions of people

When Dharavi residents took to the streets to protest the DRP's flaws on June 18, 2007, Black Flag Day served as a reminder of the hatred they felt toward the idea. The demonstration kicked off at the Dharavi T-junction and finished in front of the MHADA (Maharashtra Housing and Development Authority) office in Bandra (East). The purpose of the event was to draw attention to the rights of the locals because the redevelopment plan did not protect their way of life or include them in the process of planning or formulation.



Fig 3.2. Article talking about the community resistance to redevelopment
Source: Times of India, 2007

June 29, 2007- Dharavi redevelopment project undemocratic
Excerpts from a letter addressed to Dr. Manmohan Singh, the Prime minister of India: Local Dharavi inhabitants know very little about the DRP... Eliminating the need that any slum rehabilitation plan receive the approval of at least 70% of the population is gravely anti-democratic. The constitutional requirement for democratic decentralization is directly attacked by this action. Participation from the community is not allowed.

June 18, 2007- Planned housing densities more than double anywhere else
A letter to the press, banks, and bidders, in part: “It is troubling that there is no opportunity for community involvement in the plan. In addition, because the plan calls for housing densities that are more than twice as high as those found everywhere else in the world, those who will be housed following the reconstruction will experience unparalleled congestion.



Fig 3.3. Residents protesting on Black Flag Day
Source: <https://www.prokerala.com/news/photos/residents-of-mumbai-slum-dharavi-showing-black-flag-and-protesting-18306.html>



Fig 3.4. Article on Dharavi project being undemocratic
Source: Mumbai Mirror, 2008

February 19, 2008- Redevelopment plan will destroy livelihoods of thousands of residents
Excerpts from a letter addressed to Dr. Manmohan Singh, the Prime minister of India: It is feared that if DRP is implemented, thousands of people’s livelihoods will be lost without any other options being provided.

June 2008- World economic crisis deters bidders
The Maharashtra government decided to move through with the distribution of tenders despite the objections and invited proposals from numerous multinational firms. Pre-qualification offers were requested for the project, and tenders were floated
There were 19 proposals submitted in all. Allied Real Estate of Bulgaria, a partnership between Indiabulls and US Shia Homes, Runwal Group and Capital Land of Singapore, Emaar-MGF and Dubai’s Expanse Constructions, a partnership between Neptune Developers and Pacifica of US, and a partnership between Lanco and Sunray City (South Africa) were among the bidders
The global economic crisis, which was brought on by the collapse of the U.S. housing market, first affected India during this bidding process. Due to the substantial initial investment required for the project, the companies were compelled to withdraw from the bidding process due to the global economic crisis. An estimate of the project’s cost was Rs. 15,000 crores. The successful bidders had to put down roughly 500 crores of rupees. In response to risk and the recession, most bidders withdrew.

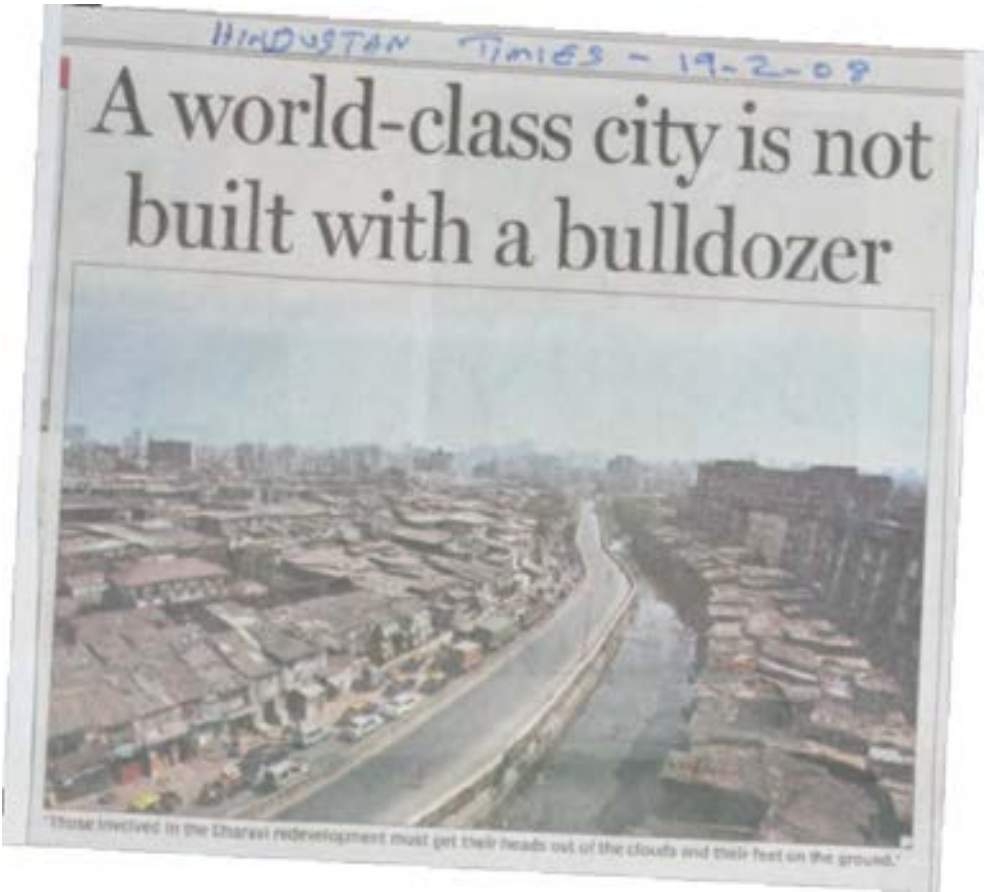


Fig 3.5. Article questioning the non-involvement of residents in redevelopment
Source: Hindustan Times, 2008



Fig 3.6. Newspaper headline when the expert’s team was set up
Source: Times of India, 2009

February 2, 2009- Experts team set up

The Maharashtra government formally formed a team of experts in February 2009 to serve as the Committee of Experts advising the government on the rehabilitation process. The group included:

- D.M. Sukthankar, IAS (Retd.), former Chief Secretary, GoM
- Shirish Patel, structural engineer, and urban planner
- Vidhyadhar Phatak, urban planner
- Chandrashekhar Prabhu, architect, and housing activist
- Arvind Adarkar, Director, Academy of Architecture
- Neera Adarkar, architect, and social activist
- Aneerudha Paul, Director, Kamala Raheja Vidyanidhi Institute of Architecture
- A.Jockin, President, National Slum Dwellers Federation
- Sheela Patel, Director, SPARC
- Sundar Burra, IAS (Retd.) and Adviser, SPARC

June 3, 2009- Aghast to find no surveys/studies done

Extract of a letter from CoE to Sitaram Kunte, Secretary, Housing Department: A comprehensive socio-economic survey of Dharavi, in addition to a plane table and topographical survey, transportation studies, infrastructure and environmental assessment studies, etc., were all fundamental prerequisites for a project of this size and complexity. When we discovered that no such surveys or studies had been carried out, we were horrified. The consultants may have encouraged bids under the false pretense that these investigations had already been done or weren't essential. (Rivas, 2021)

June 15, 2009- FSI 4 will cause unviable density

Extract of a letter to Johny Joseph, Chief Secretary, Govt. of Maharashtra: "...Accommodating the free sale FSI up to a limit of 4 makes the resulting density in Dharavi untenable, in that the needs for roads, open spaces, social amenities, and facilities to ensure a bare minimum acceptable quality of life cannot be met. Additionally, it is impossible to maintain the basic minimum of ventilation and light between buildings... There are approximately 57,000 families in Dharavi that qualify for rehabilitation, according to the thorough socioeconomic survey that was conducted there. The households residing on the top storeys, whether as tenants or members of extended families, have been left out of this study, however, in accordance with current government policy. We recognize that, given that such households are now ineligible for rehabilitation, their removal from the project will inevitably cause considerable dissatisfaction and jeopardize its very execution.



Fig 3.7. Article highlighting that redevelopment is about the land and not about people
Source: DNA Mumbai, 2009

July 7, 2009- A sophisticated land grab

Extract of a letter to Ashok Chavan, Chief Minister of Maharashtra: "The DRP is an artful land grab. Over the years, the government has forced residents from different sections of the city to dwell there while taking great care not to grant them the required legal rights of occupation. The right to this legalization should have been guaranteed to the locals when they initially came there, but it was obstinately withheld from them at the time..."
"In exchange for moving into an area that is less than half the size of the land they currently inhabit, they are now being provided in-situ free pucca homes. The remaining land will now be commercially utilized, and both the government and the developers in charge of the project are anticipated to make large profits. Instead of being motivated by the wellbeing of the Dharavi population, the initiative is driven by individual avarice".

July 30, 2009- Bidding Suspended

The decision was made to open the final bids for the project on July 30 after the original deferral of the bid opening from June 20, 2009. Only 14 of the original bidders remained. But on the morning of July 30, the procedure for receiving and opening the bids was put on hold indefinitely. Although there are many rumors, it is unclear why the indefinite suspension occurred.

August 24, 2009- Consultant not equipped to handle such a vast project

Extract of a letter to Sitaram Kunte, Housing Secretary: "According to what was heard, the Cabinet decided to name Shri Mukesh Mehta as Project Advisor. He was given the title of Project Management Consultant by the Empowered Committee, which was led by the Chief Secretary and went well beyond the Cabinet's decision..."
"To put it mildly, our assessment after meeting with the Consultant is that he was insufficiently qualified to manage the project of this complexity..."



Fig 3.8. Article calling redevelopment plan a 'sophisticated land grab'
Source: Times of India, 2009

Fig 3.9. Newspaper headline with controversial viewpoints
Source: Mumbai Mirror, 2009

November 2, 2009- Alternative approach needed

Extract of a letter to Ashok Chavan, Chief Minister of Maharashtra, highlighting key flaws in the formulation of the Dharavi Redevelopment Project:

- i) Lack of public involvement in the development and design of the DRP
- ii) The lack of competition and the requirement for updated bids.
- iii) The lack of entitlements for some categories and the low number of Dharavi inhabitants who were deemed to be eligible.
- iv) A different strategy is required for the development of Dharavi.

January 16, 2010- Infrastructure still lacking

Extract from a letter to Swadheen Kshatriya, Municipal Commissioner, Mumbai:
“Since there is a chance that the DRP bidding process will be restarted, I would strongly advise you to find out whether the DRP has actually received approval from the MCGM (Municipal Corporation of Greater Mumbai) for all of its infrastructure ideas. I should emphasize here that some of the transit camps in Dharavi that MHADA erected over 20 years ago still don't have water connections today, and they still get their water from tankers. Tens of thousands of residential and commercial buildings should not be constructed under the DRP without a guarantee of the required infrastructure, especially water”.

January 16, 2010- Transparent Procedures a must

Extract of a letter to Shri Vinod Rai, Comptroller and Auditor General of India:
“Shri Mukesh Mehta's appointment as a consultant was made without adhering to the open, accepted practices for such appointments. Aside from his lack of credentials and expertise, his performance has been subpar, and his compensation was set arbitrarily and on an illogical scale. If the market could spend 8 to 10 times that amount, what justifies putting the premium at Rs. 450 sq. foot is not at all evident. Although we oppose the notion of treating DRP as a cash cow for government revenue, if such a strategy is in fact implemented, there must be a fair, open, and publicly disclosed procedure for determining the premium amount. How, therefore, can such a figure remain constant in the face of market fluctuations?”



Fig 3.10. Article showing disconnect between community & government
Source: Times of India, 2009

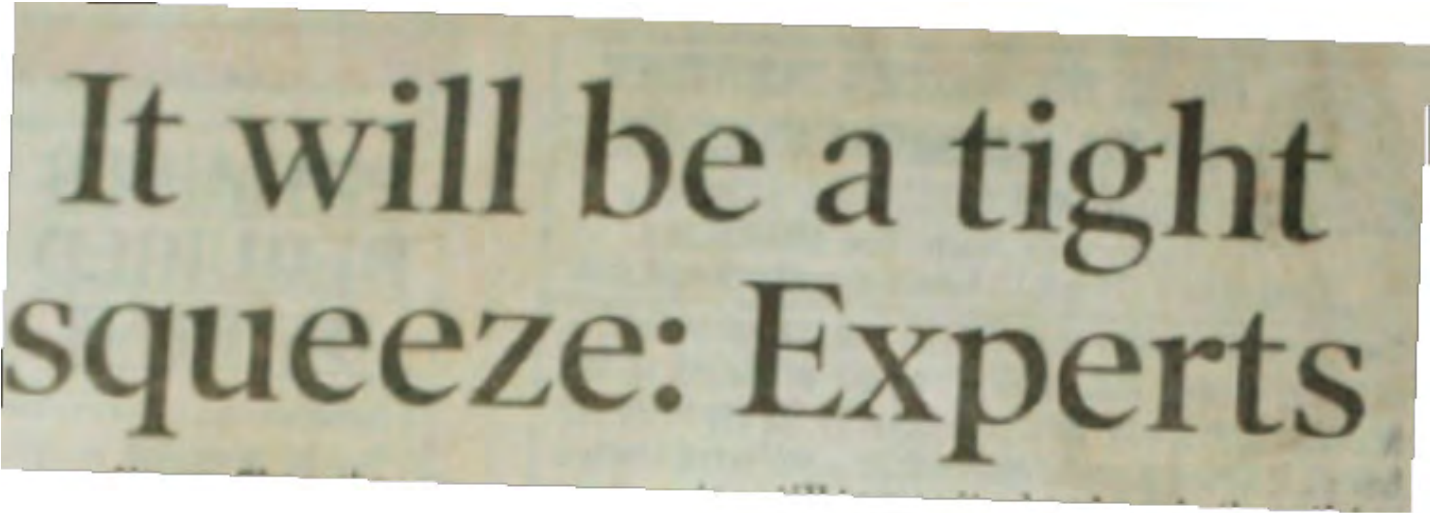


Fig 3.11. Newspaper headline on experts deliberating over the infrastructure
Source: Times of India, 2010

March 23, 2010- Objections to SRA notification

Gist of points made in a letter to Gautam Chatterjee, Officer on Special Duty, DRP:
For Dharavi, which the SRA is obligated to draft in accordance with Section 21 of the MRTP Act, no Development Plan has been created. The development plan must be backed by a report that details its goals, including whether they are to assist locals or generate revenue for the government and developers. No such justification has been given. Development Control Regulations cannot be framed in the absence of a Development Plan. There is no justification given for selecting FSI 4... The population density that will live in the development is not considered. 50 to 80 percent of the floor space for renovation is provided as an incentive in urban renewal plans. This incentive has, however, been unjustifiably enhanced to 133% in the Draft Regulations. The date by which residents are eligible has changed...All residents who were residing in Dharavi at the time of Mashal Shul's biometric census should receive rehabilitation here. There should also be room for residents who live on upper floors, including mezzanines and lofts.



Fig 3.12. Newspaper headline indicating failed redevelopment attempt
Source: Times of India, 2010

Details on the latest Redevelopment proposal

The government of Maharashtra accepted architect Mukesh Mehta's proposal to redevelop Dharavi under Slum Rehabilitation Authority (SRA). The proposal was expected to be implemented soon and global tenders were invited to bid. Because of political controversies, debates over the scheme and high cost, nobody came forward to invest money. (Sabu, 2016)



Fig 3.13. Masterplan showing Dharavi segregated into 5 sectors
Source: Urban Design Research Institute

Details on the latest Redevelopment proposal

There were a lot of issues with this proposed model of redevelopment.

Profit
The goal of redevelopment, which is to improve the conditions of slum people, is neglected by developers

Creation of alien surroundings
In the context of India, the kind of top-notch infrastructure and zoning does not appear to enable many utilitarian uses of public places. The local culture has not been attempted to be preserved. It creates a sense of alienation and dislocation from one's own environment. Despite being physically in the same location, the cloth they will be placed in will be very different and will prevent them from feeling at home.

Access
The layout tried to limit access from the rail and road system to the settlement's exterior while keeping pedestrian-only walkways within. The preservation of the current layout is crucial for keeping the streets in this crowded area pedestrian-friendly and free of traffic jams.

Economic Barriers
The population that is no longer eligible for rehabilitation is forced to remain in other slums.

Redundant Scheme
The rehabilitation plans put out by developers like MM Architects sound promise in terms of amenities, lodging, and hygiene, but they eliminate people's means of subsistence and cultural traditions, rendering the plans ineffective.

Housing and Economy
Their needs are not met by the type of home that is offered. It is suggested to build "vertical slums." People desire homes that will allow them to continue operating their modest cottage businesses. They also require areas where they may perform other crucial tasks for their own domestic industries. There are about 15,000 one-room factories in Dharavi. Products are sold all over the world. The high-rise plan is unable to preserve this. The system does not allow for separate industries to thrive independently as it does in the current environment because it is often like an EWS dwelling that stacks people one on top of the other.

Social Acceptance
The acceptance of slum inhabitants near the intended high-class infrastructure community is in doubt since those who use such services—whether they be for offices or housing—are not used to such integration. Slum dwellers frequently claim that the redevelopment plan is an attempt to remove them from power.

Key takeaways
Due to most residents and their solid economic foundation, Dharavi's reconstruction is still in the books. They have the power to compel the authorities to adopt strategies for seeing the settlement from their perspective. The development plan for this area must consider the interdependence of the local economy and culture. The plan appears to be nothing more than a callous game of chance devoid of the locals' humanistic and cultural touches. (RTF, 2021). Densification of the ground necessitates vertical growth, but the needs of the populace cannot be met by traditional high-rise structures. For the plan that provides both residential and commercial areas to every dwelling unit in the same location, low rise multi-platform innovative design interventions must be embraced. Moving here and uprooting yourself is inappropriate. High-end commercial and residential centers shouldn't be promoted because they would create additional socioeconomic acceptance issues in the region. Therefore, development must focus on enhancing the whole quality of life for every resident of the area, not simply the GDP numbers. (Saigal, 2022)



Fig 3.14. Redevelopment is a utopia for the residents of Dharavi
Source: <https://theoryofsettlementsyear4.wordpress.com/2016/11/18/dharavi-redevelopment-scheme-gagandeep-singh-sanmadi-k-r-shyam-sunder-meher/>



Fig 3.15. Proposed road infrastructure in the latest scheme
Source: <https://www.systematica.net/project/dharavi-redevelopment-project/>



Fig 3.16. Vertical slums being proposed
Source: <https://www.thepolisblog.org/2010/04/contested-urbanism-reclaiming-right-to.html>

Impact of the Redevelopment Proposals

The Dharavi Redevelopment Project was distinctly impacted by the process of public involvement and resistance. Even though there were other factors at play, the Committee of Experts’ close cooperation with officials at all levels of government also contributed to changes:

- The DRP officials understood the importance of community involvement.
- A survey of Dharavi’s physical and socioeconomic conditions was ordered by the authorities.
- They also paid for research to create a thorough transportation plan.
- The government was informed of the potential negative repercussions of excluding families who reside on mezzanine floors from the scope of resettlement and rehabilitation. (Udas-Mankikar, 2020)
- The DRP did not apply to some areas including Gaothans, Kumbharwada, and private properties.
- Since the initial plan had been thoroughly examined and its flaws meticulously listed, the government started to consider alternate options, such as MHADA taking over a certain area.
- It was discovered that slum communities might collaborate with experts and academic institutions for the purpose of self-development.

Conclusion

But the difficulties of rebuilding also have two benefits for the Dharavi neighborhood. First, the Alliance was established by a group of organizations that includes the National Slum Dwellers Federation of India (NSDF), the International Slum Dwellers (ISD), Mahila Milan, and the Society for Promoting Resource Centers because of the ongoing failure of rehabilitation efforts (SPARC). Second, the Slum Upgradation Programme of 1985 mandated that slum residents form housing cooperatives to receive solid tenure rights for a low monthly rental cost. These cooperatives are still in operation today and are becoming more powerful. While these cooperatives do exist in the slums, they are not always successful in working with local and state government representatives to accomplish their objectives. Third, even though the redevelopment has not yet occurred, “squatters” are included in the census as “residents of a notified slum” when the slum is being mapped and surveyed during the preliminary phase. Squatters can benefit from this formal documentation by receiving ration cards from the government, which entitles them to social welfare programs like food distribution. (Yuvaonline, 2020)



Fig 3.17. Series of articles showing the impact of the resistance to redevelopment
Source: Times of India, Mumbai Mirror, India times 2010

No	Project	Year	Salient features	Reasons for failure
1	Slum Improvement Project	1972	Provision of basic amenities to the slum like water, electricity, latrines, and sewage disposal	No comprehensive census on number of households and residents obtained
2	Slum Improvement Project (contd.)	1976	Granting the residents “Legitimate status”; Photo identities were issued to establish legitimacy of residents	Administrative issues due to lack of census; de facto control of slumlords over parts of the slum in which no redevelopment could take place
3	Slum Upgradation project	1985	Leasing out existing slum land to cooperative groups of slum dwellers at affordable rates; granting loans for environmental and housing improvements	A large portion of the area to be redeveloped was held by private developers and could not simply be redeveloped without adequately compensating them

No	Project	Year	Salient features	Reasons for failure
4	Slum Rehabilitation Scheme	1995	Transferable Development Rights introduced to attract private developers; central monitoring and clearing agency developed; redevelopment efforts more decentralized than before	The scheme required consent from 75% of the slum dwellers, which was not obtained due to fragmentation of slum communities
5	Dharavi Redevelopment Project	2004	Rehousing Dharavi's residents into tower block apartments; freed up slum land would be sold for private development	Unacceptable to Dharavi residents since livelihood depended on Dharavi's sprawling layout

Table 3.1. Summary of various Dharavi Redevelopment Plans (Page 2 of 2)
Source: Author

Thus, the process of resistance by the residents severely impacted the redevelopment proposals. It was thus clear that the community of Dharavi has a very strong economic foothold over the region, and it is vital to have community participation so that maximum people agree to the proposal.

PRECEDENT STUDY

To understand how informal settlements operate throughout the world as well as in the Indian sub-context, I looked at 5 different case-studies to derive the tools for my designing.

Quinta Monroy, Elemental, Chile (2003)

Quinta Monroy is in Sold Pedro Prado, Iquique, a Chilean desert in Chile. The area located in the center of Iquique with approximately 5000 square meters was illegally occupied by about a hundred families for thirty years. The idea was to settle those families on the same site. Architect Alejandro Aravena was hired to propose a design strategy for the area. (Quillbot, 2019)

Parameters	Statistics
Number of families	93
Location	Iquique, Chile
Site Area	5722 Sq.m
Built Area	3620 Sq.m
Building budget	Approx \$750,000
Cost of site	Approx \$150,000
Cost per family	Approx \$8,300
Housing Committee	Quinta Monroy

Table 4.1. Statistical data for Quinta Monroy
Source: Author

Parameters	Areas
Initial built area	25 Sq.m
Potential final area	72 Sq.m
Green areas	2.215 Sq.m
Governmental subsidy	3620 Sq.m
Building budget	\$1069,500
Initial built area per house	36 Sq.m
Potential final area per house	70 Sq.m

Table 4.2. Construction analysis of Quinta Monroy
Source: Author



Fig 4.1. Quinta Monroy housing
Source: <https://www.archdaily.com/10775/quinta-monroy-elemental>

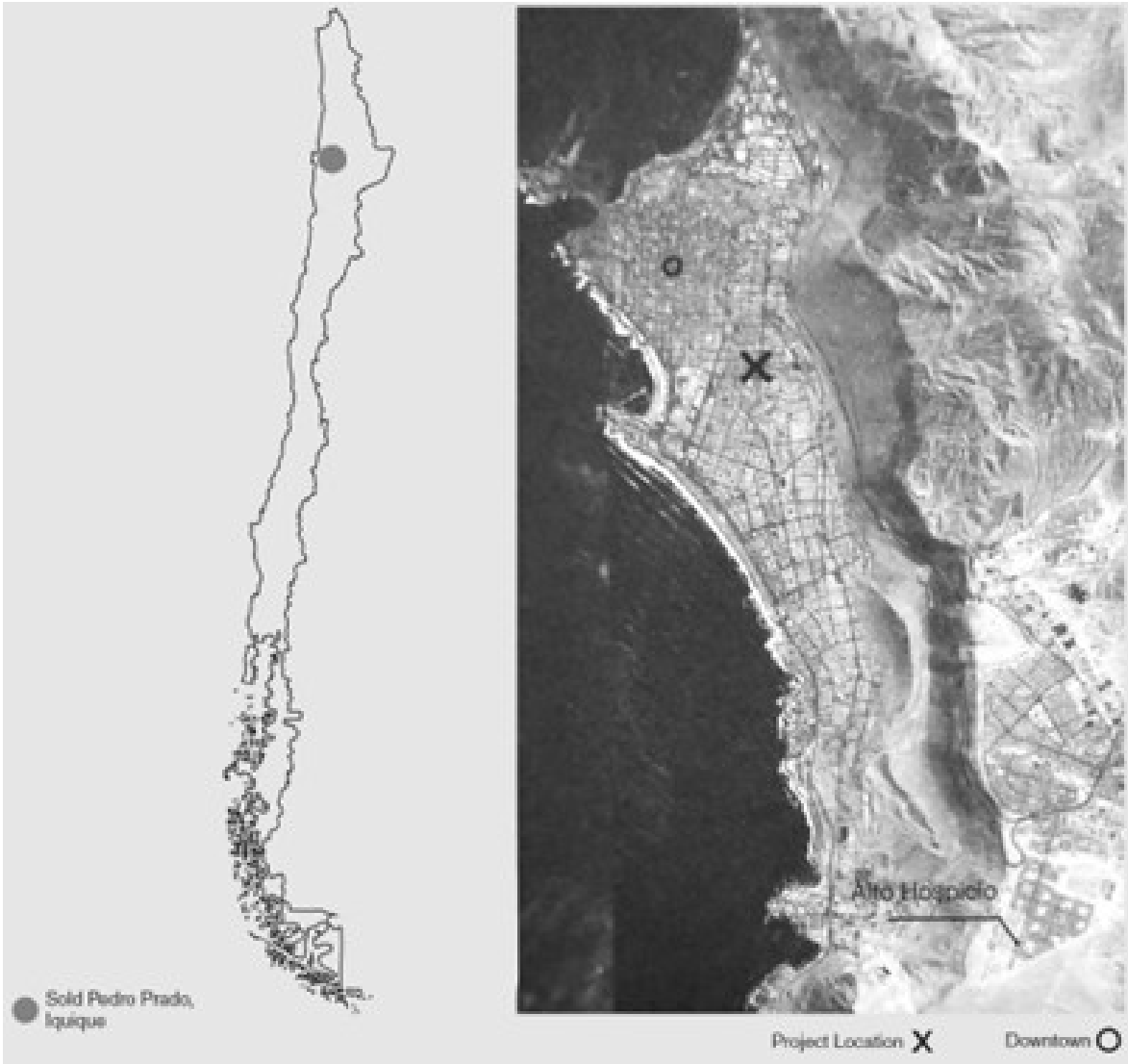


Fig 4.2. Quinta Monroy site location
Source: <https://www.archdaily.com/10775/quinta-monroy-elemental>

Site Conditions

Most of the spaces, approximately 60%, lacked any direct light and ventilation. There was no system of sewage or running water as the area was encroached and lacked the basic facilities. The families on average occupied about 30 square meters of area per family which on average comprised of four members. So, each family member had an area of approximately 8-meter square. But the quality of the structures was bad as they were made from scrap materials and had a labyrinth-like feel to it. Housing subsidies approval: September 2002; Building permits approval: December 2003; Construction start date: January 2004; Construction end date: December 2004.

Challenges

The families were willing to accept only detached house typology and, in this case, if one house was placed on one lot, only 32 units could be accommodated on the site and the rest would have to relocate. Also, the families wouldn't be able to pay for the cost of land by the subsidy that was being provided, that was US \$7500, for the land, infrastructure as well as the houses. Not only that but detached houses typology didn't favor any future growth. So, it would ultimately end up with informal additions over time therefore, the architects persuaded the inhabitants to look for another solution. (Gray, 2014)

The other solution that was tried was to place the houses in a row where the width of the house would be the same as the width of the plot. This provided much better results as compared to the last approach as the architects were able to house sixty families on the site, but the rest would still need to relocate to Alto Hospicio, another site where more social housing projects were ongoing. But even this solution did not satisfy the needs. The future expansion was still a problem as construction of new rooms would block the light and ventilation of the existing spaces and lead to overcrowding. (Busta, 2021).

The last solution tried was to build upwards utilizing the apartment block typology. This was the best solution out of all the three as it was very efficient in land use and all the families could be easily accommodated in this area. But the inhabitants did not like the solution and the families threatened to go on a hunger strike if this proposal was accepted. Also, there were conflicts among neighbors regarding the maintenance of the common areas which would ultimately lead to the deterioration of these spaces. It also didn't leave any room for future expansion and the apartments would only be 30 sq.m.

As none of the existing typology's solutions worked out, Elemental decided to come up with a unique solution that would meet the needs of the residents but would also be architecturally and environmentally efficient. They decided to use low rise structures which would avoid overcrowding and leave some scope for incremental construction without deterioration of the area. The main idea was to design with the involvement of the community using safe and economic solutions. And all of it was done within the guidelines and the provided subsidy of US \$ 7500 per home. (Gray, 2014)

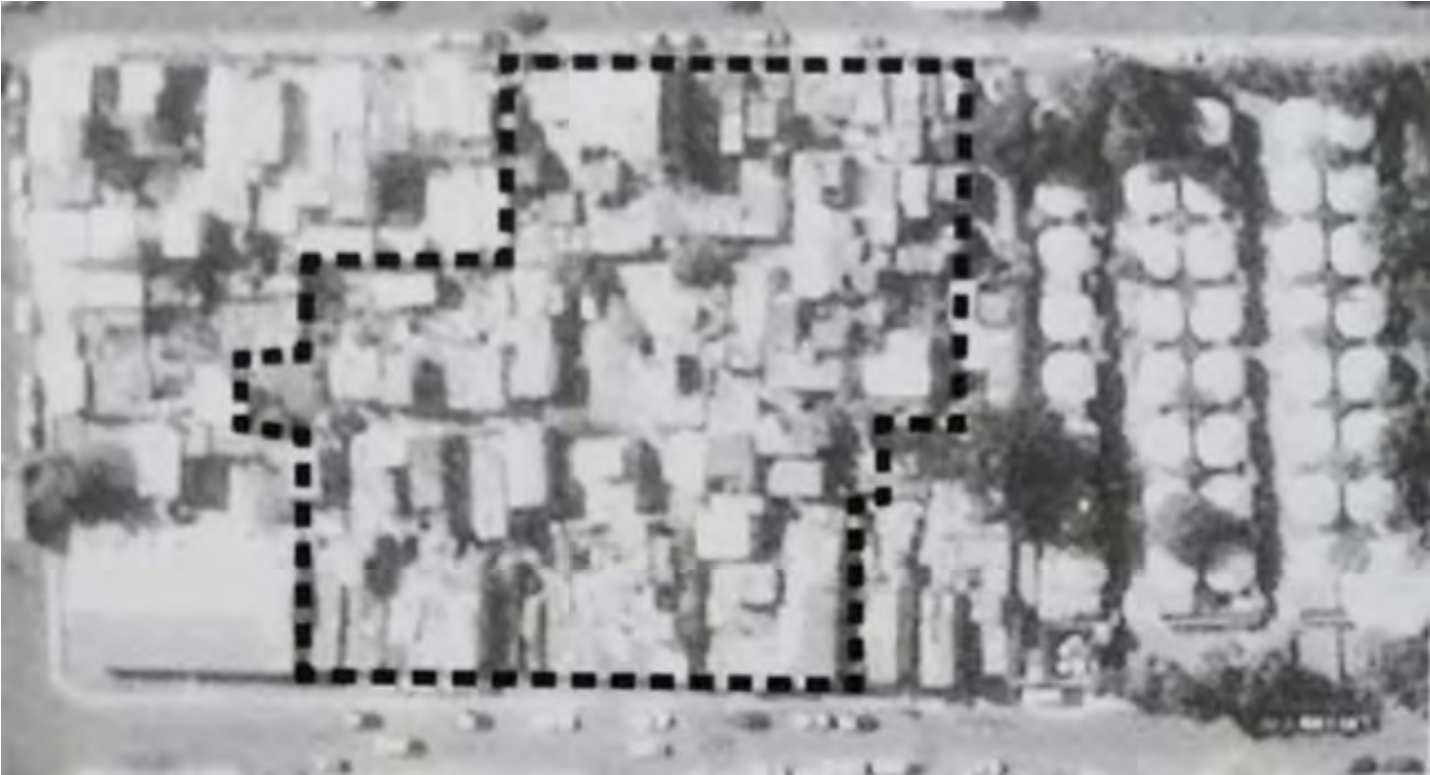


Fig 4.3. Site plan of Quinta Monroy before intervention
Source: <https://www.archdaily.com/10775/quinta-monroy-elemental>

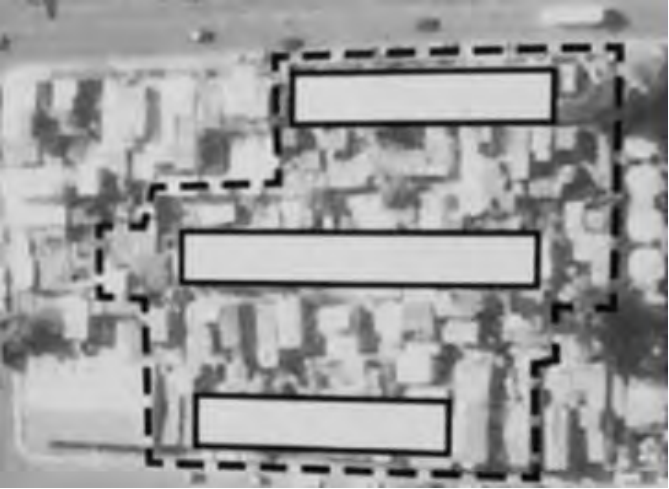
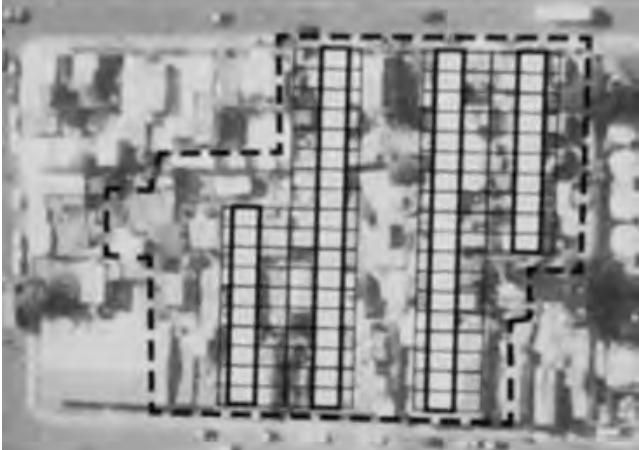
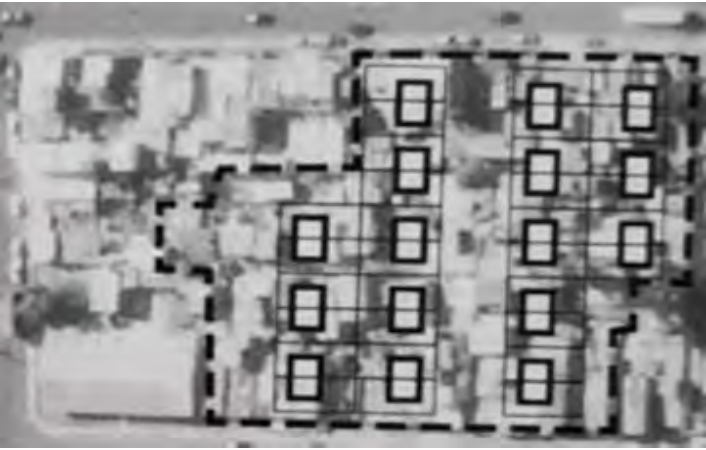


Fig 4.4. Different typologies for Quinta Monroy
Source: <https://www.archdaily.com/10775/quinta-monroy-elemental>

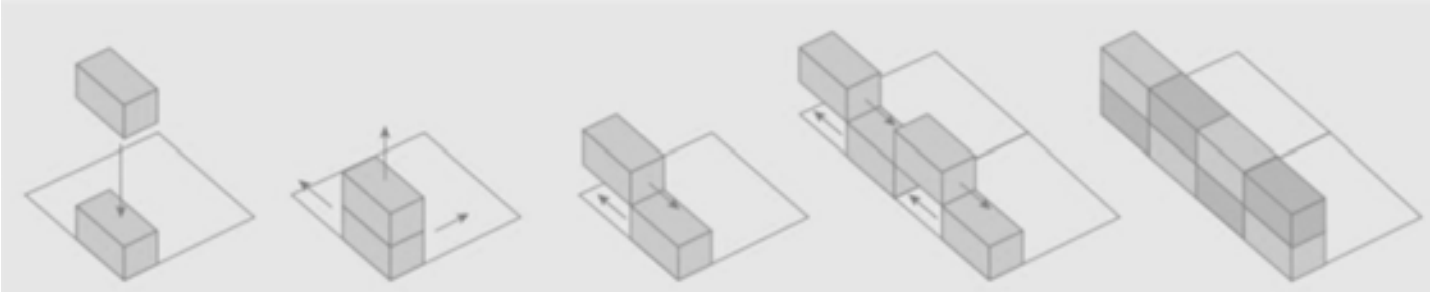
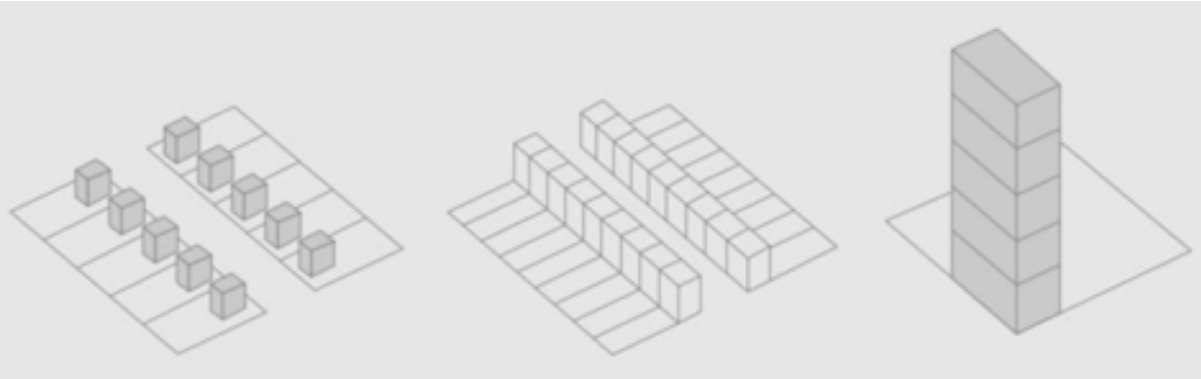


Fig 4.5. Stacking typologies
Source: <https://www.archdaily.com/10775/quinta-monroy-elemental>

The Construction Process

The main part of the construction process was the participatory workshops that were held at every stage. This included the initial workshops discussing he needs and the design with site visits, the workshops regarding the future expansion explaining everyone where and how should the future construction be done, the facade workshop discussing the facade of all the houses and the most important was the workshop regarding the collective spaces and their use and maintenance. It took almost two years for the architects to complete the site work ranging from surveys and requirements to the negotiations for the right to the property of the residents.

The evictions of families began by gaining the trust of families and by taking house by house inventory of everything that was required to be dismantled, moved, and recycled.

To calculate the costs of the temporary location camps specific lists were made of the transportable materials, bathroom fixtures and items that needed dismantling and refitting.

After the surveys, it was concluded that 46rooms of size 3X3m were required for emergency shelters. During the dismantling, the inhabitants helped in dismantling their own homes and a new temporary camp was built. It was proposed that once the construction gets completed, the temporary camps will be used for another project’s relocation.

Once the architects received the permits for construction, the process began. But it was really a challenge to find contractors as due to the new 2001 policy, most contractors were bankrupt. They were able to find one who was willing to do the construction. And finally, the bid was won by Loga Building Company for US \$ 7200 per house. The project was taken on as an apprenticeship as the company didn’t earn a profit. The construction finally began in January 2004 with reinforced concrete once the people were relocated to Alto Hospico. The inauguration ceremony took place on December 14, 2004, when most of the families moved from the relocation camps to their homes. Only a few families preferred to move once the expansions were completed. (ELEMENTAL, 2008)



Fig 4.5. Structure development
Source: <https://www.archdaily.com/10775/quinta-monroy-elemental>

Fig 4.5. Quinta Monroy in December 2004 (Top) and June 2016 (Bottom)
Source: <https://www.archdaily.com/10775/quinta-monroy-elemental>

Empower Shacks, South-Africa (Urban Think-Tank) (2016)

In the year 2004, the post-apartheid Constitution of South Africa gave the “Right of Access to adequate housing” but the country and its poor population (almost 75 million people) are still locked out of the houses or the formal property in the market due to the prices that keep escalating every year, limited or no access to financing due to their lack of knowledge as well as money and the rigid land regulations creating a barrier for the ow income groups (Block, 2017).

Due to these circumstances approximately 2700 informal settlements are caught in the poverty trap and lack basic services as well as infrastructure leading to poor housing quality in turn develops the environmental risks and dangers due to ad hoc developmental patterns. Even though the present government is trying to meet these needs through social housing schemes, these schemes seem to fail due to the increasing demand as well as their inability to cater for diversity. Also, the fact that the problem exists at a very huge scale and the government’s efforts have failed to integrate these informal settlements into the formal frameworks. (Editorial, 2016).

The main challenge, therefore, is “to develop a scalable, transparent and replicable model that addresses both real demand, market dynamics and municipal planning objectives”. This was achieved by the designers by offering an inclusive methodology that is also innovative to ensure that everyone receives access to public space and basic services and provides a safer environment both on the scale of the residence as well as the urban scale with new economic possibilities that the government fails to address in the repeated attempts. The idea was to focus on the living environments and their construction which will integrate the livelihoods of people and their programs including micro-financing, water access and management, renewable energy, and skills training.



Fig 4.8. Empowered Shacks

Source: <https://www.dezeen.com/2017/12/28/empower-shack-urban-think-tank-low-cost-housing-khayelitsha-south-africa/>



Fig 4.9. Site Location for Empowered Shacks

Source: <https://www.dezeen.com/2017/12/28/empower-shack-urban-think-tank-low-cost-housing-khayelitsha-south-africa/>

Parameters	Statistics
Number of shacks built	20
Location	Khayelitsha, Cape Town, South Africa
Year	2013-ongoing
Local Architect	Design Space Africa

Table 4.3. Statistical data for Empowered Shacks
Source: Author

Once the prototype was built and along with it four more shacks were built after discussions with the community. The next step was to involve and train the community so that they can build these shacks themselves in the future, according to their needs. Workshops were organized with the people to teach them how to do the construction along with the final planning regarding the placements of their shacks along with the choice of their neighbors. This was done through block models where the people would sit in a group along with the professionals and arrange the blocks according to their needs.

The step that needed to be taken further was to explain and teach the residents about financing and how to attain funds for the construction. This was done by the financial and affordability assessments workshops where the residents were taught about how to use the affordable materials, where to attain it and how to get micro-financing through loans to upgrade their single-story buildings to double story. (Block, 2017).

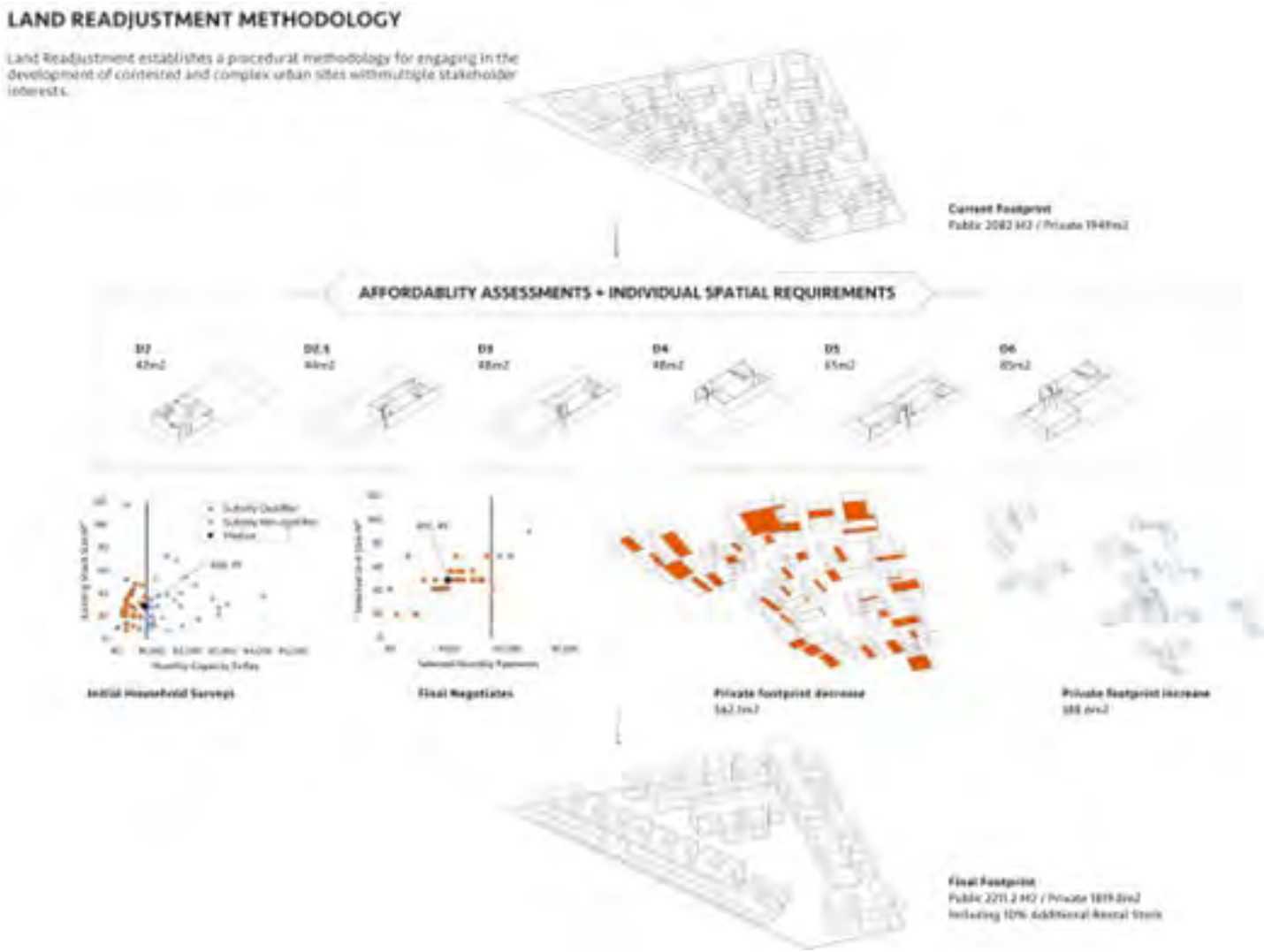


Fig 4.10. Land Readjustment Methodology

Source: <https://www.dezeen.com/2017/12/28/empower-shack-urban-think-tank-low-cost-housing-khayelitsha-south-africa/>

The resulting structures would also help the people to give the upper floors on rents and thus, receive income or get involved in the sale of stock. These would help the residents to achieve sustainability and the right to stay on the land. (Quillbot, 2019).

Once the workshops were over, construction started where the residents constructed their own homes with the low-cost or affordable materials as proposed by Urban Think Tank. Almost 20 shacks have been constructed when the last years and more are constructed every year, giving the freedom to the residents to build as per their comfort and needs. The residents can even build up to three or four story's in the future if they feel the need or the requirement.

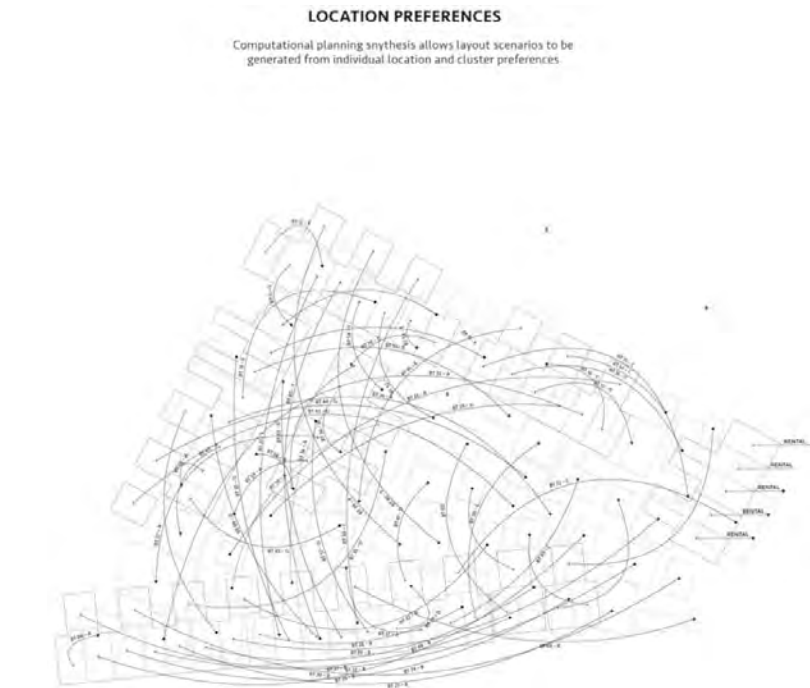


Fig 4.11. Location preferences
Source: <https://www.dezeen.com/2017/12/28/empower-shack-urban-think-tank-low-cost-housing-khayelitsha-south-africa/>

Today the units are available in 6 sizes ranging from 38 q.m to 84 sq.m with an average construction price of 9660 € (ZAR 160,000). Not only that, but due to the workshops offered by UTT, the residents pay an average of 14% of the cost through micro financing.

Urban Think Tank has also developed software which is able to model various urban layouts for different Empowered Shacks that need to be built. The user can input various requirements including the needs of both individual as well as community combined with the frameworks of Municipal Planning. The software then generates various urban layouts that can be easily replicated. The software is also capable of generating 3D models online that can be accessed by the residents to view layouts and the residents in turn can send their feedback regarding the same. As the software is free and open to everyone, it can be easily accessed by the people and urban planners to help communities like these to construct affordable housing for themselves.

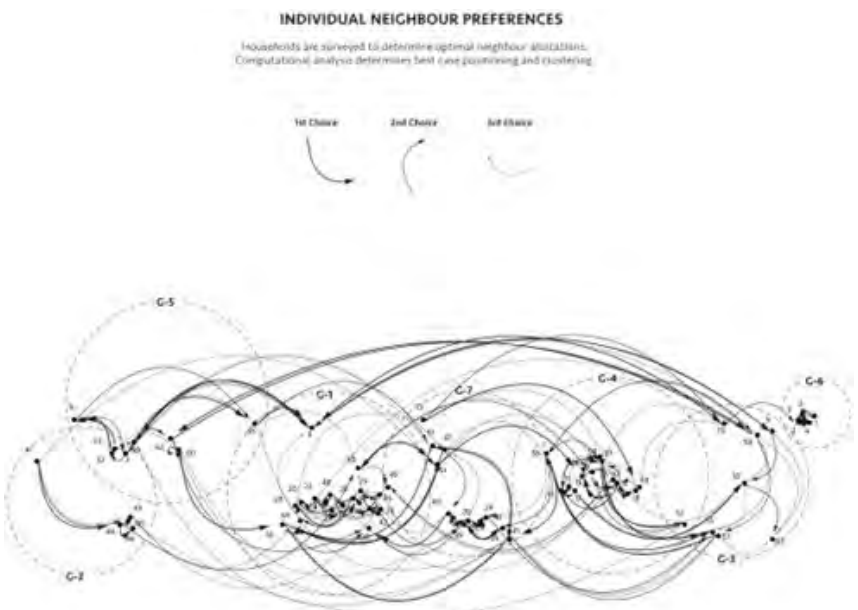


Fig 4.12. Individual neighbor preferences
Source: <https://www.dezeen.com/2017/12/28/empower-shack-urban-think-tank-low-cost-housing-khayelitsha-south-africa/>



Fig 4.13. People arranging blocks of houses in Land Readjustment Works
Source: <https://www.archidatum.com/gallery?id=9254&node=9239#>



Fig 4.14. People constructing their homes
Source: <https://www.archidatum.com/gallery?id=9254&node=9239#>



Fig 4.15. After Empowered Shack Construction
Source: <https://www.archidatum.com/gallery?id=9254&node=9239#>

Santo Domingo Savio, Medellin, Colombia (2013)

Medellin Diagram

The Medellin diagram was developed by Teddy Cruz and Fonna Forman for the informal settlement of Medellin, Colombia. The city of Medellin was one of the most dangerous, a battleground of drug lords and severe poverty and unemployment. But in just 20 years, it became an egalitarian city with dramatic urban transformation

They focused on two cities: Moravia and Santo Domingo to close or reduce the gap between wealthy and poor. They selected the important sites of both wealthy and the poor and performed integration with these sites. The Medellin diagram was “a project of political and civic renewal through which institutions were transformed and cross-sector collaborations facilitated a new interface between the top-down and bottom-up”.

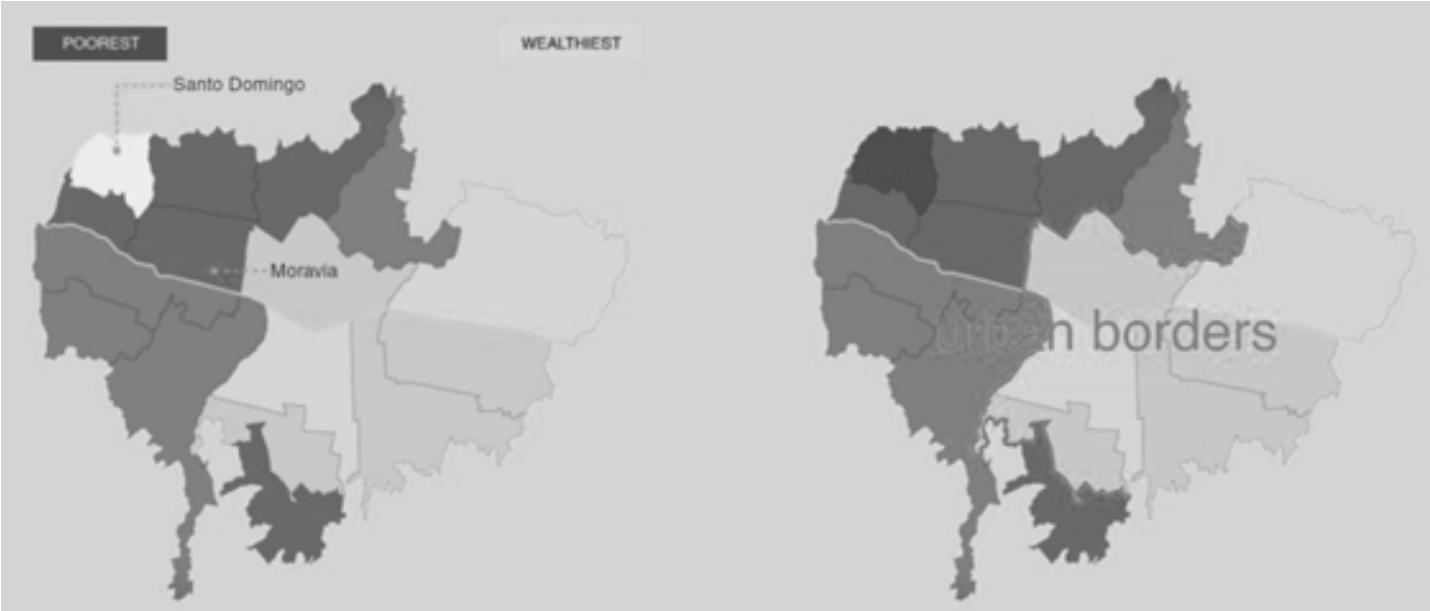


Fig 4.16. Site Location for Santo Domingo
Source:

Few observations can be made with these approaches that lead to certain results or solutions which if integrated in the other informal sites can bring huge urban transformations. (Chaney, 2016. Fragmented institutions ultimately lead to the fragmentation of the city. Therefore, the collaboration of these institutions is of utmost importance. Integration of knowledge and resources and their redistribution to different sites of poverty can lead to drastic results

The most important is to mediate the top-down and bottom-up using design strategies to achieve desirable results and dissipate inequality. The meeting of formal and informal participants and their curation. The co-production of public policies with the help and participation of communities

Public spaces are a generator of knowledge and play a huge role in threading the community. Hence special attention needs to be paid to these areas. And lastly, mobility including transportation is the most important tool that if used efficiently can produce great results. (Chaney, 2016)

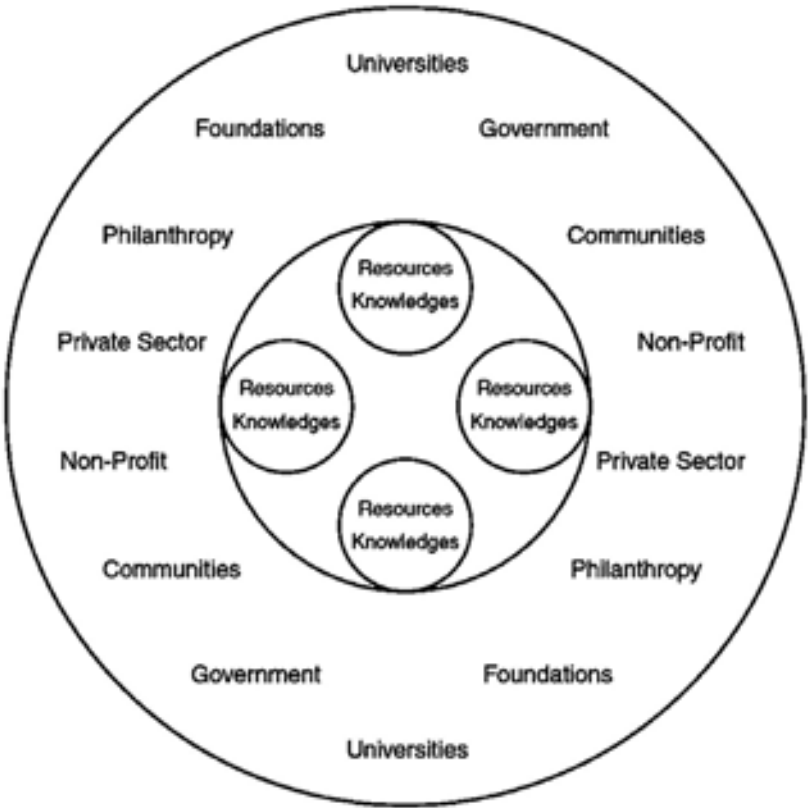


Fig 4.18. Integration of Knowledge and Resources
Source:

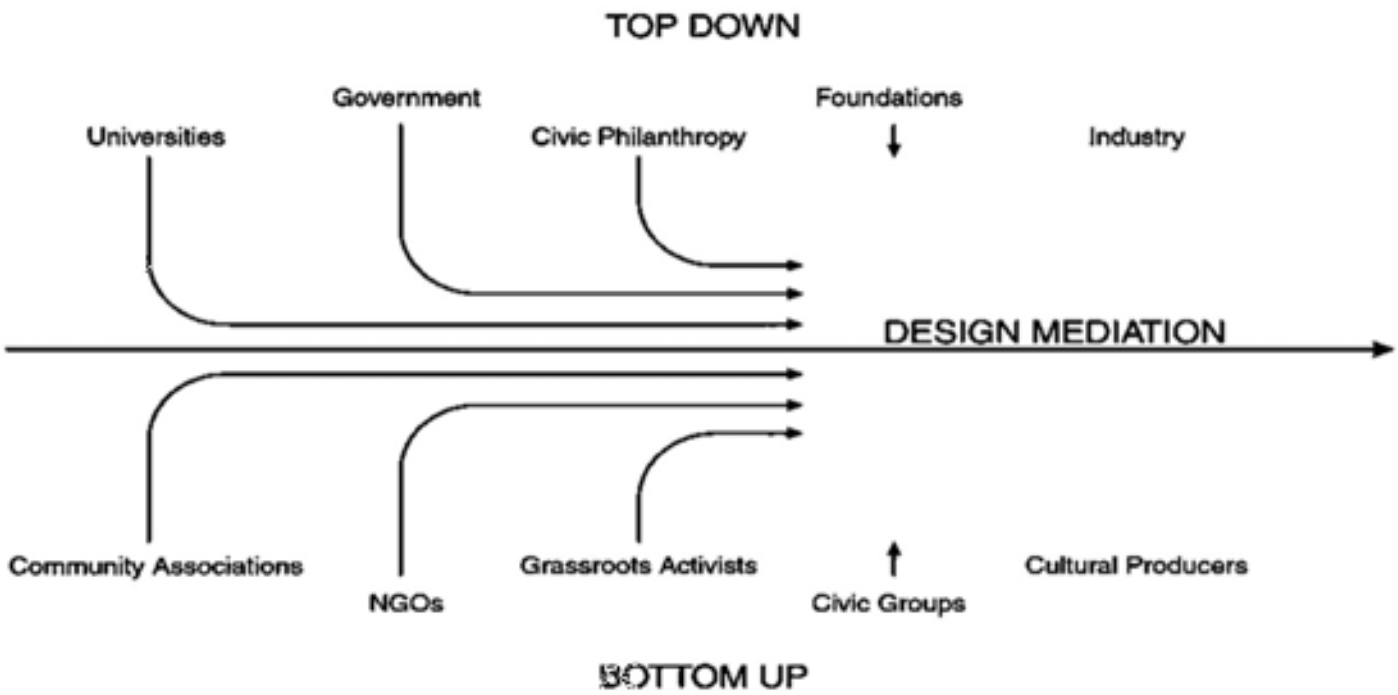


Fig 4.17. Medellin Diagram: Design Mediation
Source:

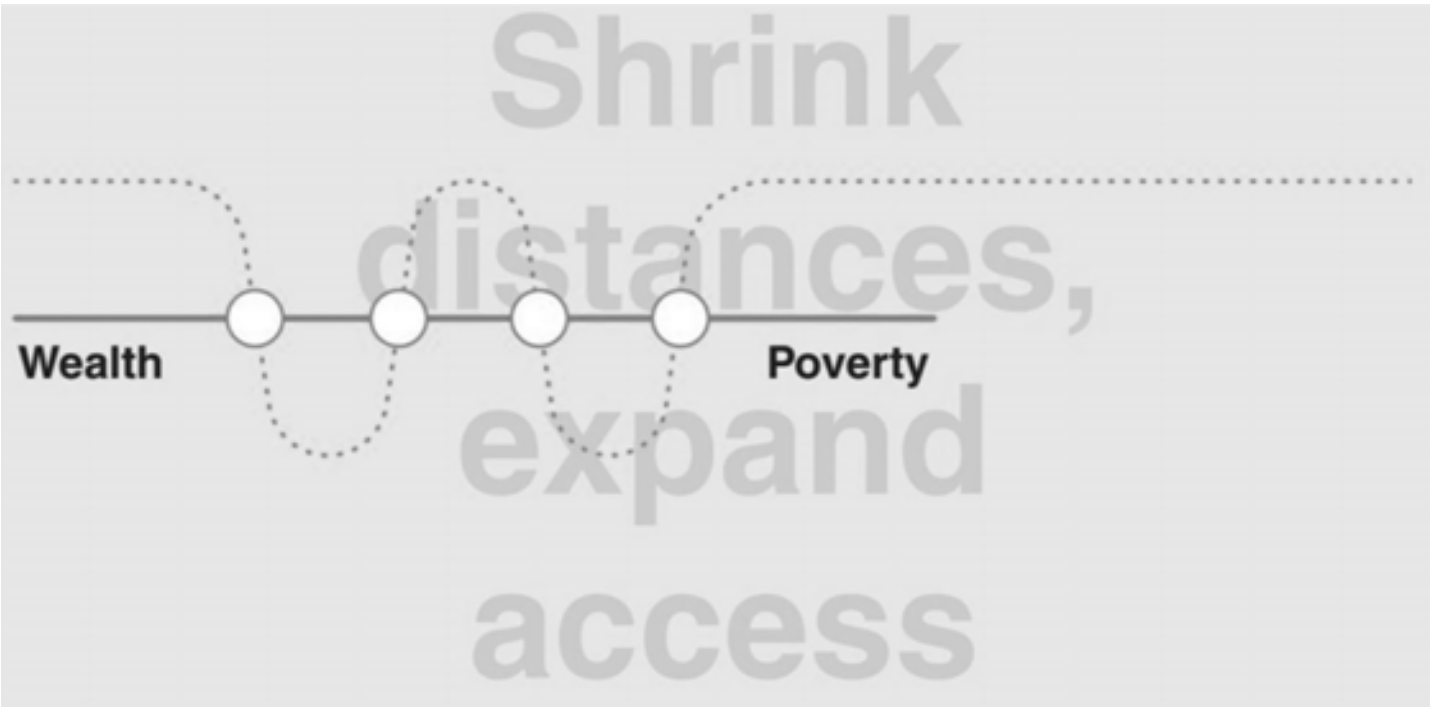


Fig 4.19. Medellin Diagram: Shrink distance and expand access
Source:

Aranya Low-cost housing, Indore, India (1989)

"They are homes where a contented community resides, not actual houses. That is ultimately what matters" B.V. Doshi, Aranya's sculptor, adds. Doshi's architecture was crucial in shaping the environment for people because of its clear understanding of the interrelationships between humans and the environment, the connection between our lives and culture, a sensitive approach to society, its traditions, and ways, and a conscientious response to the ecosystem. (Doshi, 2019)

His architectural idea remains self-evident and expressive of its aim since functionality and a lyrical aspect coexist. Numerous housing developments were created beginning in the 1970s because of the growing need to shelter large populations of people. Doshi created two of his most well-known initiatives, Aranya Low-Cost Housing in Indore in 1989 and Life Insurance Corporation in Ahmedabad in 1973. In terms of strategy, design ethos, and how it affected consumers' lives, this project was very original. (Foundation, 2021)



Fig 4.20. Street façade decorated by the residents in Aranya housing
Source: <https://www.re-thinkingthefuture.com/case-studies/a5290-aranya-low-cost-housing-by-b-v-doshi-low-cost-urban-housing/>



Fig 4.21. Plots with planned framework to grow
Source: <https://www.re-thinkingthefuture.com/case-studies/a5290-aranya-low-cost-housing-by-b-v-doshi-low-cost-urban-housing/>

Social Context

Aranya In order to address the lack of housing and shelter at the time, the Indore Development Authority commissioned Low-Cost Housing, which was co-funded by the World Bank and India's Housing and Urban Development Corporation (HUDCO).

It was designed for Economically Weaker Sections of Society. It became crucial for the government to address this issue in an inexpensive and well-served manner as the number of slums, illegal settlements, and shelter shortages increased. Numerous case studies were conducted, and the housing scenario was then considered to be one in which obtaining inexpensive housing equated to repeating a series of modules that served as a conventional answer. Aranya Housing is made up of six independent neighborhoods over an area of 86 hectares.

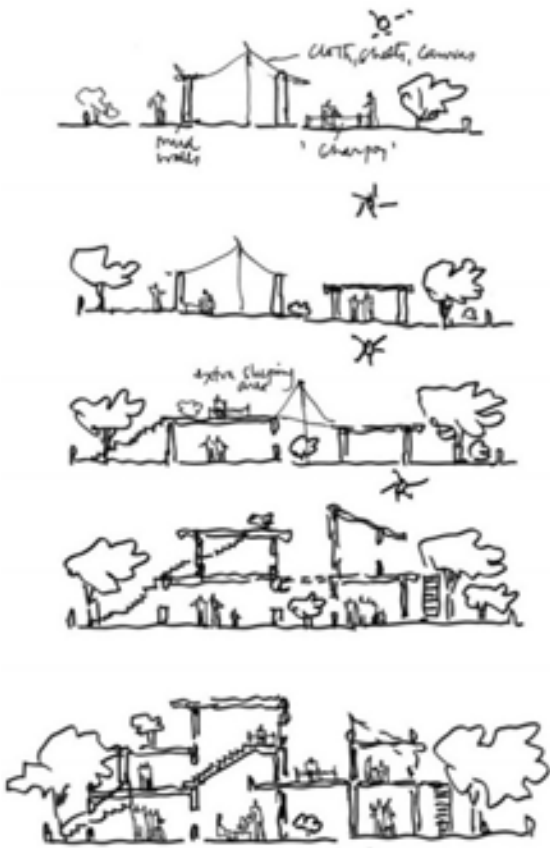


Fig 4.22. Street house relationships
Source: <https://www.re-thinkingthefuture.com/case-studies/a5290-aranya-low-cost-housing-by-b-v-doshi-low-cost-urban-housing/>



Approach

The goal of Doshi's integrated strategy for mixed-income groups was to bring them together. The reaction resulted only from a comprehension of people's basic requirements and the realization that space and design ultimately revolve around the user. The housing township contains more than 6500 homes over six sectors. Doshi made the decision to give the families a framework of things, with the plot including provisions for services, a restroom building, and a brick plinth to construct on. (Doshi, 2019).

This framework gave people the freedom to design the space to suit their basic needs and way of life rather than giving them a finished dwelling. With the help of this concept, the house was also able to subjectively develop in terms of connectedness, human adaptation, private and public buffers, indoor-outdoor linkages, and space movement.

Hierarchy and Connectivity

The street network clearly separates traffic for vehicles and pedestrians, with the vehicular route pointing toward the outlying major road and pedestrians or unofficial pathways. The streets and pathways between the houses are changed as a result, rendering the busy traffic dormant and encouraging unofficial public gatherings and encounters. The scale of these paths in relation to the built spaces allows ease to the spaces and functionality, as do the entrances, spaces that act as a barrier between public and private, streets, and space between two houses.

The public spaces in the community are organized in a hierarchy, with an even distribution of neighborhood sub-centers that see frequent foot traffic. Due to the primary public space's connections to all auxiliary areas, a connection is kept throughout the township. Sector roads join a central spine, which is spaced apart at two points to lessen traffic. (Ansari, 2021)

Planning, Masses and Form

here was a courtyard shared by ten groups of dwellings. There was one main public area for the entire township and six smaller green spaces for each sector. The user had the option to decide how much and with which combinations to develop. Every house had a distinctive personality because every entrance, staircase, verandah, and buffer had variances.

Doshi had a kit of components that allowed families to customize it to meet their needs, including a staircase, openings, railings, overhangs, and verandahs. For the families, he designed and built 80 model homes that varied in size, spatial layout, mass, and movement. Depending on their financial bracket, residents might pick from houses with one bedroom or more. The shared courtyard between the two residences is shaded by the nearby houses. (Ansari, 2021)



Fig 4.23. Interconnectivity and space interactions (Part 01)
Source: <https://www.re-thinkingthefuture.com/case-studies/a5290-aranya-low-cost-housing-by-b-v-doshi-low-cost-urban-housing/>



Fig 4.24. Interconnectivity and space interactions (Part 02)
Source: <https://www.re-thinkingthefuture.com/case-studies/a5290-aranya-low-cost-housing-by-b-v-doshi-low-cost-urban-housing/>

Construction and Materials

Doshi employed shallow concrete piles for footing because the soil at the site was black cotton soil. A base, load-bearing brick walls, and concrete slabs were all included with each house. This also brought down the overall cost of construction. The lengthier side of the house was exposed to the heat since it was situated along a north-south axis. The house's decor, furnishings, and color scheme were entirely at the discretion of the owners.

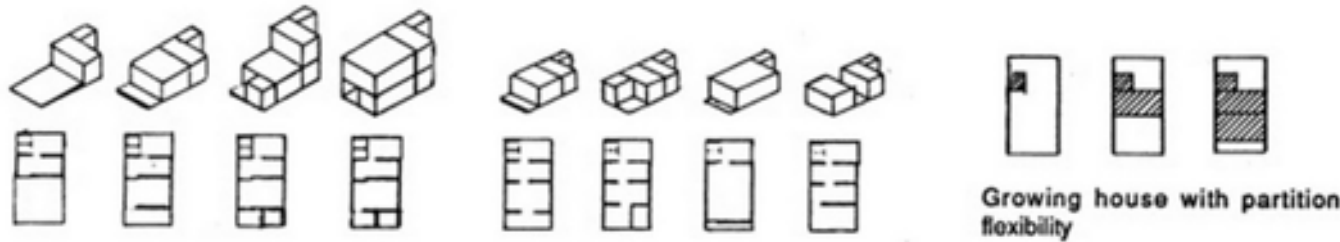


Fig 4.25. House Variations
Source: <https://www.re-thinkingthefuture.com/case-studies/a5290-aranya-low-cost-housing-by-b-v-doshi-low-cost-urban-housing/>

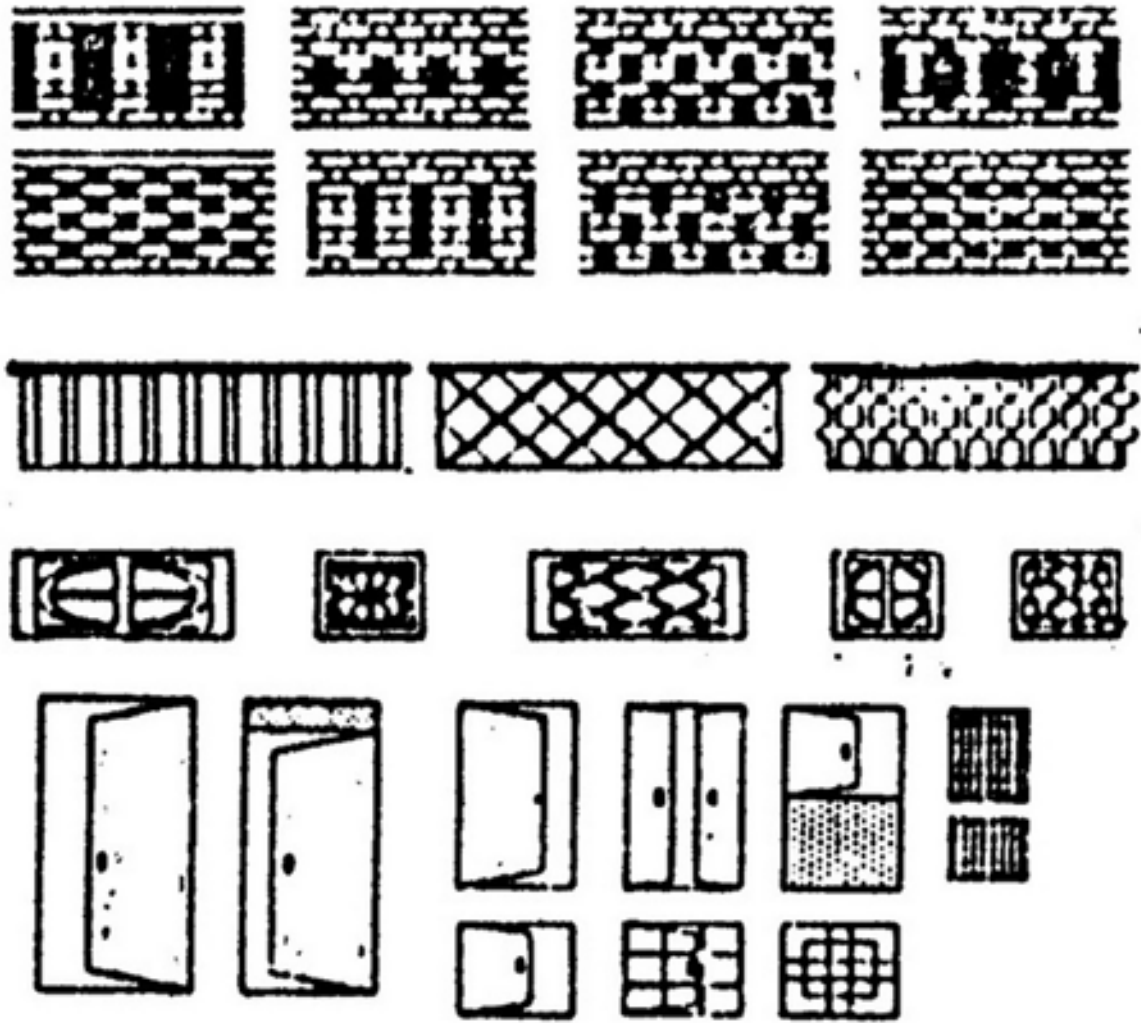


Fig 4.26. Kit of Elements
Source: <https://www.re-thinkingthefuture.com/case-studies/a5290-aranya-low-cost-housing-by-b-v-doshi-low-cost-urban-housing/>

Doshi's goal of creating a feeling of community by involving the user in the design process was a successful and useful strategy for the housing. Purely out of necessity and the way of life of families, houses grew, and spaces changed. Hierarchy, movement, and scale were important design factors in the Master Plan. There are informal paths and tiny public areas connecting each neighborhood. (Foundation, 2021).
public areas connecting each neighborhood. (Foundation, 2021)
This framework gives users room to develop and move naturally. This initiative is regarded as one of the best efforts made to achieve its goals. This strategy improved the weaker class in addition to maximizing functionality and space usage.
"Simple material knowledge, a knack for finding solutions, and reverence for the natural world are all that design is." - B.V.Doshi



Fig 4.27. Play of color, material, spatial features, light and shadow

Source: <https://www.re-thinkingthefuture.com/case-studies/a5290-aranya-low-cost-housing-by-b-v-doshi-low-cost-urban-housing/>



Fig 4.28. Load-bearing masonry

Source: <https://www.re-thinkingthefuture.com/case-studies/a5290-aranya-low-cost-housing-by-b-v-doshi-low-cost-urban-housing/>



Fig 4.29. View of Aranya Housing

Source: <https://www.re-thinkingthefuture.com/case-studies/a5290-aranya-low-cost-housing-by-b-v-doshi-low-cost-urban-housing/>



Fig 4.30. Present Scenario of Aranya Housing

Source: <https://www.re-thinkingthefuture.com/case-studies/a5290-aranya-low-cost-housing-by-b-v-doshi-low-cost-urban-housing/>

Belapur Incremental Housing, Navi Mumbai, India (1986)

One of the few truly indigenous architects, Charles Correa blended global modernism into the Indian environment without emulating it. He was an Indian native, hence he was better equipped than any top-tier international architect to comprehend the complex interaction between culture, traditions, and design.

Bombay is a magnificent city but a terrible location, he famously remarked. Today, there are more than 2 crore residents of Mumbai. Its major issue has been a lack of space for years. Ar. Charles Correa has made significant contributions to the growth of India's financial hub, ranging from minor interventions like altering pavement usage in accordance with its activities to developing the entire city of New Bombay. Mumbai's main issue has always been a lack of available roofing due to the city's excessive population. Correa researched the issue and offered useful, original answers. His work on affordable housing is extensive and outstanding. (Davey, 1985)

The Belapur Housing Project is one of these initiatives. Belapur lies in the Thane district, which is then just outside of Navi Mumbai. It is a node in Nerul, around two kilometers from New Bombay's center. The 5.4-hectare property was designed to accommodate 500 individuals (or around 100 families) per hectare. The first residents moved in by 1986 after three years of construction work that started in 1983.



Fig 4.31. View of a cluster in Belapur Housing with nearby hills in the background

Source: <https://www.re-thinkingthefuture.com/case-studies/a3735-belapur-housing-by-charles-correa-a-sense-of-home-and-community/>

The Concept

The project's foundation is built on a single idea: a separate plot for each residence to allow for future growth as and when necessary. The program was created to serve a range of socioeconomic classes, including upper, middle, and lower. The plot sizes vary slightly, from 45 square meters to 75 square meters, despite the broad variation of socioeconomic groups (a ratio of 1:5). Originally, Ar. Correa planned every plot to be the same size, however this had to be changed owing to affordability and lending agency regulations. (Associates, 2021).

Because no two houses are connected by a common wall, each one has its own plot and modest open area. This has two advantages: it gives residents a sense of individuality even in a densely populated area, and it allows for development as and when the owners see appropriate. This was praised as a quality of smart modern planning.

Planning

The total project is low-rise and dense, like most of Correa's past affordable housing initiatives. According to plot size, there are five different types of homes; the simplest is a single room with a bathroom, and the most sophisticated is a two-story tenement. The toilets of two adjacent homes share a wall, but the residences do not, to make plumbing services easier.

Charles Correa cleverly employs his distinctive hierarchical open spaces in this instance as well to evoke a sense of home and community. The individual yard of each house is the smallest expanse of open land. A small, 8 m x 8 m courtyard surrounds a cluster of seven homes.

A larger area measuring around 12 m by 12 m is surrounded by three such clusters. Around the main communal area, which measures 21 m x 21 m, three of these clusters come together. A seasonal stream (nullah) that runs through the center of the site and transports runoff after heavy rains is visible from the common spaces.

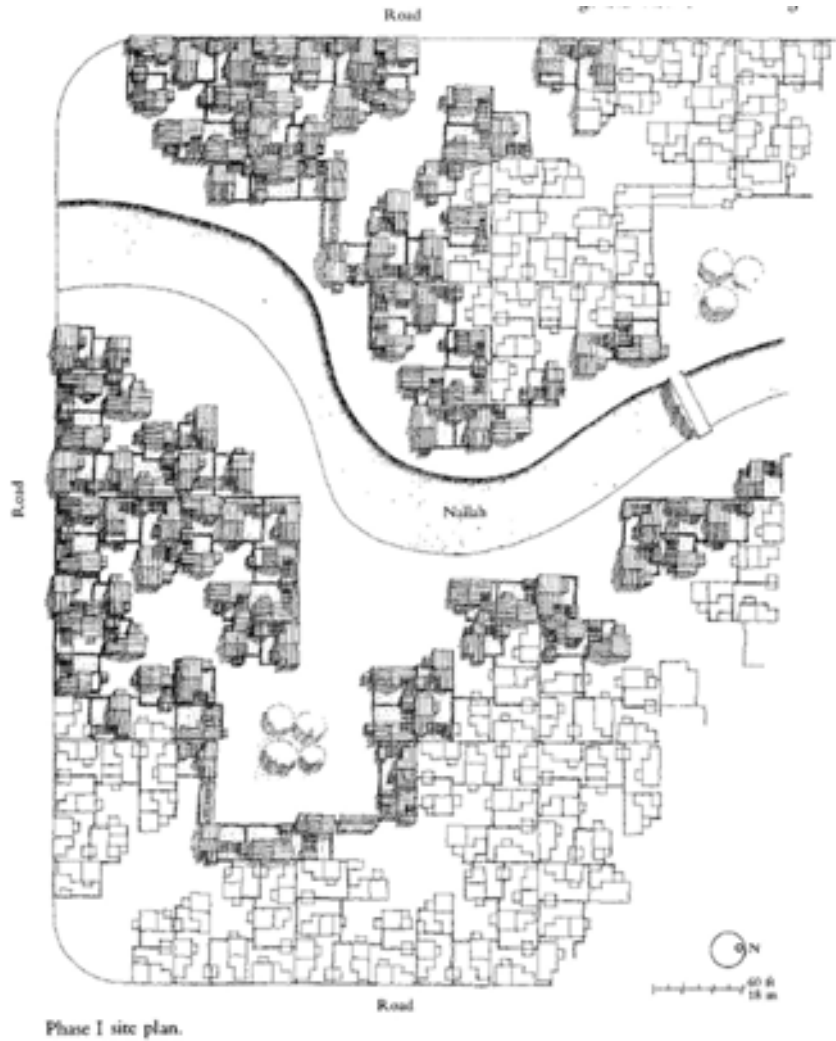


Fig 4.32. Phase I Site Plan

Source: <https://www.re-thinkingthefuture.com/case-studies/a3735-belapur-housing-by-charles-correa-a-sense-of-home-and-community/>

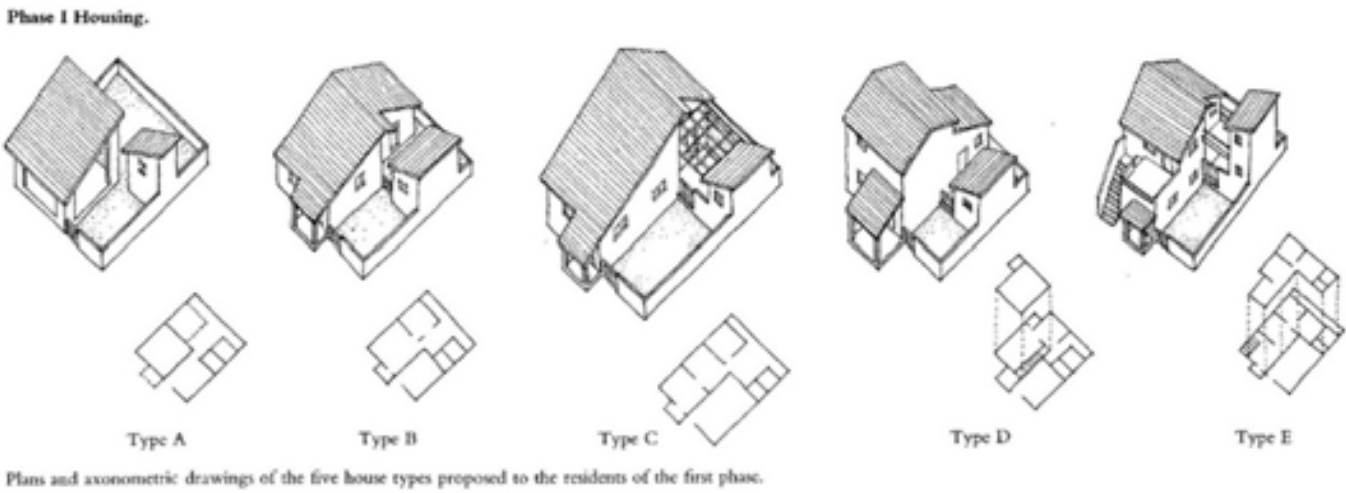


Fig 4.33. Five types of dwellings were designed based on plot sizes from 45-75 Sq.m

Source: <https://www.re-thinkingthefuture.com/case-studies/a3735-belapur-housing-by-charles-correa-a-sense-of-home-and-community/>

One house may adjoin the boundary of two additional homes thanks to the way that all the units are set up. To protect the residents' privacy, no windows are installed on those sides. All the units have been painted, plastered, and built with load-bearing brickwork. Simple construction methods were used so that even inexperienced local laborers could build the homes, providing employment for them.



Fig 4.34. Hierarchical Open Spaces

Source: <https://www.re-thinkingthefuture.com/case-studies/a3735-belapur-housing-by-charles-correa-a-sense-of-home-and-community/>

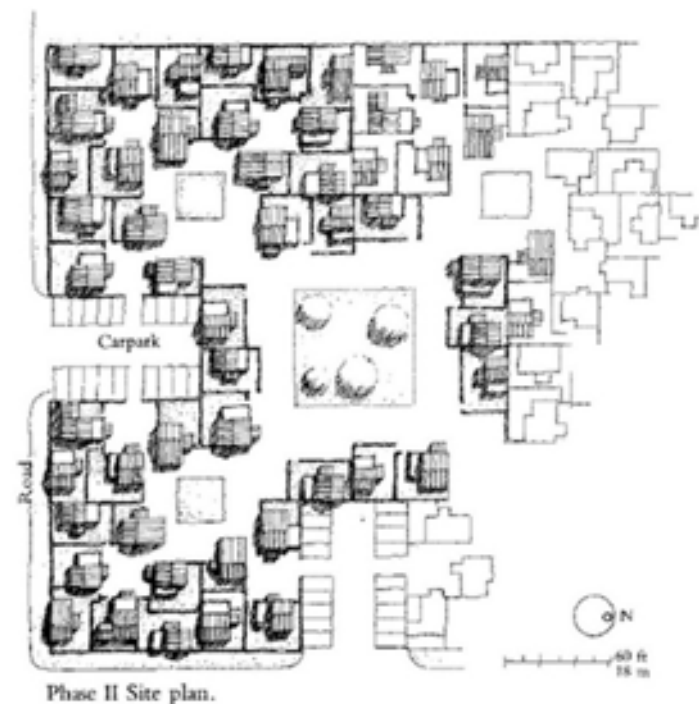


Fig 4.35. Phase II Site Plan

Source: <https://www.re-thinkingthefuture.com/case-studies/a3735-belapur-housing-by-charles-correa-a-sense-of-home-and-community/>

Aesthetic

The settlement has a charming village-like atmosphere because of its clusters of compact, white-washed homes that are covered in Mangalore tiles. The most common stone used in angans, Shahabad, is utilized to pave the front yards.

The genius of Correa's planning, particularly for housing, is that it is usually simple to comprehend but extremely challenging to come up with. The ideas of incrementality, open-to-sky spaces, equity, and a strong sense of community are all combined in Belapur housing to make a statement about what Correa thought to be the most important aspects of housing. Parking spaces are provided on the edges, however all mobility within the property is pedestrian-only. The open spaces easily transition from a tiny court to a huge community space. (Davey, 1985)

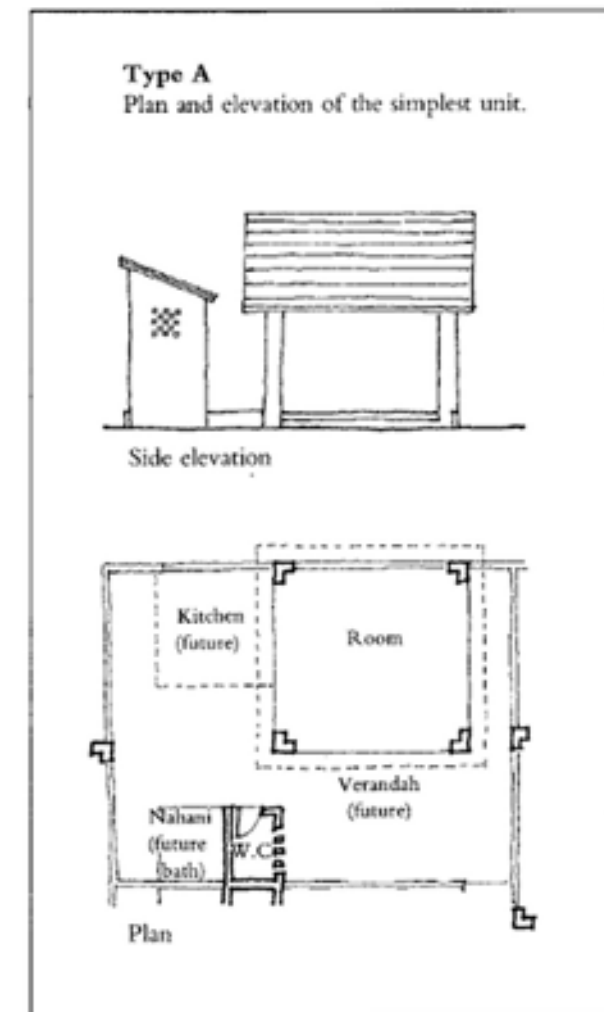


Fig 4.36. Plan & Elevation of Type A unit- simplest unit consisting of 1 room & toilet

Source: <https://www.re-thinkingthefuture.com/case-studies/a3735-belapur-housing-by-charles-correa-a-sense-of-home-and-community/>

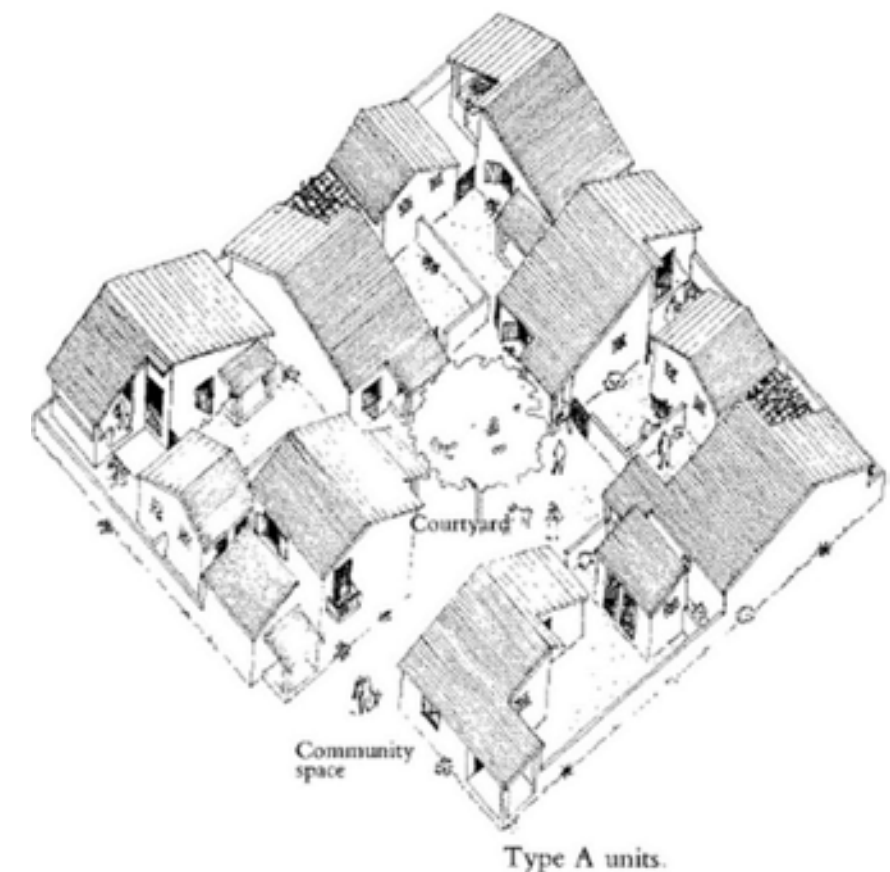


Fig 4.37. Cluster of type A unit

Source: <https://www.re-thinkingthefuture.com/case-studies/a3735-belapur-housing-by-charles-correa-a-sense-of-home-and-community/>



Fig 4.38. Plan and elevation of Type E unit – largest unit

Source: <https://www.re-thinkingthefuture.com/case-studies/a3735-belapur-housing-by-charles-correa-a-sense-of-home-and-community/>

The Project Today

The Belapur of today, however, is substantially different from the Belapur of the 1980s. It is now a well-known commercial area. It has surpassed the trend of high-rise dwelling projects. Considering this scenario, the low-rise dense village would appear to be using far too much space while providing housing for far too few people.

Nowadays, even low-income families have at least two two-wheelers, thus parking spaces are scarce. Since this is an incremental housing plan, it is envisaged that the owners will adjust as necessary. But because of the dramatically increasing demands, many homes had to be renovated or rebuilt. The locals themselves disliked the “village” atmosphere. If we look at Belapur housing today, it appears to be balancing between high-end low-cost housing and shoddy unplanned growth.

Does this imply that the strategy is no longer applicable in the present situation? No, because Charles Correa developed the best answer at the time after thoroughly analyzing the needs of the suburb at the time. And although Correa had little influence over how his design would be altered by users a few years later, the overall atmosphere of the space has not changed thanks to careful planning.

There are plenty of open and green places right in the center of a busy city in today’s era of “compact” and “space-saving dwellings.” Despite today’s apartment culture that encourages people to be confined within their own small boxes, the sense of community still exists among Belapur incremental housing inhabitants. And that’s what makes Ar. Correa successful.

Nobody could have predicted the recent growth spike that Mumbai experienced. The city is still developing and evolving. Nobody can foresee a city’s future ten years from now. The best thing that the designers, users, and public can do is adapt and work to make their beautiful city even better. (Associates, 2021)



Fig 4.39. The settlement today has grown to be very different from the 1980s

Source: <https://www.re-thinkingthefuture.com/case-studies/a3735-belapur-housing-by-charles-correa-a-sense-of-home-and-community/>

Precedents	Quinta Monroy	Empowered Shacks	Medellin Diagram	Aranya Housing	Belapur Housing
Participatory planning	✓	✓		✓	✓
Government participation	✓		✓	✓	✓
Incremental development	✓	✓		✓	✓
Bottom-up and top- down integration	✓	✓	✓	✓	✓
Low-cost materials	✓	✓		✓	✓

Table 4.4. Comparison of tools used for designing
Source: Author

Precedents	Quinta Monroy	Empowered Shacks	Medellin Diagram	Aranya Housing	Belapur Housing
Increase in social interactions	✓	✓		✓	✓
Inclusivity	✓		✓	✓	✓
Better infrastructure	✓	✓		✓	✓
Right to the area	✓	✓	✓	✓	✓

Table 4.5. Comparison of the results
Source: Author

Comprehensive Analysis of Dharavi

It might sound contradictory, but Dharavi is one of the richest slums in Asia. Around 1958, when multiple factories were established in the neighborhood, the current Dharavi population flocked to Mumbai from various parts of the country. Small residential colonies were established by traders and artisans, including potters, leather workers, tanners, and embroidery workers. In the slum, these settlement patterns are still present.

Study of existing zones and industries

Leather Industry

One of the first industries to emerge in Dharavi was leather manufacture. Muslim tanners arrived in Mumbai from Tamil Nadu in the middle of the nineteenth century, but they were forced to relocate to the swampy outskirts since south Mumbai's expanding business district deemed leather manufacturing methods unacceptable. So, in 1887, the first tannery was built in Dharavi. The company steadily expanded as migrant laborers entered the city in search of employment. (Saglio-Yatzimirsky, 2016.

The majority of those working in the leather business are Muslims from Uttar Pradesh or Muslim-Hindus from Tamil Nadu. Numerous Maharashtrians also produced wallets and bags. The Parsi, Darab Pedar, who owns his own tannery in Aurangabad, stands out as an exceptional case. Salted hides are bought from Deonar and sent to Aurangabad. He calculates that the Dharavi raw leather industry generates an annual revenue of roughly Rs. 60 crores. Due to the restriction on tanneries due to pollution, Dharavi now mostly deals in finished goods, with roughly 30 major leather goods firms and about 5,000 people employed in the industry

Before the leather is used to create the finished goods displayed in Chamda (leather) Bazaar, it must first be chemically tanned or cleaned before being dyed. In 1996, the city banned tanning due to the harm it caused to the air and water. Even though 27 out of 39 tanneries received new locations near the abattoir in distant Deonar, mostly the larger ones relocated. Though a few tanneries continue to run despite the prohibition, the days of leather tanning in Dharavi are largely over. From Deonar, the industry now purchases its tanned hides

Damodar Kamble spent 15 years working in the leather industry here after moving to Dharavi at the age of 15 because being a cobbler didn't pay enough. His company is currently the only one producing "uppers" for shoes, which are exported to Australia and Japan after soles are placed elsewhere. He started off with no employees; today, he has 20 employees and a revenue of more than one crore rupees. Today, the primary leather business is finished leather items. Many of the items on exhibit are either overstocked or returnable from export orders placed with Dharavi-based leather goods producers.

Customers from throughout the city come here in growing numbers, and the leather sector is growing on a global scale as well. Men who labor over these exquisitely finished and manufactured leather goods do so in claustrophobic lofts or workshops with insufficient lighting, inadequate ventilation, and oppressive heat. Despite difficult circumstances, the leather industry is the principal industry with which Dharavi is connected. (Mahawar, 2020.

Textile and Tailoring industry

There are many rags to riches tales in this industry, such as Waqar, who used to sell bananas but now employs twelve people and operates three stores selling shirts across India, or Mustaqeem from Uttar Pradesh, who began working as a cleaner and tea boy in a factory at the age of 13 before starting to export clothing to the United States. The origins of the textile industry, another important Dharavi company, can be linked to the collapse and downfall of Bombay's textile industry in the 1950s and 1960s. This resulted in the growth of an unofficial textile industry in Dharavi, where various businesses (such as those engaged in weaving, printing, tailoring, etc.) collaborated to create clothing. Dharavi manages a significant amount of outsourced work from global manufacturers of jeans and apparel.

The average annual income of smaller jobbers is roughly Rs. 7 lakhs, whereas that of larger ones is typically double that amount. A unit typically produces 500 to 600 shirts each day, with some workers coming from Tamilnadu and others from Bihar. Workers are often paid on a piece-rate basis and can make up to Rs. 150 per day. A single large unit may generate close to Rs. 70 lakhs in revenue annually. 500 people are reportedly employed in Dharavi's textile and tailoring industries, with an additional 100 individuals (mostly young boys from Bengal and Bihar) working in machine and hand embroidery and zardozi (done mainly by Bengalis). Additionally, many people work in related fields like machine embroidery (primarily for the local market) and hand embroidery, or "zardosi."

Food industry

All delicious, savory snacks made from rice flakes or chickpea flour, such as chivda, boondi, sev, and gathiyas, are produced in the primarily domestic food sector of Dharavi. Not to mention the salty khari cookies, delectable butter biscuits, fresh sliced bread, and buns baked in Dharavi's 25 bakeries, the sweet chikki, composed of peanuts and jaggery, is a highly popular delicacy.

Papads are a type of cracker or flatbread that are made by drying wet papads in open areas like courtyards or terraces. However, clever housewives in crowded Dharavi can maximize even the smallest of spaces. The largest candy manufacturing facility in Mumbai, and possibly all of India, is the Punjabi Ghasitaram Halwai Karachiwala factory in Dharavi. It is claimed to consume 800 liters of cow milk and 2,000 liters of buffalo milk per day. The production of regional sweets by employees from various regions is very intriguing; for example, Bengalis make chamchams and rosgollahs, Punjabis make ladoos and gulab jamuns, Maharashtrians make kaju katri and barfis, and bhaiyyas (migrants from Uttar Pradesh) make samosas.



Fig 5.1. "Dharavi's leather industry products: by Keegan Everitt licensed under CC
Source: <https://www.pexels.com/photo/shallow-focus-photography-of-assorted-color-leather-belts-1023937/>, License: <https://www.pexels.com/license/>



Fig 5.4. "Textile and Tailoring industry workers working in the factory" by BBC World Service licensed under CC by NC 2.0.
Source: <https://www.flickr.com/photos/bbcworldservice/3494488125/>, License: <https://creativecommons.org/licenses/by-nc/2.0/>



Fig 5.2. "Workers prepare goods in cramped up spaces" by Linda de Volder licensed under CC by NC-ND 2.0
Source: <https://www.flickr.com/photos/lindadevolder/49017612717/>, License: <https://creativecommons.org/licenses/by-nc-nd/2.0/>



Fig 5.5. "Embroidery work being done" by MM from Switzerland is licensed under CC by SA 2.0. Source: https://commons.wikimedia.org/wiki/File:An_embroidery_unit_in_Dharavi,_Mumbai.jpg, License: <https://creativecommons.org/licenses/by-sa/2.0/deed.en>



Fig 5.3. "Leather chemically tanned and cleaned before dying" licensed under CC0 1.0.
Source: <https://www.rawpixel.com/image/6052307/free-public-domain-cc0-photo>, License: <https://creativecommons.org/publicdomain/zero/1.0/>



Fig 5.6. "Textile and Tailoring factory" by BBC World Service licensed under CC by NC 2.0.
Source: <https://www.flickr.com/photos/bbcworldservice/3496320319/in/photostream/>, License: <https://creativecommons.org/licenses/by-nc/2.0/>

Food industry

The shops offering sweets and savory items made in the homes directly behind the shops are not very far away, at the crossroads. One of the 27 Tamilnadu chikki-makers, Ramaswamy is a Muslim and is regarded as the tribe’s founding father. However, their daughter is pursuing a career as a chartered accountant. Ramaswamy’s wife only speaks Tamil. We appreciate the yummy chikki earnings

Mumbai is home to the most intricate lunch delivery system in the world, which involves an intricate choreography of the collecting, delivery, and return of more than 200,000 tin lunch boxes to the office and other workers throughout the city. According to a recent survey, the system is so effective that there is just one error out of every 16,000,000 deliveries. Dharavi also operates a thriving dabbba kitchen. When India was ruled by the British in the 1880s, the idea of the lunchbox courier (Dabbawala) first emerged. Many Britons who preferred home-cooked meals to local fare employed this service to have lunch delivered to their desks.



Fig 5.7. “Bakery in Dharavi” by Rajesh_India licensed under CC by NC-ND 2.0.

Source: <https://www.flickr.com/photos/pamnani/24386604255>, License: <https://creativecommons.org/licenses/by-nc-nd/2.0/>



Fig 5.8. “People in the market” by Trishik Bose licensed under CC

Source: <https://www.pexels.com/photo/people-in-the-market-13403804/>, License: <https://www.pexels.com/license/>



Fig 5.9. “Lunchbox (Dabbawala) services in Dharavi” by Joe Zachs licensed under CC by 2.0.

Source: <https://commons.wikimedia.org/wiki/File:Dabbawala.jpg>, License: <https://creativecommons.org/licenses/by/2.0/deed.en>

Recycling industry

The NSDF survey indicates that Dharavi’s plastic recycling sector, which employs more than 5,000 people, is the biggest in India. In 1986, it was believed that there were Rs. 60 lakhs in sales annually; today, it should be much larger. At least 3,000 sacks of plastic depart this region each day. For eleven hours of work, recyclers are paid a daily pay of 40 to 45 rupees. There are about 722 small and large businesses, 359 of which have licenses. (Quillbot, 2019)

The blue caps are removed after the paper labels on the water bottles have been manually peeled off and shipped for further recycling into rough paper. The plastic from the bottles is then transported to small facilities where it is cut into tiny flakes that can either be exported or melted into pellets for use in the plastic industry. This type of unfinished plastic is often purchased in China.

New products are created for global exportation. The stuffing of products we use to sit on, such cushions and soft chairs as well as blankets, is manufactured from these recycled bottles. Additionally, these bottles are the source of the common fabric known as “polyester fleece” for warm apparel. Did you know that every 150 fleeces created from plastic bottles prevents about 500 kg of harmful air pollution and conserves nearly a barrel of oil (about 160 litres)? Few people are aware that Dharavi is home to India’s largest plastic recycling sector (National Slum Dwellers Federation poll). There are around 700 small and large businesses, employing over 5,000 individuals; in 1986, the estimated yearly turnover was Rs 60 lakh. (Sharma, 2000

Most of the trash produced by Mumbai’s consumer-focused residents travels to Dharavi in large bags or containers. Waste pickers collected garbage from all around the city, and after some rough sorting, it is now carefully sorted for further processing at the 13th Compound, which is located where the 60 Feet Road and the Mahim-Sion Link Road intersect. This is the renowned 13th Compound, which recycles everything. Anything, including oil cans, plastic drums, chemical drums, cotton or iron trash, empty tins or bottles, or empty plastic drums. At least 3,000 sacks of plastic depart this region each day. And what isn’t recycled is cleaned up and sold used, like chemical drums that make excellent water containers later. Will the Recycling Compound go like the tanneries...to make room for yet more homes on excellent real estate? is a worrying query.



Fig 5.10. “Dharavi’s recycling industry” by Linda de Volder licensed under CC by NC-ND 2.0.

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Fig 5.11. “Inside the recycling industry” by Linda de Volder licensed under CC by NC-ND 2.0.

Source: <https://www.flickr.com/photos/lindadevolder/49017340336>, License: <https://creativecommons.org/licenses/by-nc-nd/2.0/>

Surgical Thread industry

Another industry born out of proximity to Deonar was the production of goat intestinal sutures. Abdul Baqua, who moved to Dharavi at the age of thirteen, experimented with a variety of trades in several locations before joining his friend in the suturing trade. Baqua started off creating sutures for large corporations like Johnson & Johnson before opening his own business in Dharavi that exports to more than 100 nations. Baqua is very proud that his Dharavi lab is WHO-certified and as clean and hygienic as the hospital where the sutures will be used, despite the unsightly outside and less-than-high-tech interior design. (Quillbot, 2019)

Kite factory

Kites are made from wood from Kolkata, used file folders, and recycled plastic, paper, and cellophane from the Dharavi recycling company. The kites are created specifically for the local market and for celebrations like Makar Sankranti and Diwali; they are not intended for export. Kite strings, also known as manja (a mixture of ground glass and chemical glue coating the thread), are produced locally and used to cut each other’s kites during kite fights.

Gold and Jewellery industry

The gold jewelry-making, polishing, and refining workshops are located down darkened alleyways; a row of glistening jewelry stores fronts them on the main road. People from a variety of States, including Maharashtra, Gujarat, Bengal, and Tamilnadu, as well as a mix of Muslims and Hindus, can be seen working in this sector.

Printing industry

The entire printing supply chain, including graphic designers, creative directors, editors, printers, paper suppliers, and die cutters, are in Dharavi and serve both domestic and foreign clients. The scale of the units here ranges from people making materials for use in Dharavi from their homes or small businesses to major corporations generating digitally printed Bollywood posters and 20-meter-wide highway advertising. The printing equipment used also varies widely, from vintage pedal-driven letterpress machines to screen printing studios and the most recent digital printers.

Conclusion

The population of Dharavi is approximately 1 million people, both at home and at work. So, just like in the official city, all types of services are accessible here, including grocery stores, barbershops, laundromats, restaurants, and craftsmen and cobblers. There are major zones like Pottery (Kumbharwada) community and Koliwada (Fishermen) community, which will be discussed further since they are major industries and critical sites for research. So, to sum up the work-culture and industries over here:

“You name anything, and Dharavi will make it happen!”



Fig 5.12. “Inside the surgical thread industry” by Adam Cohn licensed under CC by NC-ND 2.0. Source: <https://www.flickr.com/photos/adamcohn/16981340326/in/photostream/>, License: <https://creativecommons.org/licenses/by-nc-nd/2.0/>

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Fig 5.10. “Dharavi’s recycling industry” by Linda de Volder licensed under CC by NC-ND 2.0. Source: <https://www.flickr.com/photos/lindadevolder/49017362511>, License: <https://creativecommons.org/licenses/by-nc-nd/2.0/>



Fig 5.11. “Inside the recycling industry” by Linda de Volder licensed under CC by NC-ND 2.0. Source: <https://www.flickr.com/photos/lindadevolder/49017340336>, License: <https://creativecommons.org/licenses/by-nc-nd/2.0/>

Site selection and analysis

To further develop my thesis, I started by selecting 3 prominent locations for my thesis. I studied all the basic parameters for those to arrive at a final site for my intervention.

SITE A: Chambda Bazar. Several industries present; Main centre of Dharavi; Proximity to Koliwada (fishermen community)

SITE B: Kumbharwada. Near Matunga Labour camp, At the periphery connected by Metro station- Mahim Station, Matunga Station and Sion Station; Located near a two-way road (60 feet road)

SITE C: Koliwada. Comprised of the fishermen community; The fishermen community is polluting the Mithi river and the mangroves. The pollution affected the aquatic life that further affected the employment of the fishermen; Near to chawls (another type of informal settlements apart from slums). A part of that area was also used for slum redevelopment projects (New houses under the Rajiv Indira project)



Fig 5.13. Dharavi's map showing three chosen sites
Source: Author



Fig 5.14. Area analysis of the three chosen sites
Source: Author



Fig 5.15. Vegetation analysis of the three chosen sites
Source: Author



Fig 5.16. Formal road network (Vehicular and pedestrian)
Source: Author



Fig 5.17. Informal road network (Vehicular and pedestrian)
Source: Author

Final Site Selection

The three sites were compared by dictating some important criteria for the same, as given in the table below. Site B (Kumbharwada) came out to be the final site for the intervention.

5-Most Ideal; 1-Least Ideal	Site A	Site B	Site C
Proximity to natural elements (water, mangrove, natural park)	1	0	3
Close to transportation systems	0	3	1
Houses several industries	3	1	1
Includes labor camps and housing settlements	1	2	2
(Slum re-development projects)			
Polluting industries	1	5	0
Industrial center of Dharavi	3	2	2
Presence of old houses and patios	1	3	3
Totals	10	16	12

Table 5.1. Site Selection Criteria
Source: Author

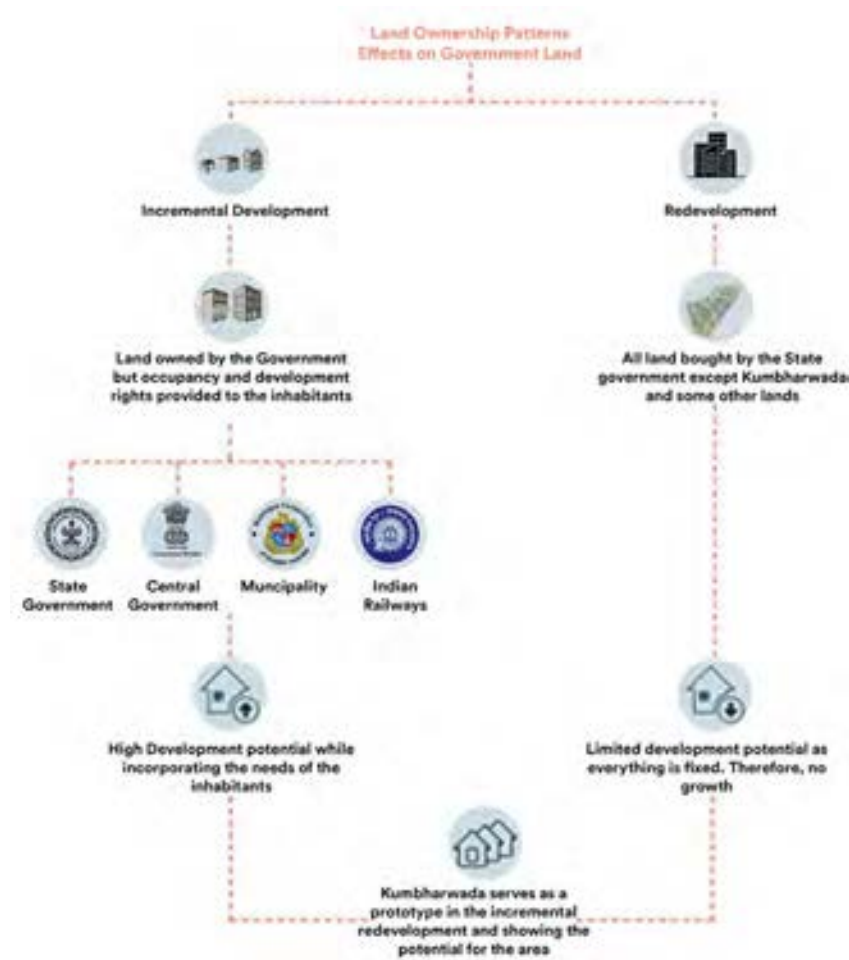


Fig 5.18. Land ownership patterns in Dharavi
Source: Author

Analysis of existing site fabric

A magnificent parcel of land measuring 12.5 acres in Dharavi is known as “Kumbharwada” (Potters’ Colony) in honor of the migratory Gujarati potters. They were relocated to Dharavi from the then-emerging city center in south Bombay, like many other populations who migrated to the island city to pursue their fortunes. The 250 resident families of potters, whose company dates to the founding of Dharavi, hold a special place in the neighborhood. Their homes, which serve as both residences and places of employment, have an intriguing design. They are long, narrow buildings with two entrances: one leads to the yard where production takes place and where the communal bhatti or kiln is located, and the other leads to the street where the finished goods are displayed and sold.



Fig 5.19. Kumbharwada's location in Dharavi's context
Source: Author



Fig 5.20. “Pottery products and process of making pots” by MM from Switzerland licensed under CC by SA 2.0 Source: https://commons.wikimedia.org/wiki/File:Dharavi_DSC3114_%285843517698%29.jpg. License: <https://creativecommons.org/licenses/by-sa/2.0/deed.en>

Analysis of existing site fabric

In the latter half of the 19th century, immigrants from Gujarat founded Kumbharwada, one of the earliest villages in Dharavi. One of the most well-known local crafts in Dharavi is pottery, which is located here. Artist’s pitch and burn pots of every size and form, from little yoghurt cups to larger water pots, decorative vases, and diya candles, in the tiny spaces between their homes. The potters, like many people in Mumbai, brought their family business with them and still have close ties to their home areas. They collaborate to enhance kiln designs, visit their towns frequently, and, most importantly, import their clay from the North where they live in constant physical and online contact with their relatives. (Balan, 2018)

The potters, who had previously established in the city’s southernmost point, were relocated to its northern edge by Mumbai’s municipal officials in the 1800s. The smoke from their kilns was regarded as an annoyance in the more affluent areas of town. Dharavi was a barren area between two railway lines when they arrived; it was sprinkled with a few villages, which have subsequently grown into one sizable settlement. Dharavi became increasingly central as Bombay expanded, generating real estate speculation as well as academic, tourist, and cultural interest in the area. (Saglio-Yatzimirsky, 2021)

As we can see in the chart on the next page , the key statistical data which stuck me is the density of the region, the ratio of open spaces to built spaces and the average number of toilets per person. Thus it is clear that there are problems associated with space and sanitation.

To put things in perspective, it is about half the size of Central Park in Manhattan and about 1 million people stay on approximately 519 acres of land: making it a highly populated region.



Fig 5.21. Dharavi's comparison with Central Park in terms of area
Source:

No	Attributes	Existing
1	Population	*Approximately 1,000,000
2	Density	1748.76 Person per Hectare
3	Number of households	45,626 (Mashal data) 50,189 (10% increase considering year 2000)
4	Ground cover (Residential)	70% (From Dharavi Base Map)
5	Built up area	1,923,181 sq.m (Assumption)
6	FSI (overall)	1.34
7	Area- roads	9.74%
8	Area- residential	75%
9	Area- commercial	5.64%
10	Area- R+C	0.28%
11	Area- industries	4.59%
12	Area- recreational/open space	3.58%
13	Public-semi public	2.80%
14	Number of floors	G, G+1, G+2 (Average 1.79-Mashal Data)
15	Average HH size	2.98 (Mashal Data)
16	Size of unit for rehab	225 and 269 sq.ft
17	Size of unit for rent	180 sq.ft
18	Toilets per person	1 toilet / 1440 people
19	Type of development	Plotted

Table 5.2. Key Demographics of Dharavi
Source: Author

User typology

With the density of Dharavi being what it is, the 1-4 story houses are a mixture of housing, commercial and industrial units. Each unit, no matter the typology, doubles up as a sleeping area for workers.

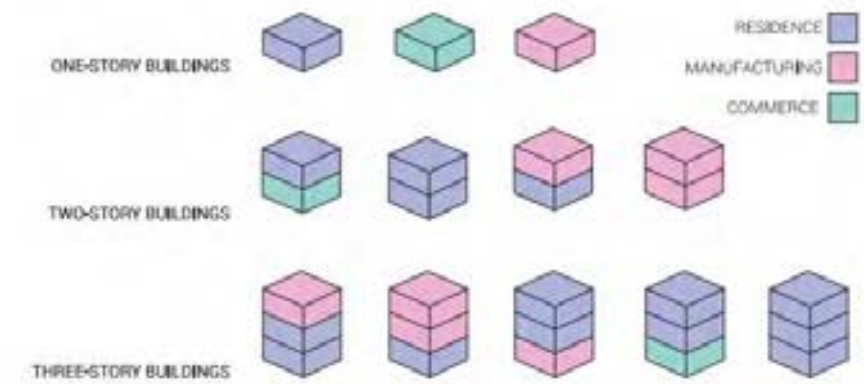


Fig 5.22. User typologies in Dharavi
Source: Author

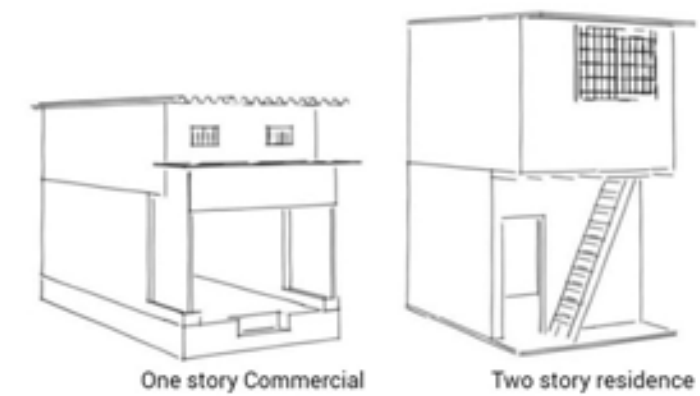


Fig 5.23. Housing typologies in Dharavi
Source: Author

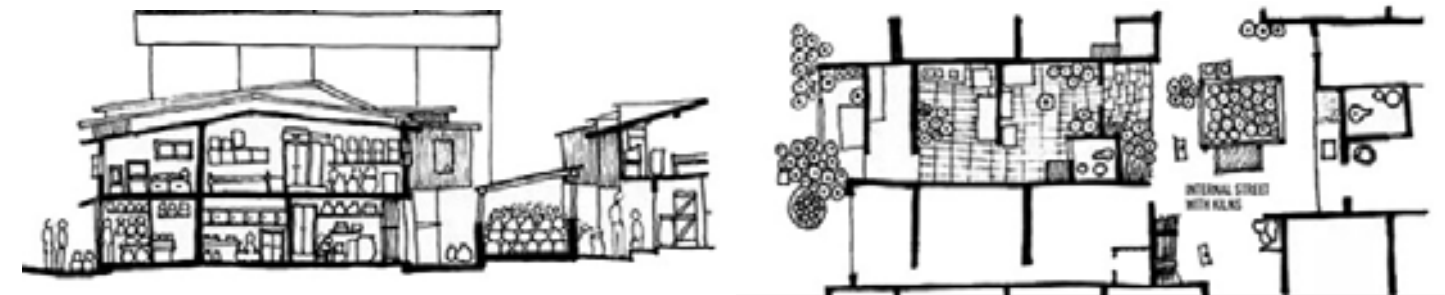


Fig 5.24. House plan and section of existing settlement
Source: Author

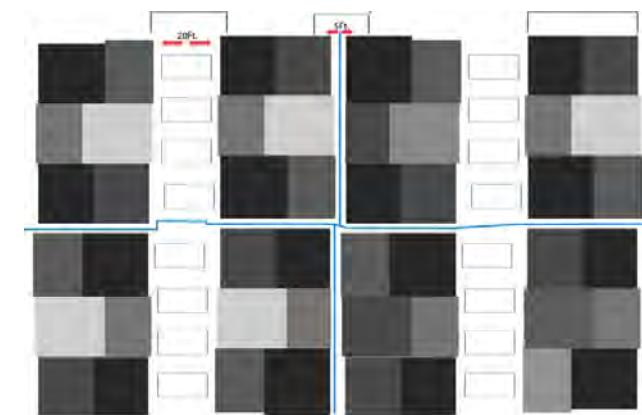


Fig 5.25. Internal street layout for Kumbharwada
Source: Author

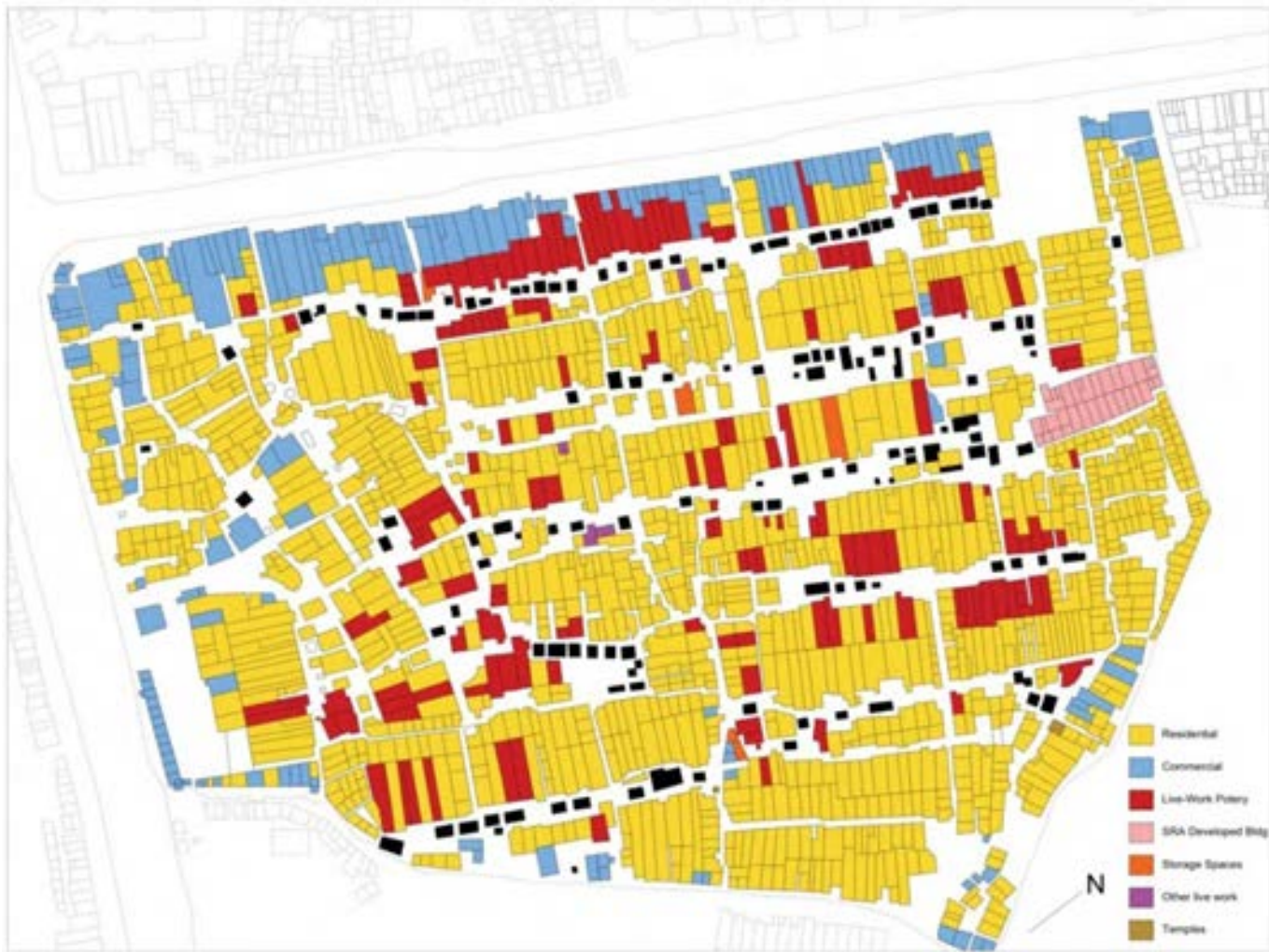


Fig 5.26. Kumbharwada Masterplan
Source: Author



Fig 5.28. "Existing site image- Area outside the houses-02" by SriChinnu is licensed under CC by SA 4.0.
Source: <https://commons.wikimedia.org/wiki/File:Kumbharwada.jpg>, License: <https://creativecommons.org/licenses/by-sa/4.0/deed.en>



Fig 5.29. "Potters selling products on the external road" by Arnova.
Source: https://commons.wikimedia.org/wiki/File:Kumbharwada_road.jpg, License: <https://creativecommons.org/licenses/by-sa/3.0/deed.en>



Fig 5.27. "Existing site image- Area outside the houses-01" by MM from Switzerland is licensed under CC by SA 2.0. Source: https://commons.wikimedia.org/wiki/File:Pottery_unit_in_Dharavi_Mumbai.jpg, License: <https://creativecommons.org/licenses/by-sa/2.0/deed.en>



Fig 5.30. Existing community drawings
Source: Author

Fig 5.31. Morning, afternoon, and evening activities
Source: Author

The activities throughout the day range between residential and social, pottery related and commercial. These can be seen spilling out onto the street and open spaces can be broadly divided into the three-time frames of Morning, Noon and Evening. Morning Activities: Domestic nature such as washing and cleaning; Movement of children going to school. Afternoon Activities: Predominantly pottery related such as making, firing of pots in the kilns, painting, transporting, and drying of pots. Evening Activities: Of social nature- the community can be seen chatting on the otlas; Children playing on the streets; Informal Street vendors such as Chaatwalas and Cotton Candy walas traverse the streets.

Design Strategies

Five lanes make up Kumbharwada, which is littered with more than 120 kilns and is crammed with residences and businesses. Many of these structures fall under the tool-house typology, which features living space above workshops on the first floor. This varied use of space, as it is throughout Dharavi, is both a blessing and a curse. Social circles, on the other hand, are highly localized and hybridized, with friends, coworkers, extended family, workers, and business rivals all in proximity. For instance, childcare might be shared by members of the extended family, or parents can work and watch their kids at the same time. Otherwise, a potter can ask a family for assistance if they need aid lighting their kiln. They can transfer an order to a cousin if they are unable to fulfill it. Everyone around here is an “auntie” or an “uncle.”

On the other hand, the locals must put up with the offensive smoke that the tiny brick kilns they have constructed next to their homes spew out. Fabric scraps from the adjacent textile industry are piled up and lit in a fire pit at the bottom of the kiln to fire pots. (Balan, 2018). This causes an unpleasant haze to be produced, choking the streets, and coating the homes in black soot. Unfortunately, this area doubles as the community’s main gathering place for the public and the open. Even though each kiln isn’t used every day, there is always one nearby, and it is obvious that this practice is unhealthy and unpleasant for the people of Kumbharwada.

Building up Dharavi: The concept and framework

The idea was to target 3S, meaning *Space, Sanitation and Smoke*



Fig 6.1. “Target 01: Space” by Meena Kadri is licensed under CC by NC-ND 2.0
Source: <https://www.flickr.com/photos/meanestindian/4220469421>, License: <https://creativecommons.org/licenses/by-nc-nd/2.0/>

The residents tend to add additional floors as the family expands or when supplemental workspace is required. Hence incremental principles at site form the basis for my intervention.



Fig 6.2. "Target 02: Sanitation" by lecercle licensed under CC by NC-SA 2.0
Source: <https://www.flickr.com/photos/lecercle/2612759786>, License: <https://creativecommons.org/licenses/by-nc-sa/2.0/>

The residents have an acute shortage of sanitation facilities, such that one toilet serves the requirement of approximately 1400 people in the area. The open sewers run along the narrow streets adjacent to the houses where children can be seen playing.



Fig 6.2. Target 03: Smoke by Cameron
Source: <https://urbz.net/articles/kumbharwada-pottery-village>

The residents tolerate the dreadful smoke that emerges out of the kilns that they have built near their home. Unfortunately, the same space is their primary communal and social space.

It is clear from the studies up till now that community participation is vital for the redevelopment to take place. Having said that, it is also essential to understand the stakeholders involved in this community to have a top-down bottom-up integration.

Dharavi's Actors

Actors are initially "diagrammed" to analyze them based on a variety of criteria, including their level of operation (local, national, regional, or international), their mandate (interests, concerns, agendas, or visions), their areas of expertise, and the connections between them that are related to these characteristics.

The first step in comprehending the internal and external forces acting on a development process is discovering and unraveling the relationships between all important "stakeholders" involved in a situation. It is predicted that a few crucial components or phenomena will show up, get attention, and be discussed. The sense of a tangible visual character that aims to be plainly instructive and stands alone in a map, regardless of how it is interpreted or how it turns out, is a hint to future action.

To underline the importance of the players involved in a situation having a clear platform and conscience from the start and to comprehend the mechanics of mapping transitory actor connections. To comprehend how working on and mapping space has dialectics, 1) Space as a tangible thing to be encountered, understood, and appropriated; 2) Space as a concept of the city, as representation through symbol and conflict—the interaction between the "physical" and the "non-physical".

It is critical to note that all the actors must be involved at different levels for the redevelopment to take place.

Design principles that are more firmly established aim to ground the rules in a way that considers the practitioner's position and moral code while highlighting innate values that are essential to support both the general will and voices of the populace and extensive top-down development plans. When a result, these human-centered principles will produce guidelines for practitioners to follow as they deliberate and bargain over design solutions. These design guidelines and principles include in some fashion, a clear division of intentions, scale, possible timeframes, and involvement of certain actor stakeholders wherever applicable. The suggestions are meant to be a legible framework for further exploration and development of design responses.

(Refer to the diagrams on the next two pages)

dharavi's actors

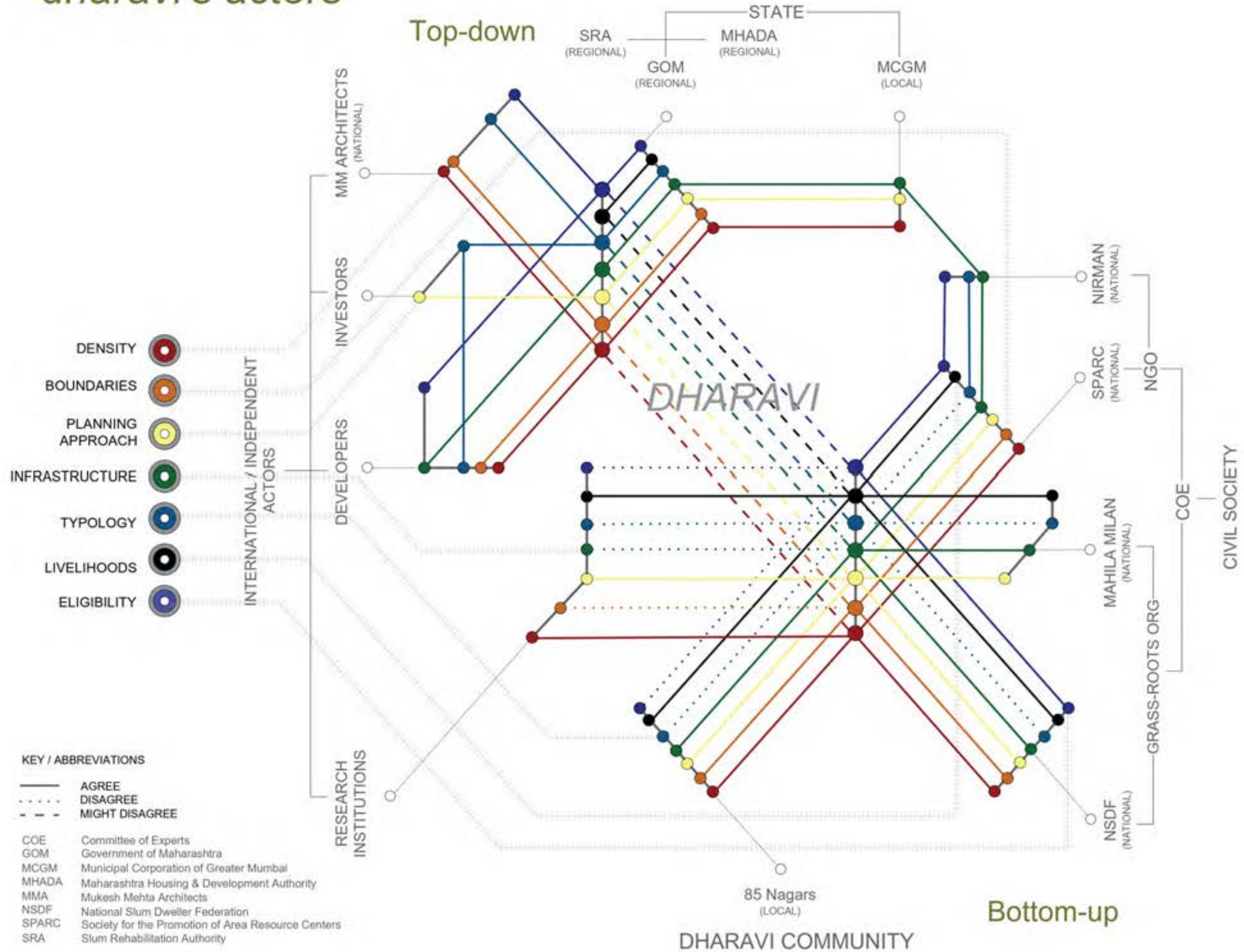


Fig 6.4. Diagram of stakeholders involved in the redevelopment process
Source: Author

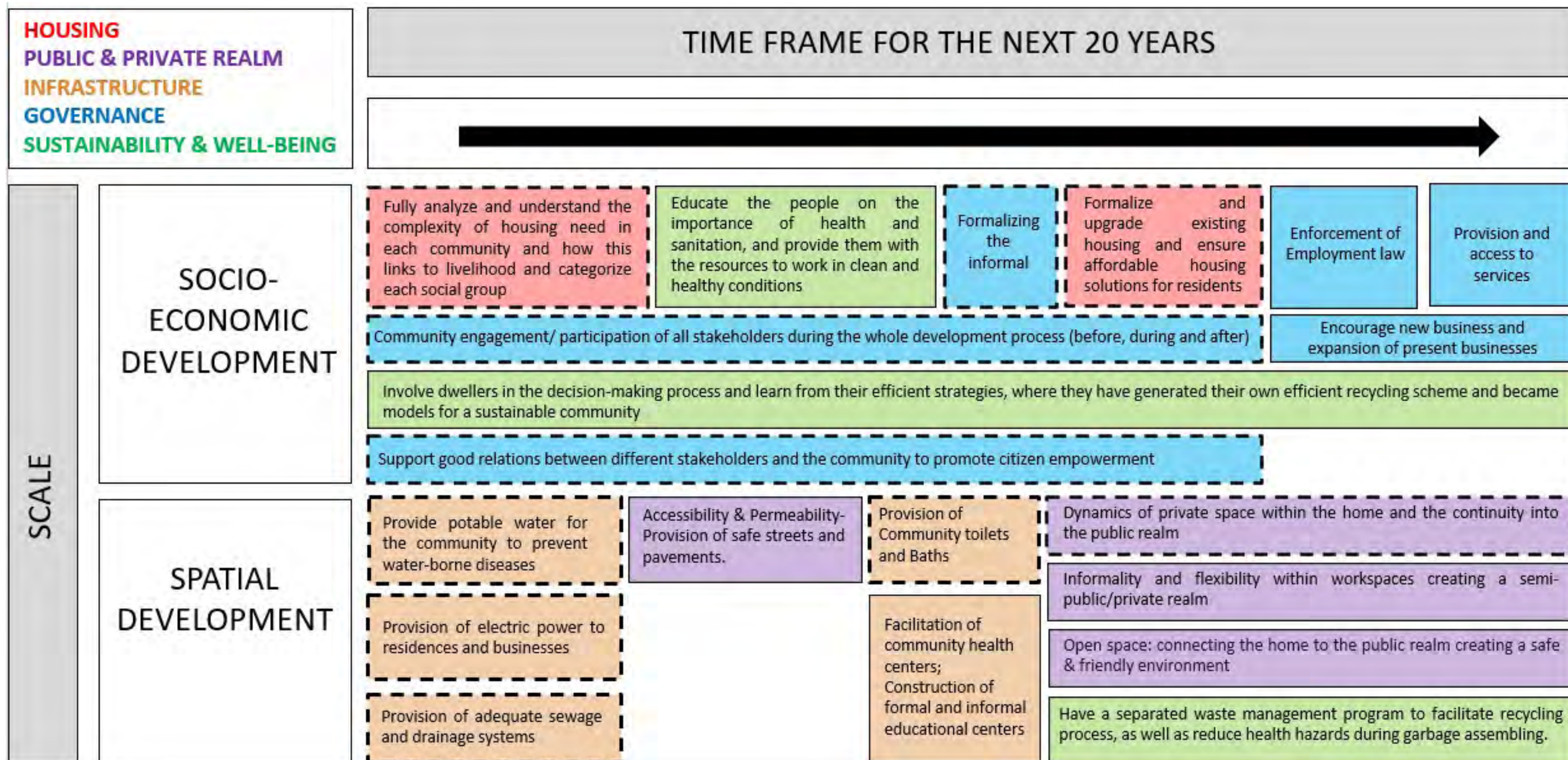


Fig 6.5. Phasing: Time and Scale of Implementation of principles
 Source: Author

dharavi's actors

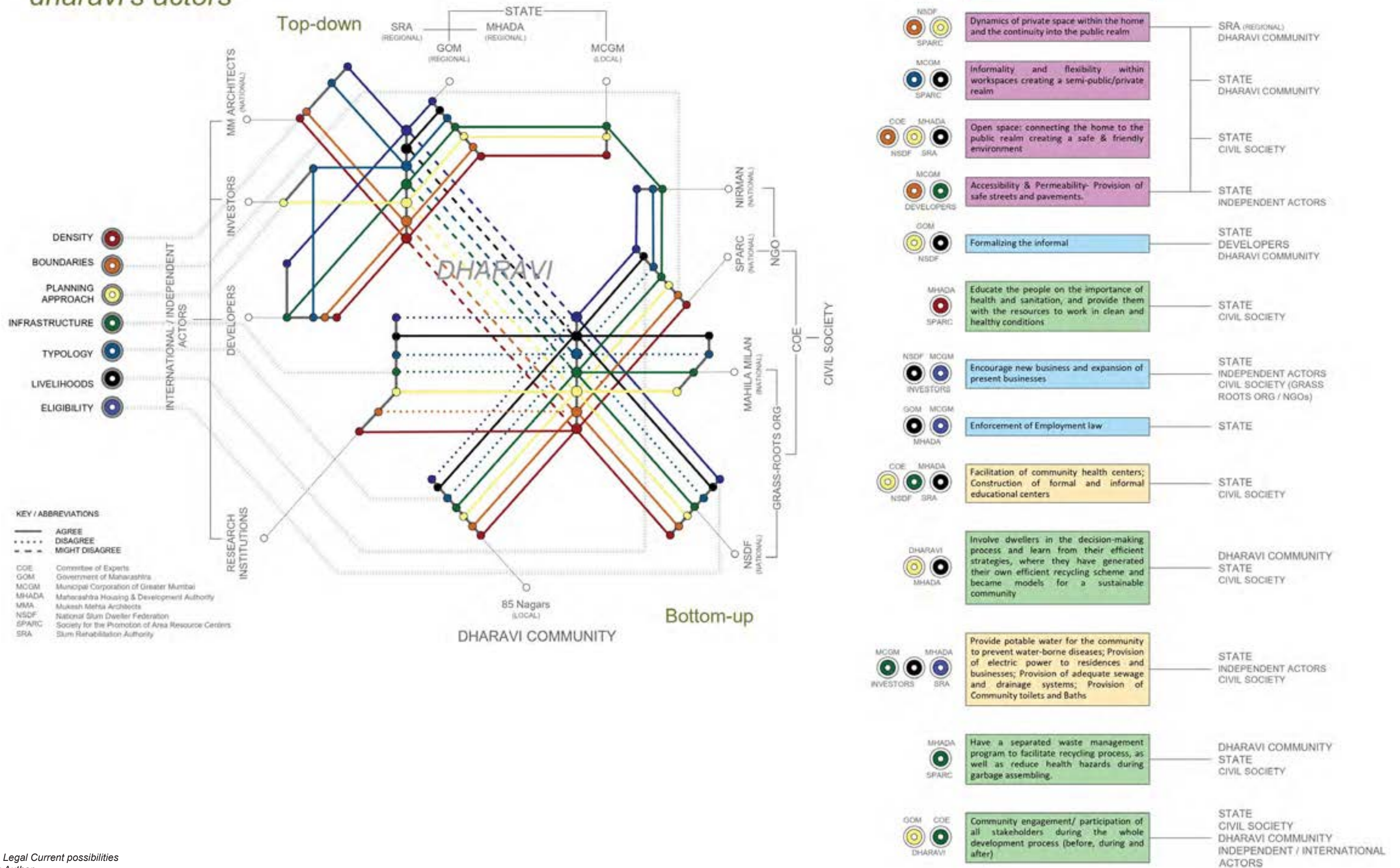


Fig 6.6. Legal Current possibilities
Source: Author

Building up Dharavi: The concept and framework

Both the diagrams were then superimposed to understand what the legal current possibilities might be for the redevelopment of the area I am dealing with. To have a top-down bottom-up approach, it is important to identify which stakeholder would be responsible for carrying out the identified principles of redevelopment. Since I am essentially dealing with space and sanitation, it is important that the MHADA (Maharashtra Housing and Development Authority) and SRA (Slum Rehabilitation Authority) get involved to solve the issue of housing and de-densification. The developers and investors can be looped in for funding the basic amenities like sanitation and provision of electricity to carry out / expand their business.

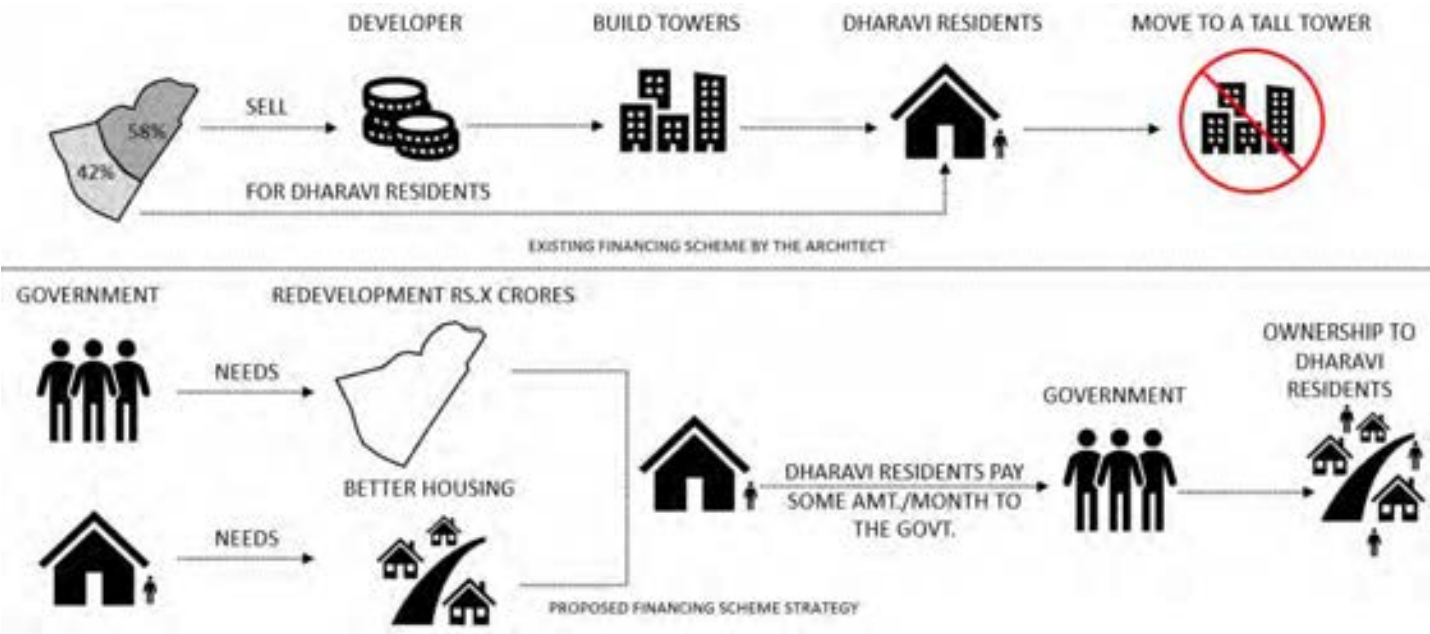


Fig 6.7. Ownership based finance scheme
Source: Author

According to the financial plan which the thesis proposes, NGOs can facilitate micro-loan to each household which could be given to Government as a rent and can be used for redevelopment of Dharavi. This will allow the inhabitant to establish better living conditions and grant land ownership rights. In terms of the site scale phasing strategy, following steps can be taken:



Fig 6.8. Revised road network along community boundaries
Source: Author



Fig 6.9. Retaining the existing 85 communities and their religious structure
Source: Author



Fig 6.10. Site scale phasing strategy
Source: Author

People would be housed in transit camps (certain open spaces which are identified in Dharavi), during the redevelopment of a particular zone, thus avoiding displacement.



Fig 6.11. Three scales for Dharavi (Micro, Mesa, Macro- from left to right)
Source: Author

Micro-scale-Realistic (Units / Individual): Community driven micro interventions at the cluster/ neighborhood level. Network of complemented public spaces managed by the community.

Mesa-scale-Possible (Group): Mesa interventions at the local level. Network of complemented public spaces with public facilities needed in Dharavi.

Macro-scale-Idealistic (Community/Government): Macro interventions at the city level. Public spaces and facilities that connect with the city.

The idea is to identify the urban acupuncture strategies on three different scales for all the communities within Dharavi.

While the outcomes of individual interventions provide useful indicators for the movement that serves as a catalyst for community transformation, there is a need for collective strategies that can match the scope of grand planning initiatives and provide the ability to represent both actual and symbolic change. The tactics will go beyond conceptual suggestions and start to show a deeper comprehension of what is possible, how it is possible, and who is possible to accomplish it with. The impact of interventions at the intersection of the micro, mesa, and macro scales must be carefully considered.

(Refer to the diagram on the next page)

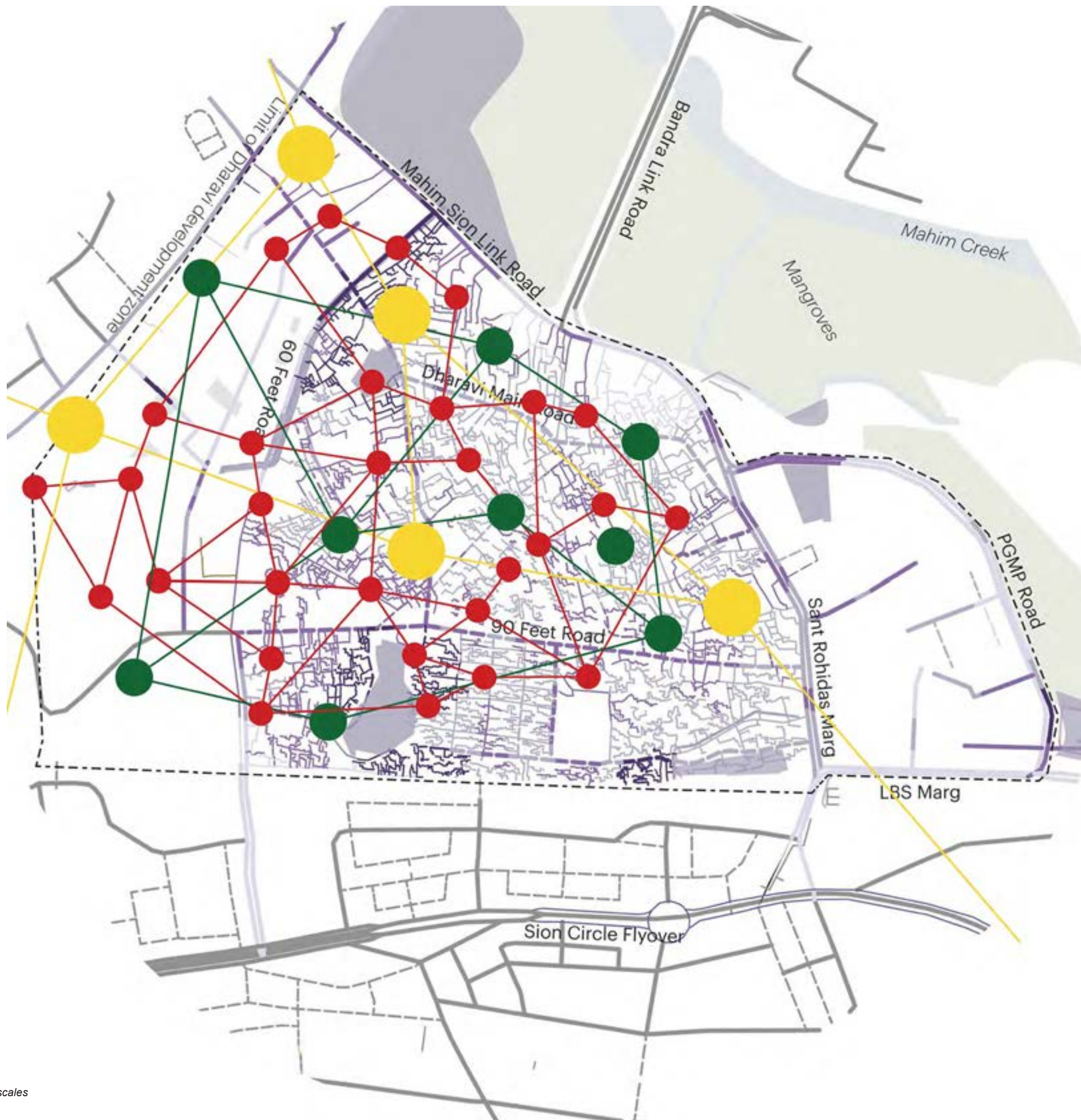


Fig 6.12. Urban context- Dharavi in terms of three scales
Source: Author

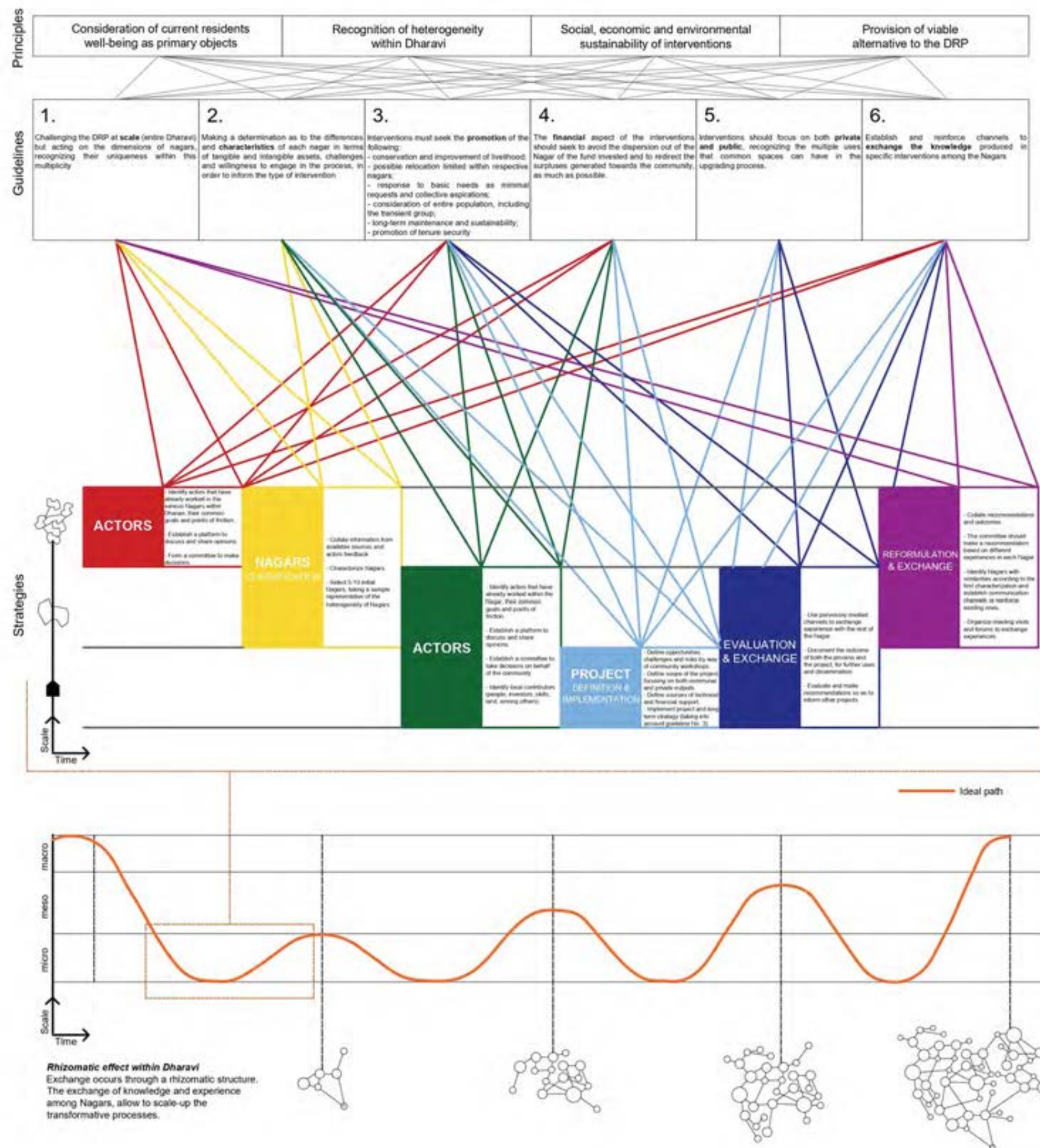


Fig 6.13. Redefining the idea of scaling-up in Dharavi
Source: Author

A toolkit of strategies for progressive self-development

Thus, it is clear from the analysis of the existing fabric and framework that community participation as well as government’s involvement is vital in the redevelopment process of the area. The incremental approach is best suited for this site since it would help the community to improvise as per the available resources.

The marked strategies can go in parallel or back and forth depending on the incrementality of the area or the requirements of the residents.

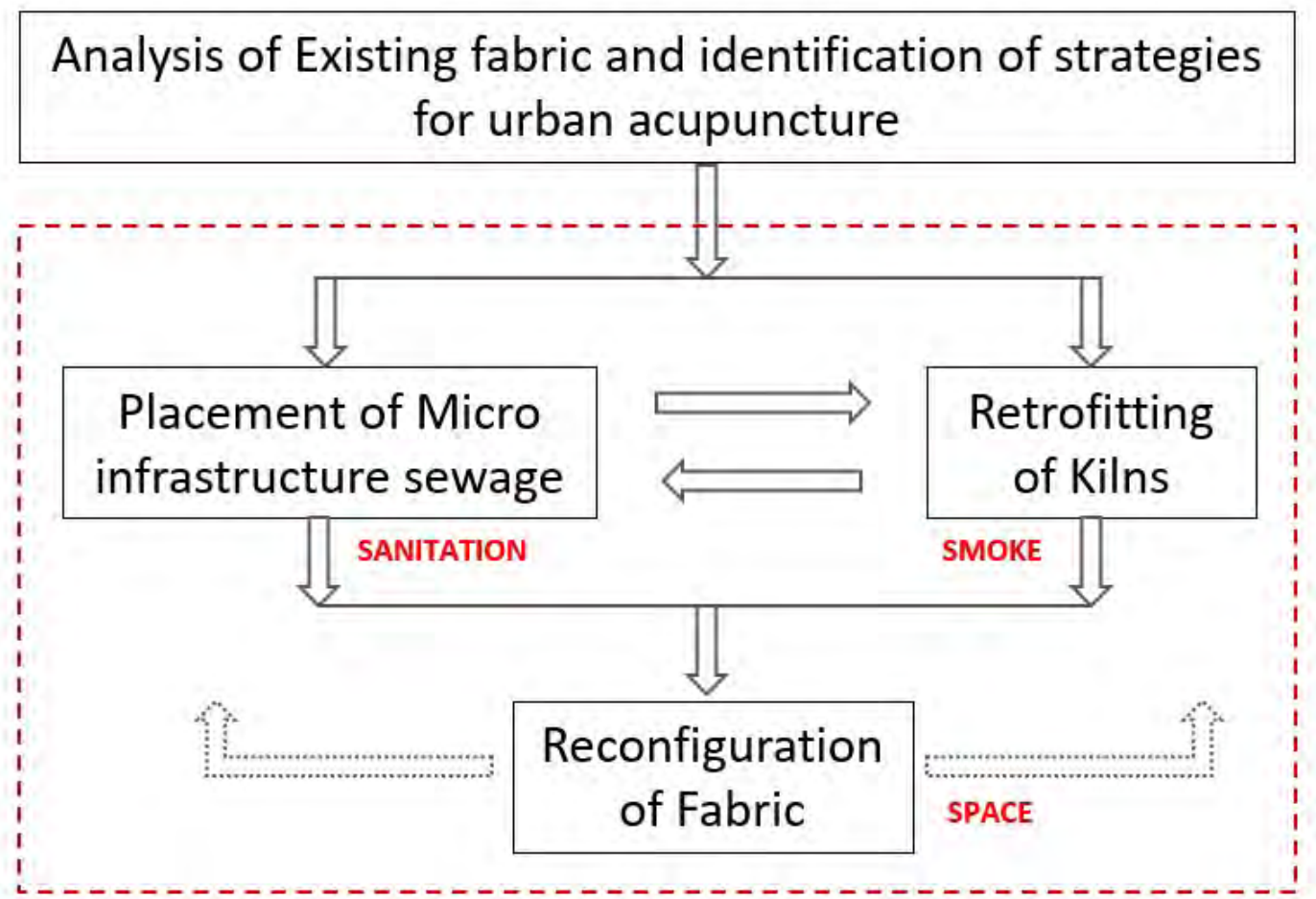


Fig 6.14. Ecological Strategies
Source: Author

Identification of strategies

Kumbharwada’s existing fabric was studied at the site level, cluster level, street level and the unit level. As per the observations, various inferences were drawn, and a possible design scheme was derived for all these levels.

Identification of strategies

Kumbharwada’s existing fabric was studied at the site level, cluster level, street level and the unit level. As per the observations, various inferences were drawn, and a possible design scheme was derived for all these levels.

Site Level



Fig 6.15. Observations, Inferences at Possible design scheme at Site level (L-R)
Source: Author

Observations:

- Commercial edge: 3 sides of the site.
- Kumbhar streets (N-S); transverse streets (E-W): form grid organization.

Inferences:

- Nodes for interaction.
- Kumbhar streets: main access streets into the site and main activity zones.
- Transverse streets: movement and transport zone.

Possible Design Scheme:

- Retain orientation of main activity streets and of transverse streets.
- Introduce more private connections between clusters.

Cluster Level



Fig 6.16. Observations, Inferences at Possible design scheme at Cluster level (L-R)
Source: Author

Observations:

- Courtyards forming connections at the edge of the cluster.
- Connections between main streets and transverse streets

Inferences:

- Constant thorough movement and dynamic activity in courtyards.
- Hence, they cannot be used by the residents as places of play or pause.

Possible Design Scheme:

- Setbacks will allow semi-private spaces for units which open into a larger communal courtyard space.

Fig 5.15. Vegetation analysis of the three chosen sites
Source: Author

Identification of strategies

Kumbharwada’s existing fabric was studied at the site level, cluster level, street level and the unit level. As per the observations, various inferences were drawn, and a possible design scheme was derived for all these levels.

Street Level

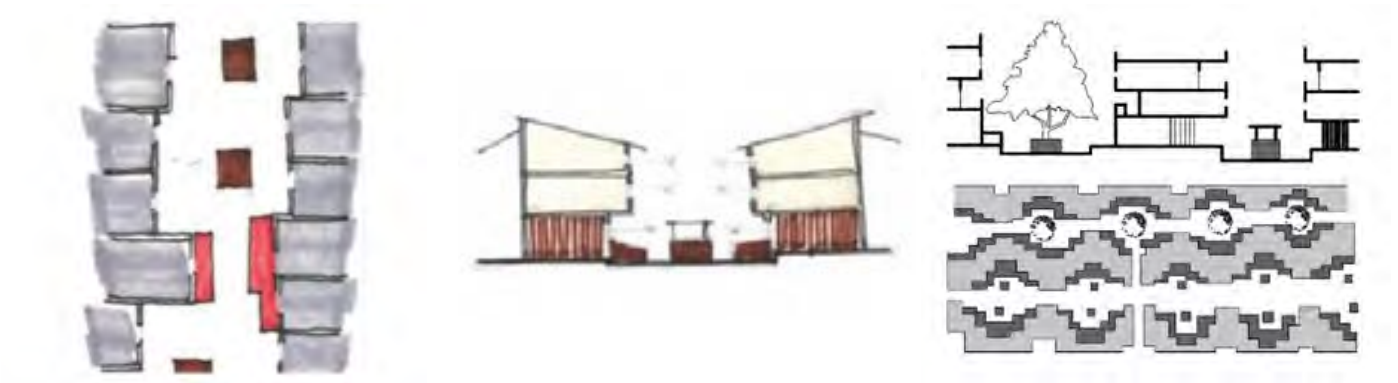


Fig 6.17. Observations, Inferences at Possible design scheme at Street level (L-R)
Source: Author

- Observations:**
- Houses which face kiln open directly onto the street-they have no threshold.
 - Hence, there is little to no activity other than pottery.
- Inferences:**
- Entire ground level: a workspace only; little social activity occurs.
 - The upper levels also do not interact as windows are kept closed due to kiln smoke.

- Possible Design Scheme:**
- Two fronts for each unit- a social front and a work front.
 - This would create more interaction thus making each unit part of 2 neighborhoods.

Unit Level 1

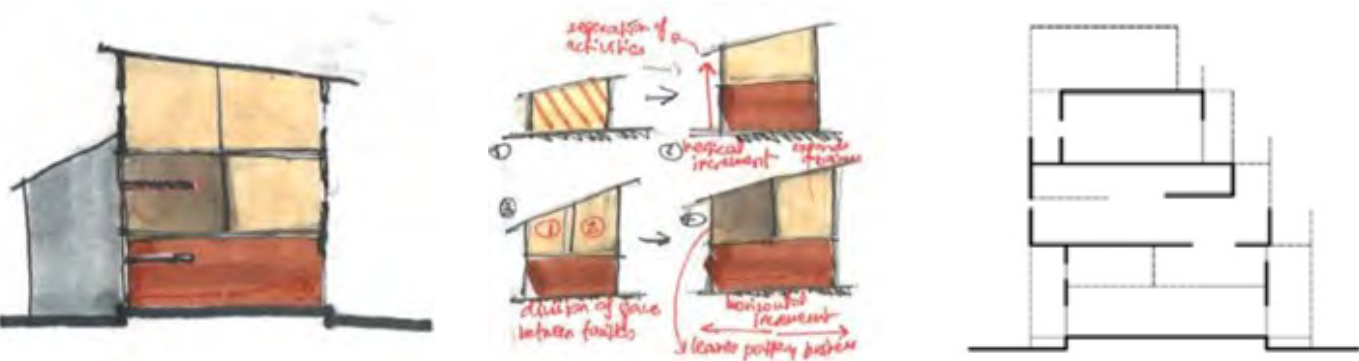


Fig 6.18. Observations, Inferences at Possible design scheme at Unit level 1 (L-R)
Source: Author

- Observations:**
- Nuclear families have been modifying structures over generations.
 - Units tend to add mezzanine mainly for storage purposes, sleeping and privacy.
- Inferences:**
- Each unit increments horizontally and vertically.
 - Some units are made larger by acquiring and combining adjoining units.

- Possible Design Scheme:**
- Units can have higher floor heights which can allow mezzanine to be used for living and sleeping.
 - Staggering the units for incremental development.

Identification of strategies

Kumbharwada’s existing fabric was studied at the site level, cluster level, street level and the unit level. As per the observations, various inferences were drawn, and a possible design scheme was derived for all these levels.

Unit Level 2

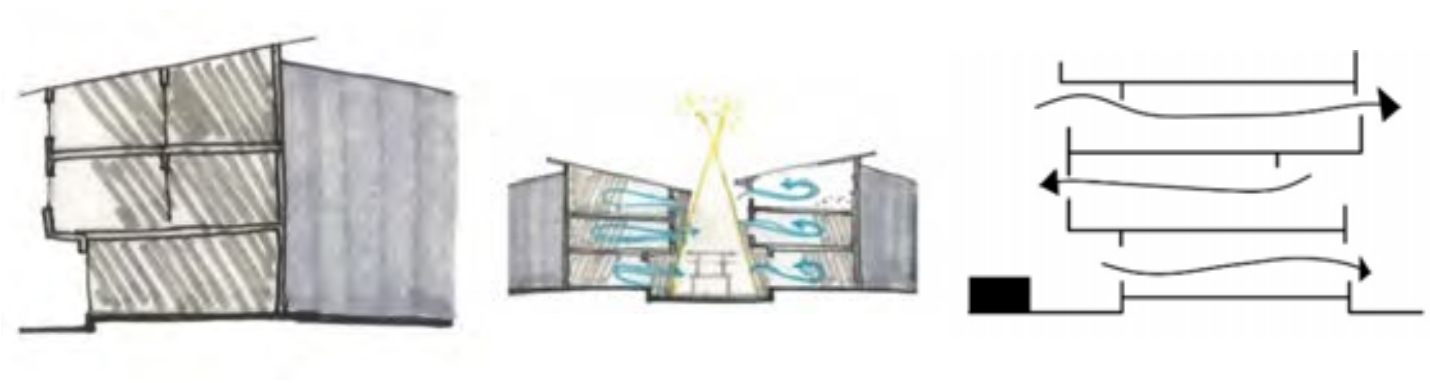


Fig 6.19. Observations, Inferences at Possible design scheme at Unit level 2 (L-R)
Source: Author

- Observations:**
- Most units share walls on the three sides leaving only one facade for openings.
- Inferences:**
- Depth of the units and street width does not allow sunlight to penetrate the spaces.
 - Constant need for artificial lighting.
 - Cross ventilation is hampered due to single side openings

- Possible Design Scheme:**
- Two sides opening, allowing better air flow and natural lighting.
 - Staggering units vertically provides shades for lower units.
 - Activities near the spill over spaces.

Thus, the idea is to assimilate through different axes of streets and spaces, which improves the live-work function, creates opportunity spaces for multiple uses, and provides belongingness, thereby synergizing the authenticity and vibe of Kumbharwada.

Concept development for housing units

The concept was to maintain the individual plot boundaries and hence took the same dimension of the existing housing blocks as a starting point. The block dimension is also retained, and it gave the dimension of the resultant cluster. The idea was to upgrade the streetscape, but a sincere attempt has been made to keep the street culture and the community vibe intact as far as possible, but at the same time radically improving the housing and infrastructure conditions of the site.



Fig 6.20. Concept development for housing units-Unit, Block, Street (L-R)
Source: Author

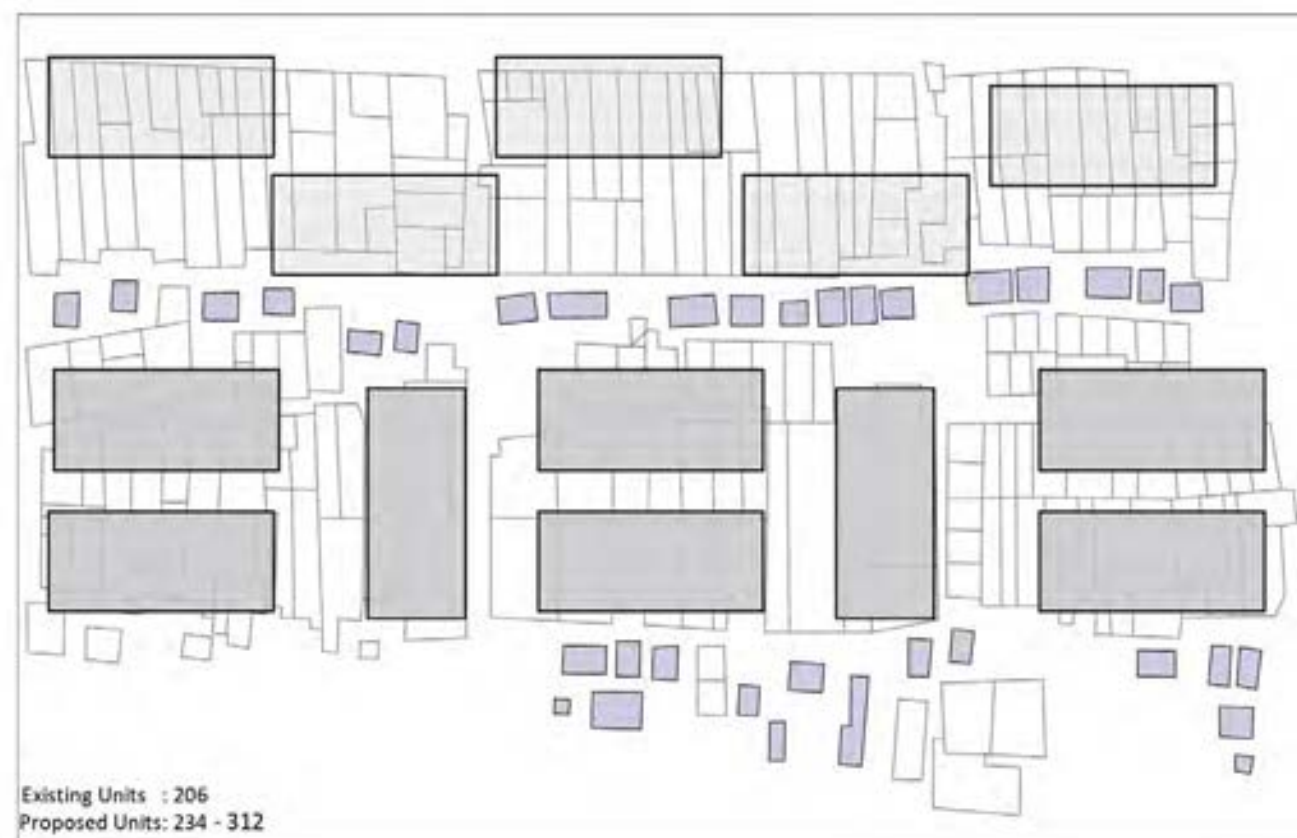


Fig 6.21. Placement of new units on existing fabric
Source: Author

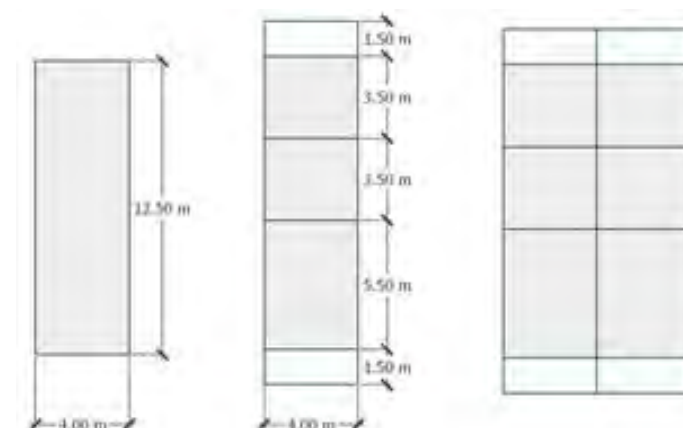


Fig 6.22. Deep housing block; RCC framework by govt., Module of two units (L-R)
Source: Author

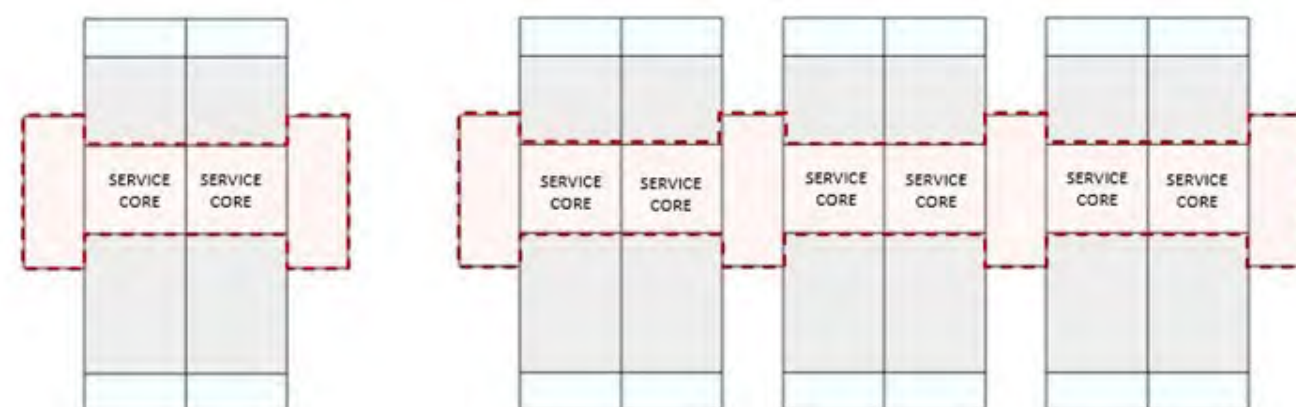


Fig 6.23. Insertion of circulation and service core in framework; Single parcel (L-R)
Source: Author

Incrementality in Housing units

A series of sketches has been added below to show how the final housing cluster was derived. The RCC framework can be provided by the housing body (government), which is the top-down approach. A module of two units has been formed using the initial deep housing block. The service core and circulation has been included in the framework. A single parcel is a combination of 6 stacks of 24 units.

There is an introduction of transient unit in the resultant cluster which would help with the incremental approach and future growth as shown in the series of sketches below. This transient unit can be occupied with the permanent unit as required. This will give the residents variations in family and unit size.

Each unit now comes with an individual toilet thus solving the issue of sanitation. Also, since the dimensions of the housing block have been derived from the existing units to keep the community vibe intact, the unit has been tested for natural light and ventilation (Refer Appendix A for further information).

Concept development overall site

The idea was to have green fringes penetrating over the site since the existing space lacked green areas. A hub of Communal areas has been formed. The junction of all working veins has been created with condensed green arteries. The final form came by having steps like staggering streets which would help in decreasing the internal street distance with the main roads. There are 2 main roads running adjacent to the site. The one on the north is called the 90 feet road and the one on the west is called the 60 feet road. It also responds to the site in a more extrovert manner along with keeping the introvert character of the community as the streets are not completely open to the main road.

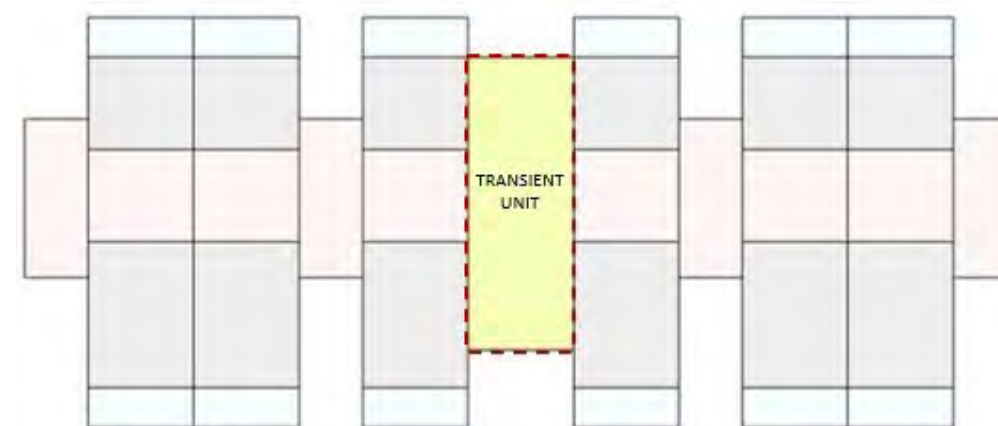


Fig 6.24. Incremental strategy-Stack of transient unit introduced in the parcel
Source: Author

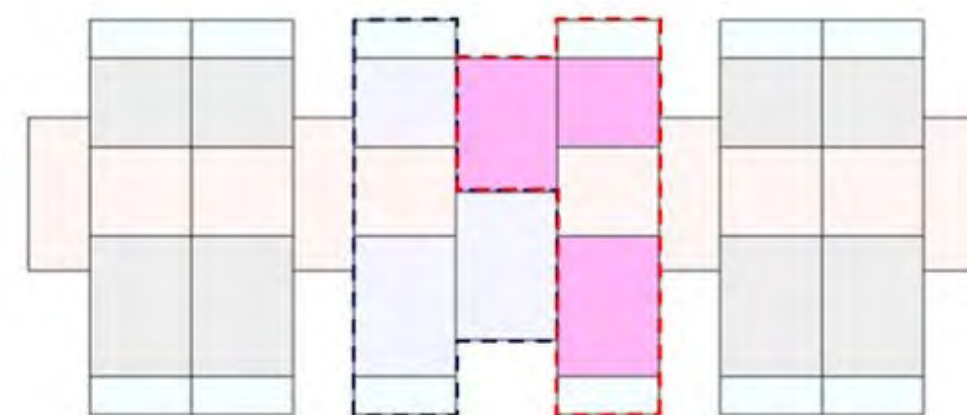


Fig 6.25. The resultant cluster-Occupation of transient unit with permanent unit
Source: Author



Fig 6.26. Layer 01- Redefining the streets
Source: Author



Fig 6.27. Layer 02- Units placed in phase 1-Incremental approach
Source: Author



Fig 6.28. Layer 03- Retrofitting of Kilns
Source: Author



Fig 6.29. Layer 04- Upgrading the fabric
Source: Author



Fig 6.30. The potential Masterplan with legend
Source: Author



RECREATIONAL AND GREEN EDGE

Fig 6.31. Recreational zones and community spaces plan
Source: Author

Concept development overall site

The residents now have better housing/living conditions. There is an introduction of recreation edge and work edge, thus creating segregation for domestic and commercial activities. Each unit comes with a toilet, hence there is wastewater management. There are smoke-free electric kilns which are taking them towards a healthier community. There are dedicated spaces for commercial activities with scope of incremental development of housing units on top (refer legend). There is now a separate loading and unloading zone for raw materials along the main roads for easy access. There are open spaces along the shop/housing fronts for expansion of activities on ground level.

Concept development overall site

The inhabitants have a higher number of recreational zones and open spaces for community gatherings in this set-up as compared to the existing situation. Children can now play in courtyards in front of the houses rather than playing along the open sewers running along the streets as per the current situation. Schematic zoom-ins of the clusters along the 90 feet road and 60 feet road are shown below.

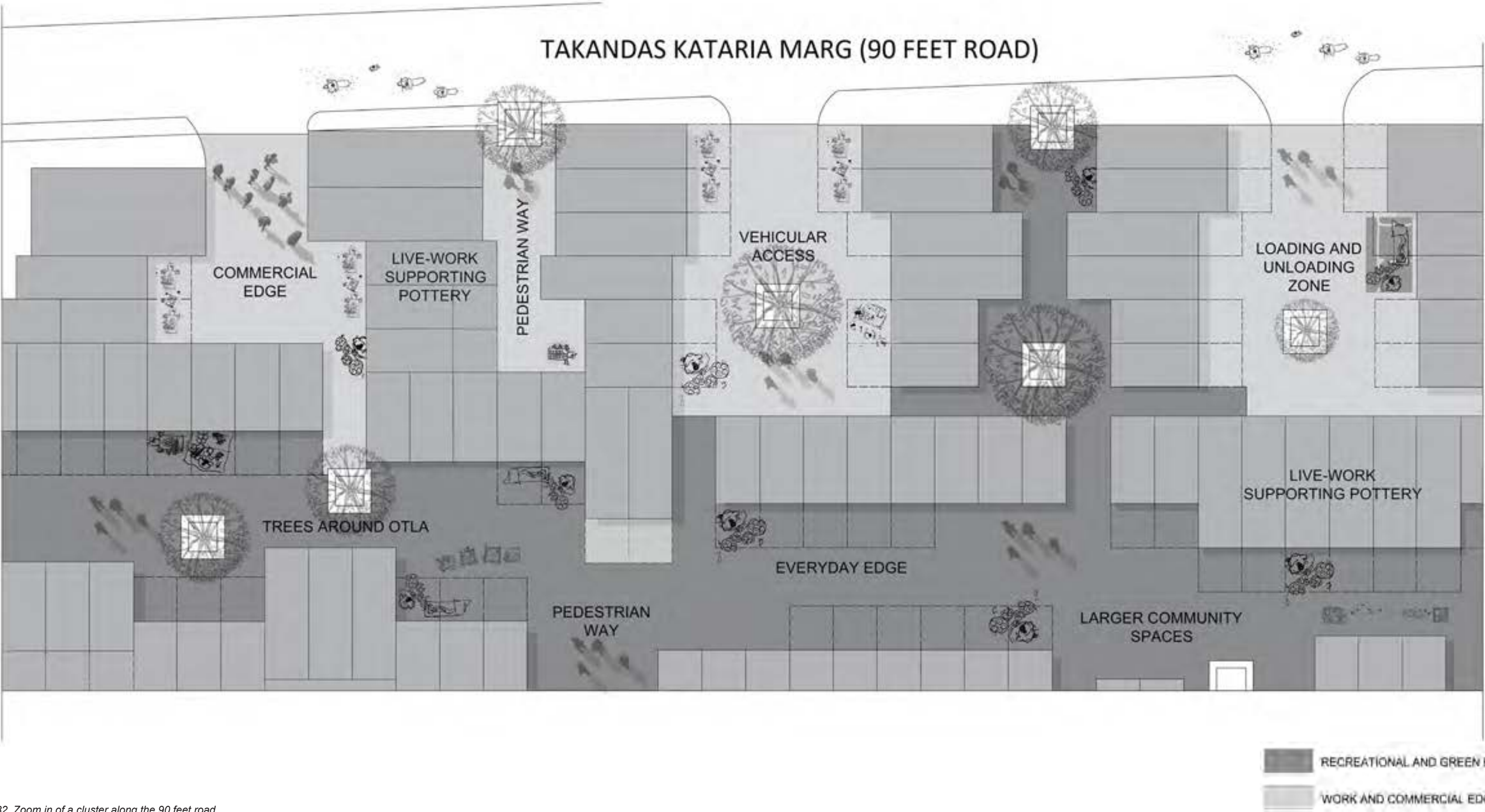


Fig 6.32. Zoom in of a cluster along the 90 feet road
Source: Author

Concept development overall site

The residents now have dedicated spaces for loading and unloading zones, vehicular access, and a commercial edge. As we move further into the site, we can find the everyday edge along with larger community spaces.

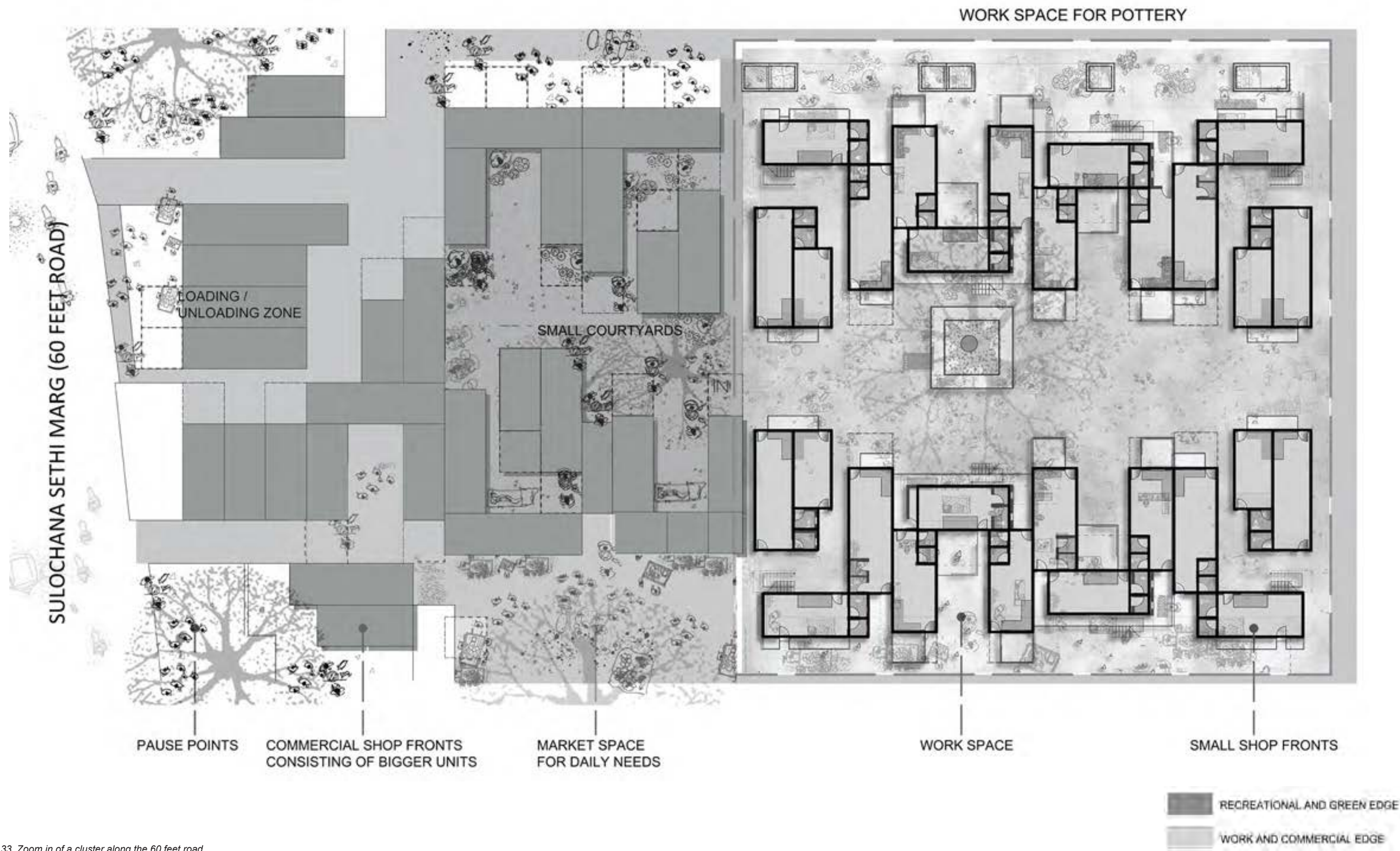


Fig 6.33. Zoom in of a cluster along the 60 feet road
Source: Author

Toolkit of elements for the residents

As mentioned earlier, the RCC framework can be provided by the housing body (government). The user then decides how much to build (incremental approach); user decides how to build (expandable approach); user makes the combination of their elements (adaptable approach).

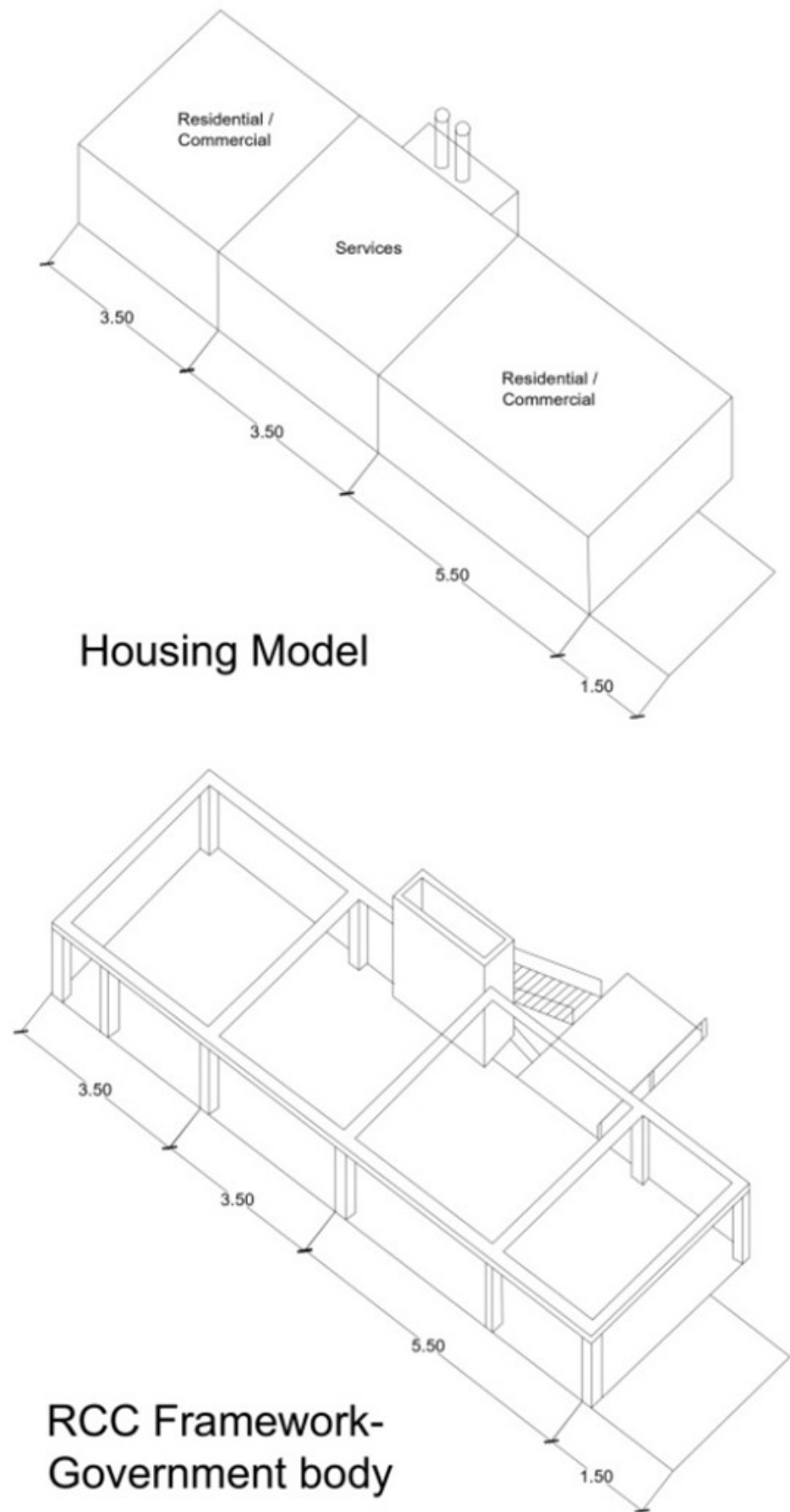


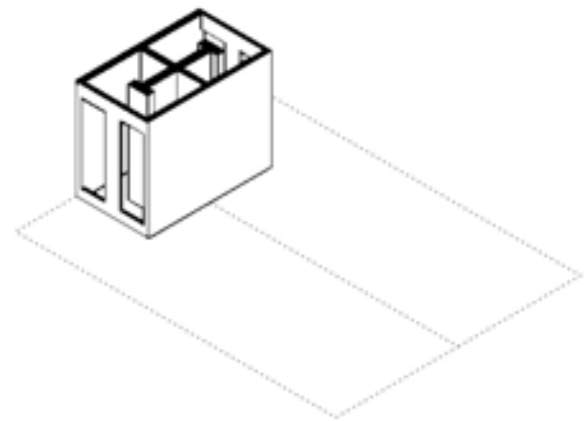
Fig 6.34. Framework given by the government
Source: Author

Toolkit of elements for the residents

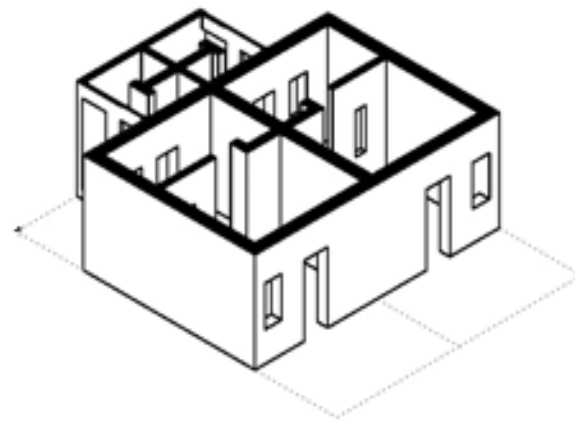
From Left to Right, tool 1 is the RCC framework; tool 2 is fly-ash bricks or plastic infill (circular economy); tool 3 is wooden doors and aluminum windows; tool 4 is steel staircase and tool 5 is outdoor spiral steel staircase (for transient units).



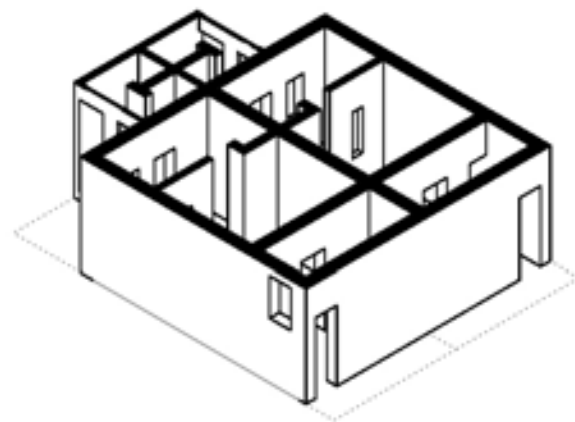
Fig 6.35. Toolkit given to the residents for incremental development
Source: Author



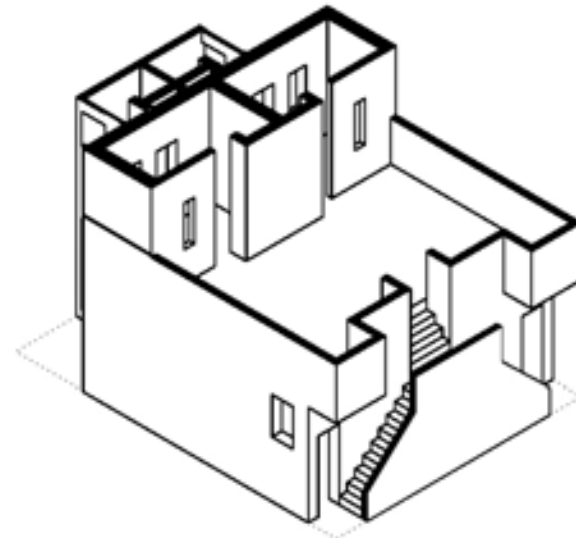
THE GOVERNMENT PROVIDES THE SERVICE CORE



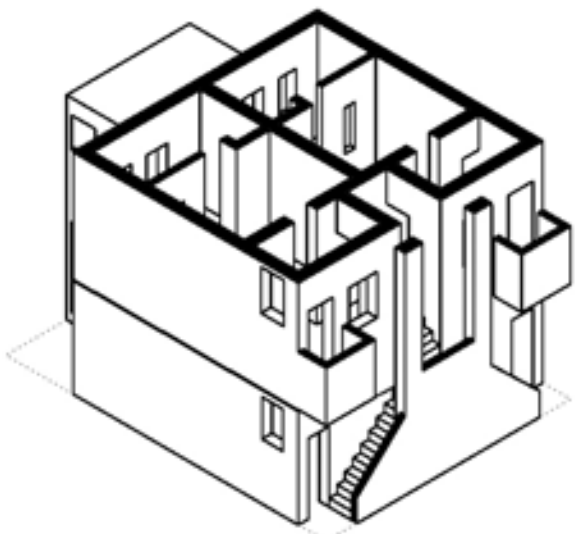
INCREMENTAL



ADAPTABLE



EXPANDABLE



THE UNITS CAN BE STACKED OVER THE YEARS AS PER THE AVAILABLE RESOURCES

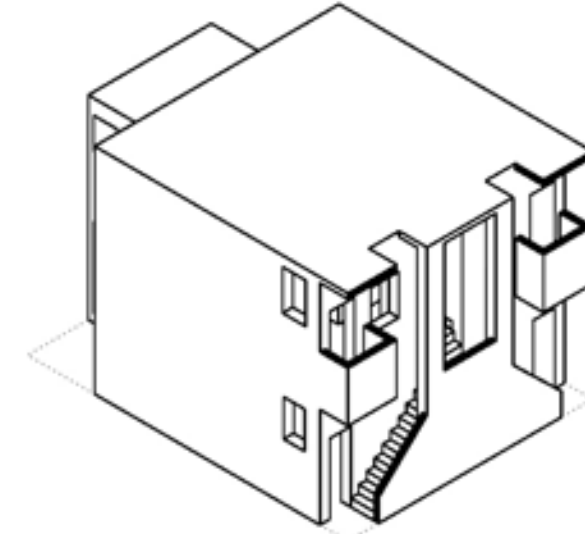


Fig 6.36. Generic incremental unit development
Source: Author (Inspired by Ar B.V.Doshi and Ar. Charles Correa)



Fig 6.37. Incremental unit development
Source: Author



Fig 6.38. Residents decorating their apartment as per the available resources
Source: Author

Toolkit of elements for the residents

From left to right, diagram shows the RCC Structural framework and services provided by housing body; Relocation of user into the housing block; User builds the house to maximum potential.

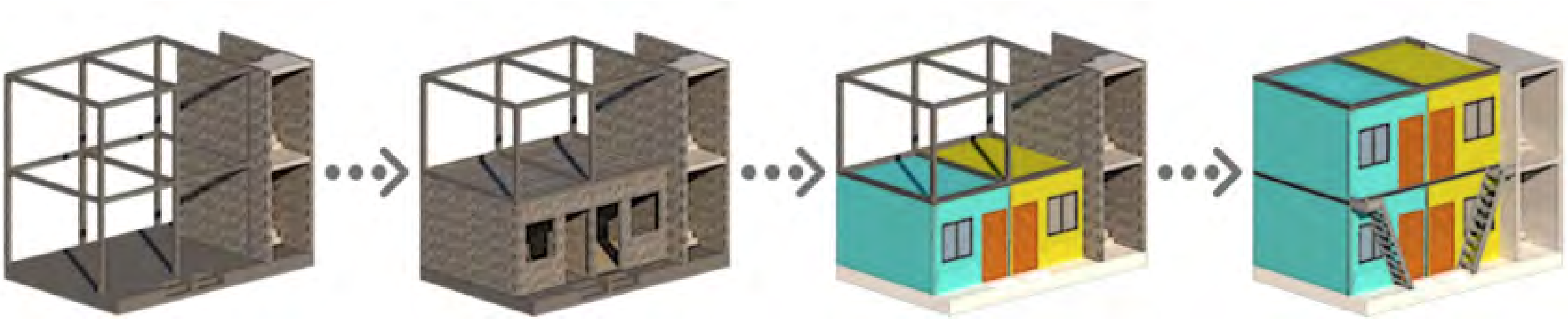


Fig 6.39. Construction stages and material palette
Source: Author

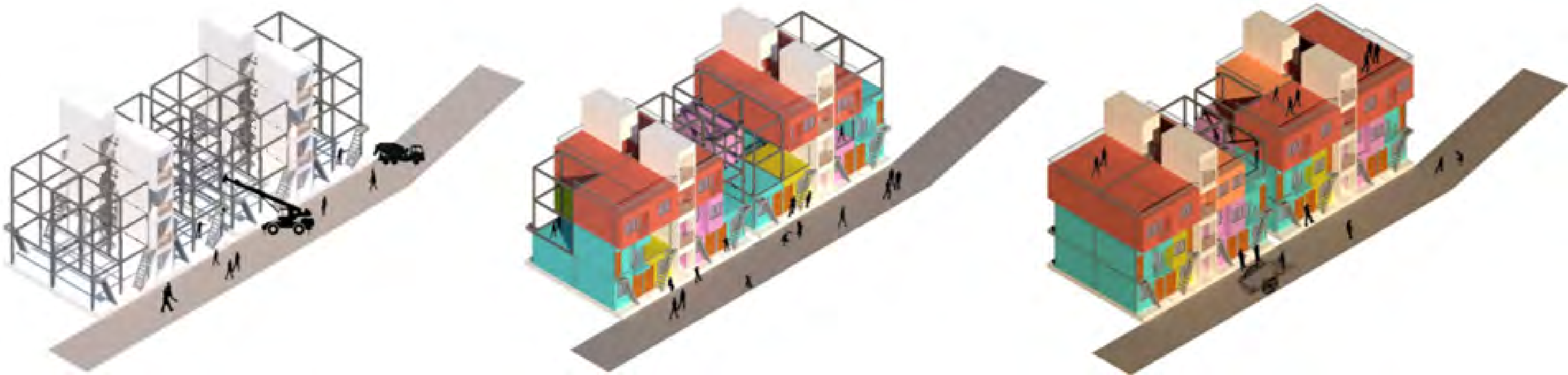


Fig 6.40. Stages for incremental cluster development
Source: Author

Toolkit of elements for the residents

The residents can transform the area eventually as per their needs and available resources, the expression of life rather than aesthetics.

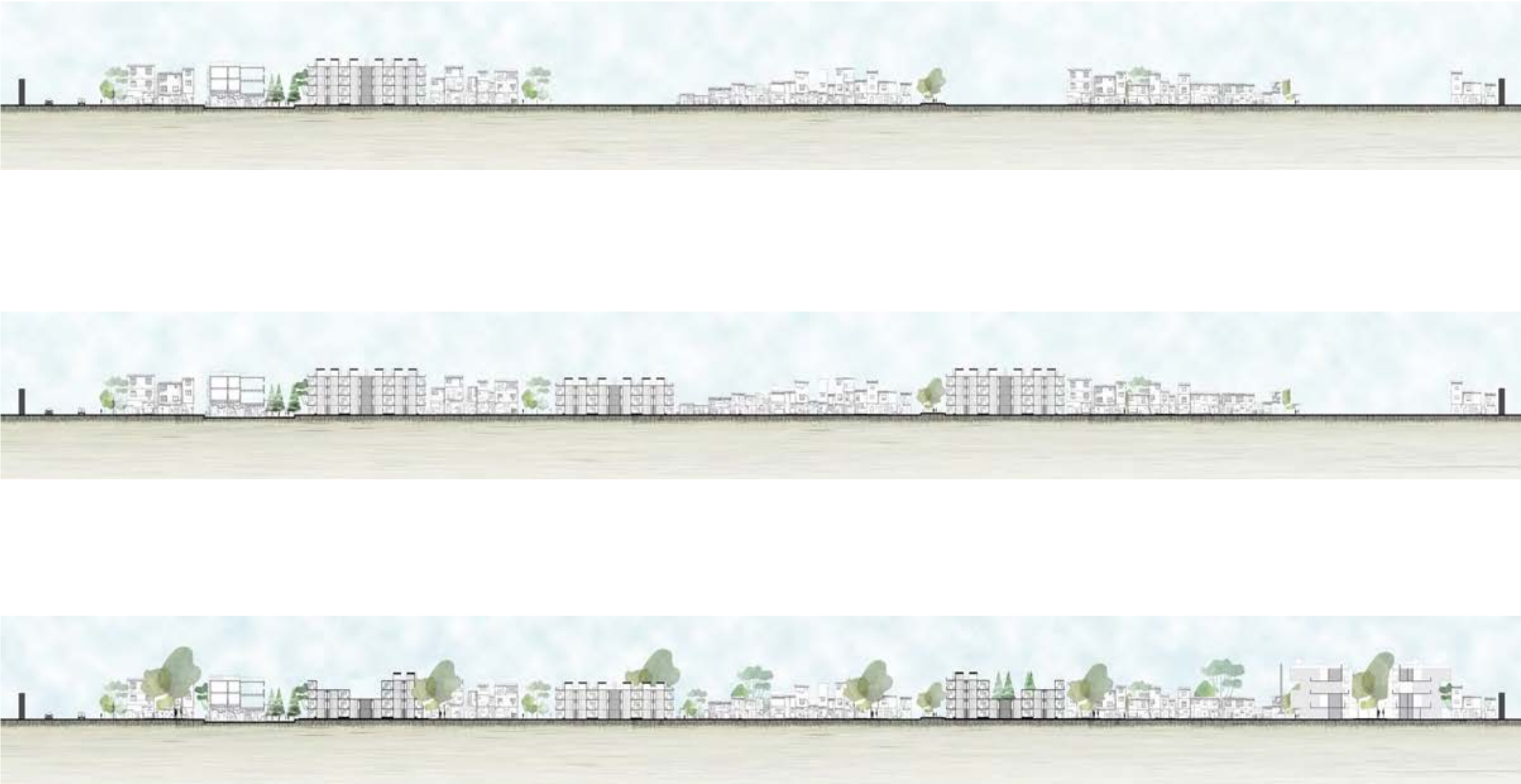


Fig 6.41. Stages for incremental site development
Source: Author



Fig 6.42. Section showing segregation between live and work typology
Source: Author

The residents now have dedicated spaces for morning activities, which are domestic in nature, within the structure; not spilling out in the public realm.

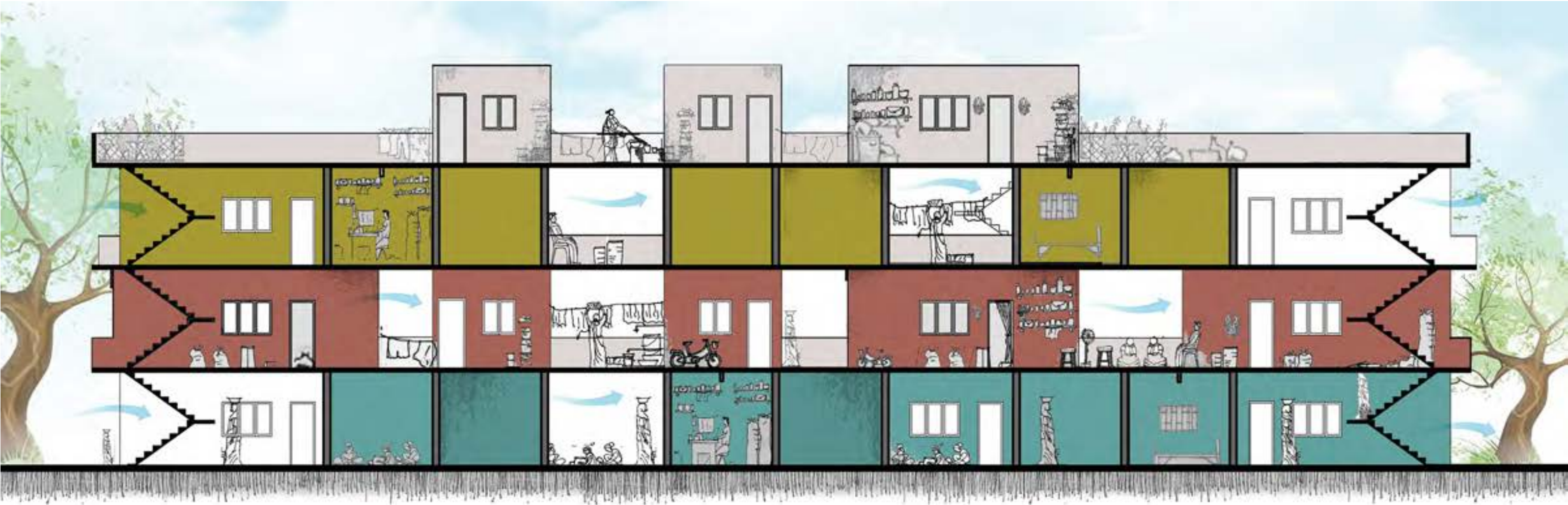


Fig 6.43. Section showing dedicated spaces for domestic activities
Source: Author

The residents also have dedicated spaces for community engagement and recreational activities, with ample natural light and ventilation.

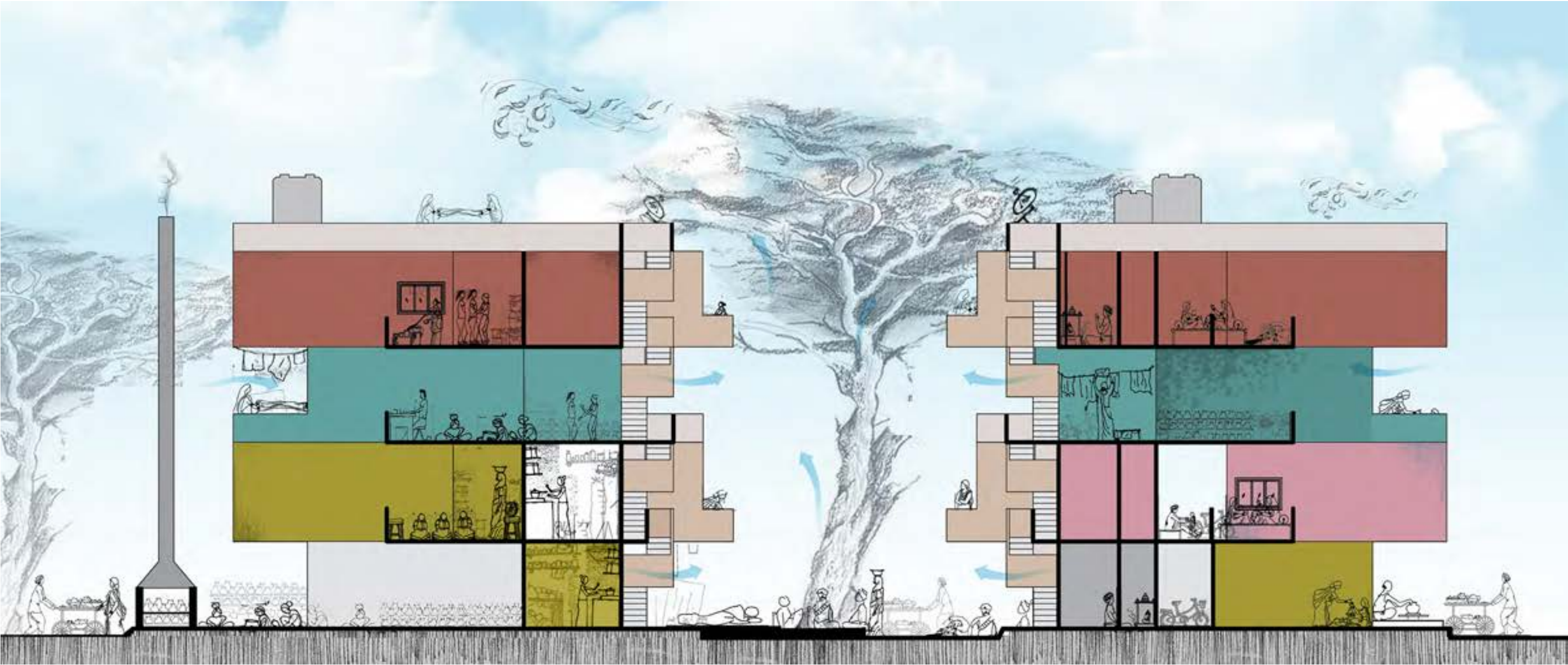


Fig 6.44. Section showing dedicated community gathering spaces
Source: Author

Proposals for improving health and well-being

Two main problems of smoke and sanitation can be taken care of in parallel. The existing kilns can be smoke-free if they are electric. To power these kilns, electricity can be generated by treating the wastewater produced by this community. Each unit has been provided with an individual toilet and a wet wall. This would help in segregating the black water as well as grey water.

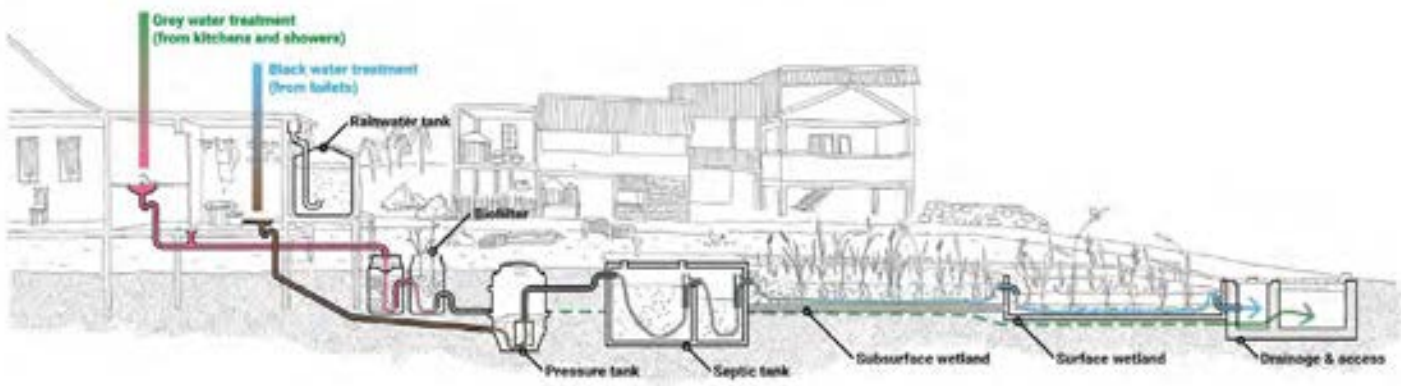


Fig 6.45. Section showing overall wastewater segregation system
Source: RISE

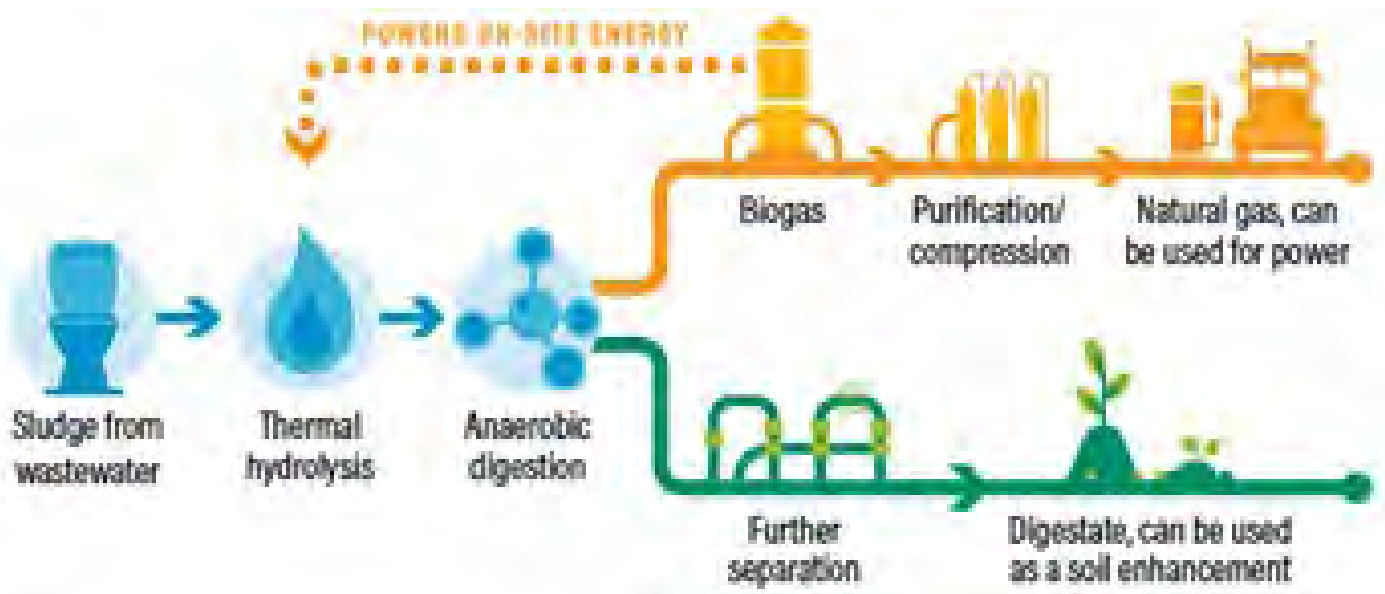


Fig 6.46. Section showing wastewater to energy system
Source:

Proposals for improving health and well-being

The technology is already used in Chennai in India. Grass-root organizations and/or investors can invest funds in such systems for the betterment of the community. The cost for one system is about \$365. I also did some technical analysis to understand if energy generated can be used to power all the kilns on the site. It is as follows

- Total number of residents in Kumbharwada: Around 10,000
- Approximate total water consumption in Kumbharwada = 150 litres X 10,000 people = 1500 KLD (or) 1.5 MLD (or) 1500 m³/day
- Average sewage generated= 85% of supplied water
- Average sewage generated= 0.85 X 1.5= 1.275 MLD = 1275 KLD (1275 kg)
- Each cubic meter of biogas contains the equivalent of 2 kWh of electrical power. Thus, an overall 114kW of energy can be produced.
- Approximately 42 kilns distributed overall on the site
- Energy required for 1 kiln = 2.5kW; Energy required for 42 kilns = 105kW

Thus, the energy generated by treating the wastewater can be used to power all the kilns on the site. This is just an attempt to understand the energy required. Kindly note that the calculations may vary as per actual technical specifications of the kilns used by the community

Apart from this, one of the terraces of each cluster can be used to place solar panels to harness solar energy for domestic purposes. The roof area can also be used for roof planting. An average Indian home requires 2.4 kW of solar power or 6 solar panels with 330 watts each. Each cluster comes with 24 panels and thus can power 3 homes in each block. Total energy produced: 121.2kW.

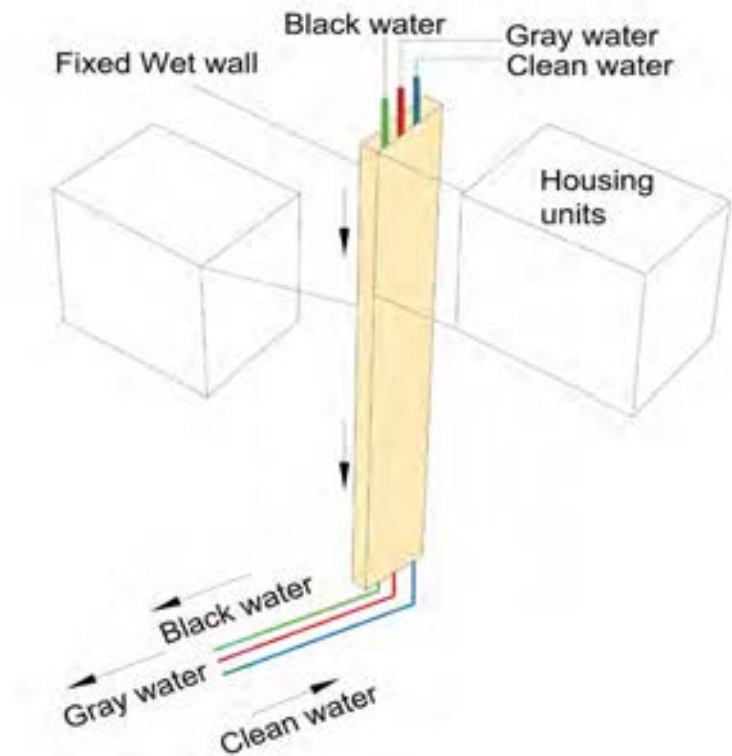


Fig 6.47. Fixed wet wall system provided in each housing unit for segregation
Source: Author

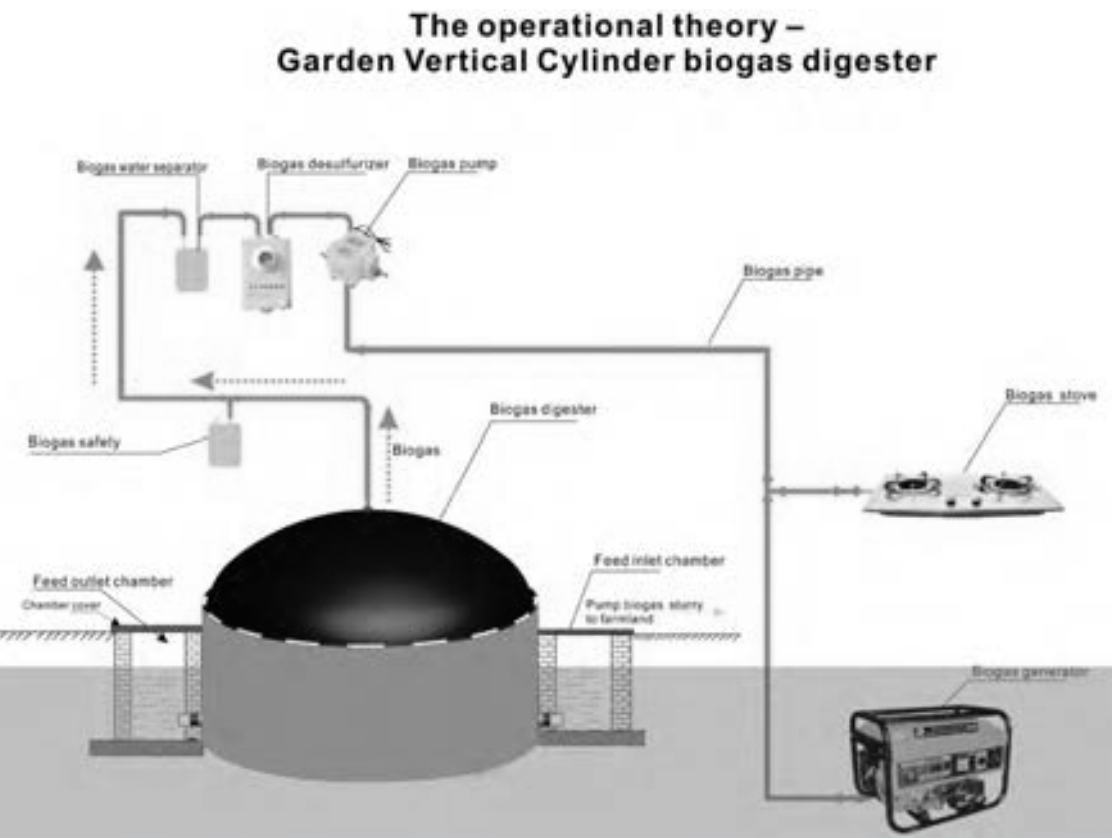


Fig 6.48. Wastewater to energy system being used in India
Source: Indiamart

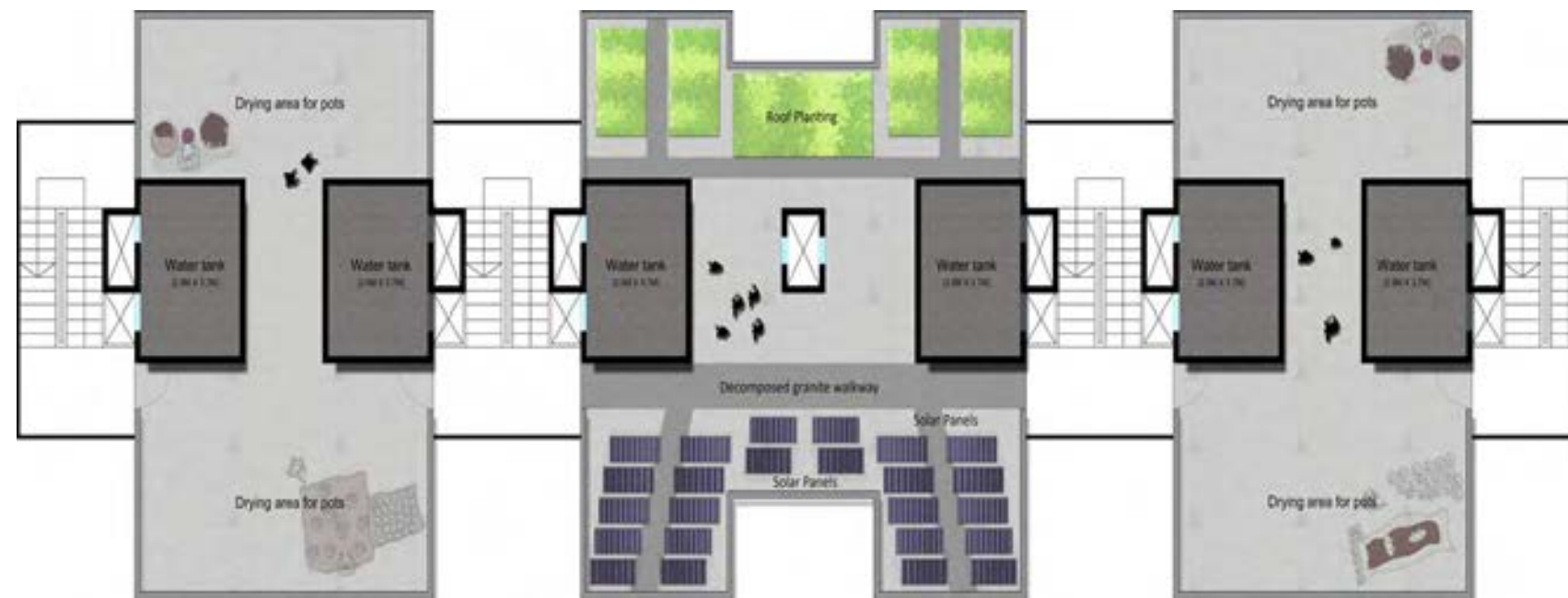


Fig 6.49. Terrace floor plan for cluster showing placement of solar panels
Source: Author



Fig 6.50. Ground floor plan for cluster
Source: Author



Fig 6.51. First floor plan for cluster
Source: Author

The idea is to formalize the informal, by allowing the structure, a form of informality to some extent; but at the same time improving the conditions of the area.

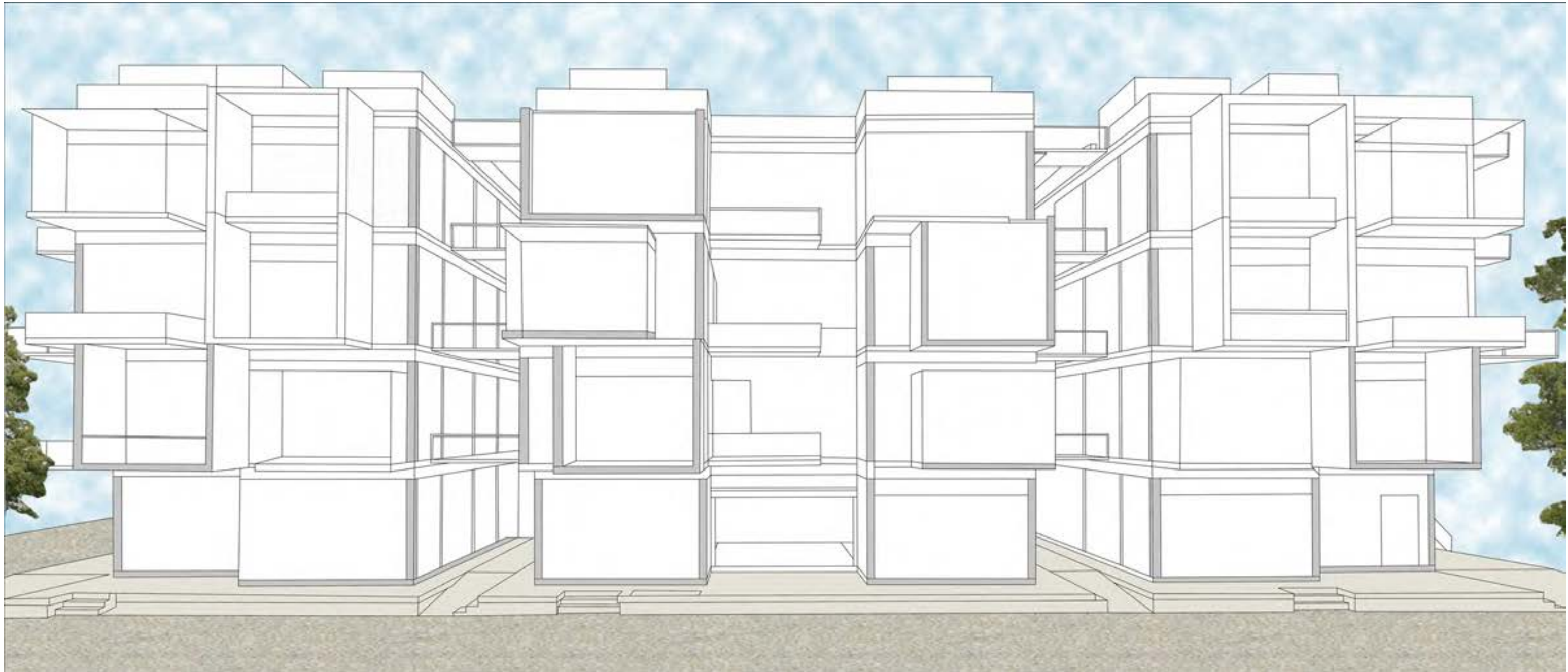


Fig 6.52. Government participation- top-down approach
Source: Author

The idea is to formalize the informal, by allowing the structure, a form of informality to some extent; but at the same time improving the conditions of the area.



Fig 6.53. Dharavi's participation- bottom-up approach
Source: Author

Conclusion

The macro, mesa, and micro scales underlying the regional situation should be considered by the interventions. The interventions should be process-driven rather than merely product-driven, and they should take the communities' involvement and impact into account. They'll probably combine elements of architectural, urban, political, and social conceptions, but they should at the very least aim to express themselves more specifically in a spatial way. This will guarantee more noticeable changes.

Kumbharwada could serve as a model to extend the principles of the outlined framework along the entire Dharavi. It can be a test ground to determine what parameters can be successful and what can be improved further for such a community. Overall, this research can be an important step towards fighting a national issue.

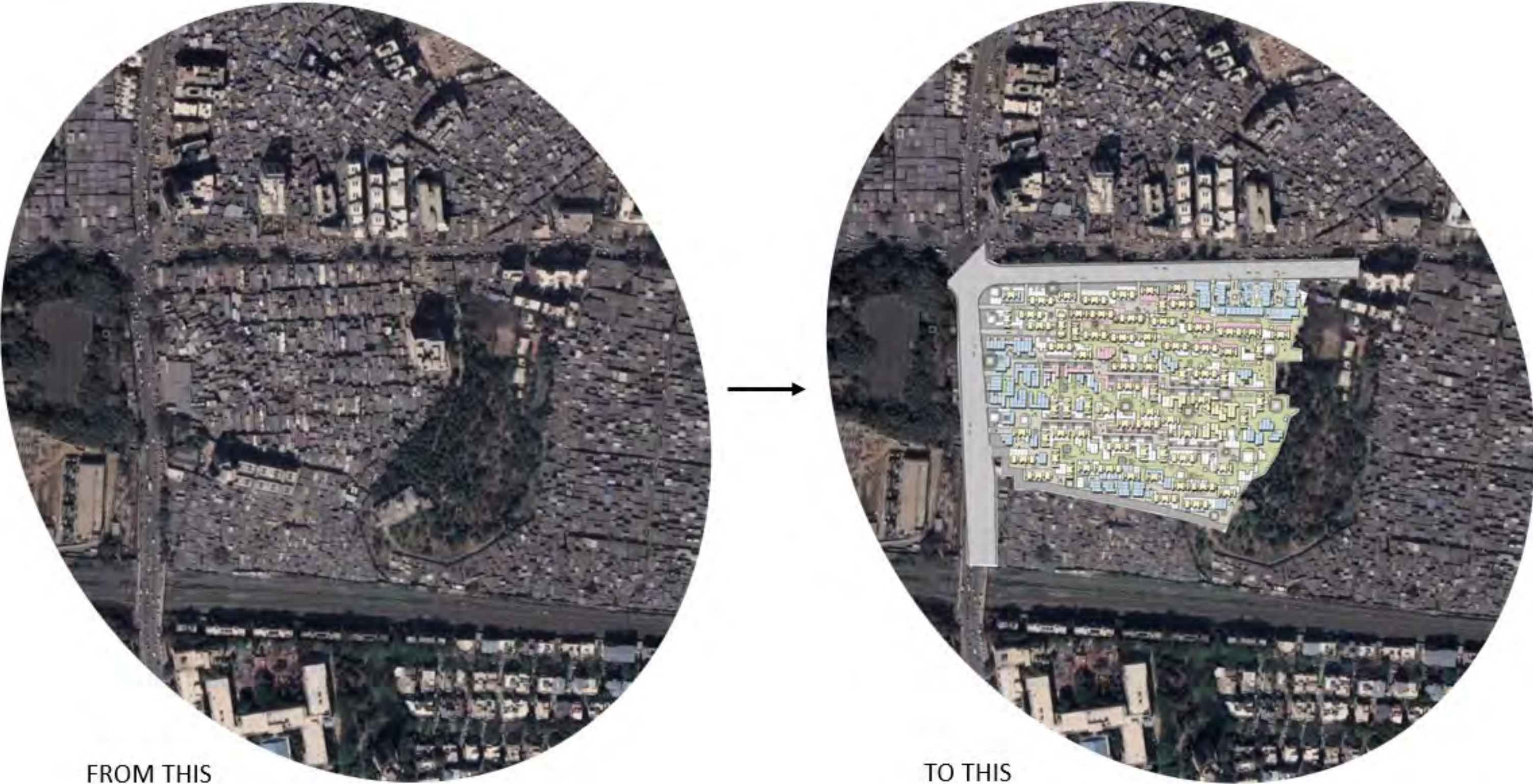


Fig 6.54. Kumbharwada as a prototype
Source: Author

Conclusion

It is said that every 6th urban Indian lives in slums unfit for human habitation. Slums are found in 65 per cent of the Indian towns. Yet slum dwellers are the most overlooked section of society. 6 out of 10 slum dwellers live close to unsanitary drains and almost 4 of every 10 do not get treated water. To solve these major issues, I believe that it is vital that we implement such a toolkit of strategies, to get this region from this (Figure 6.57) to that (Figure 6.58). Thus, my thesis culminates with me being the architect of this system and being an advocate for such communities.

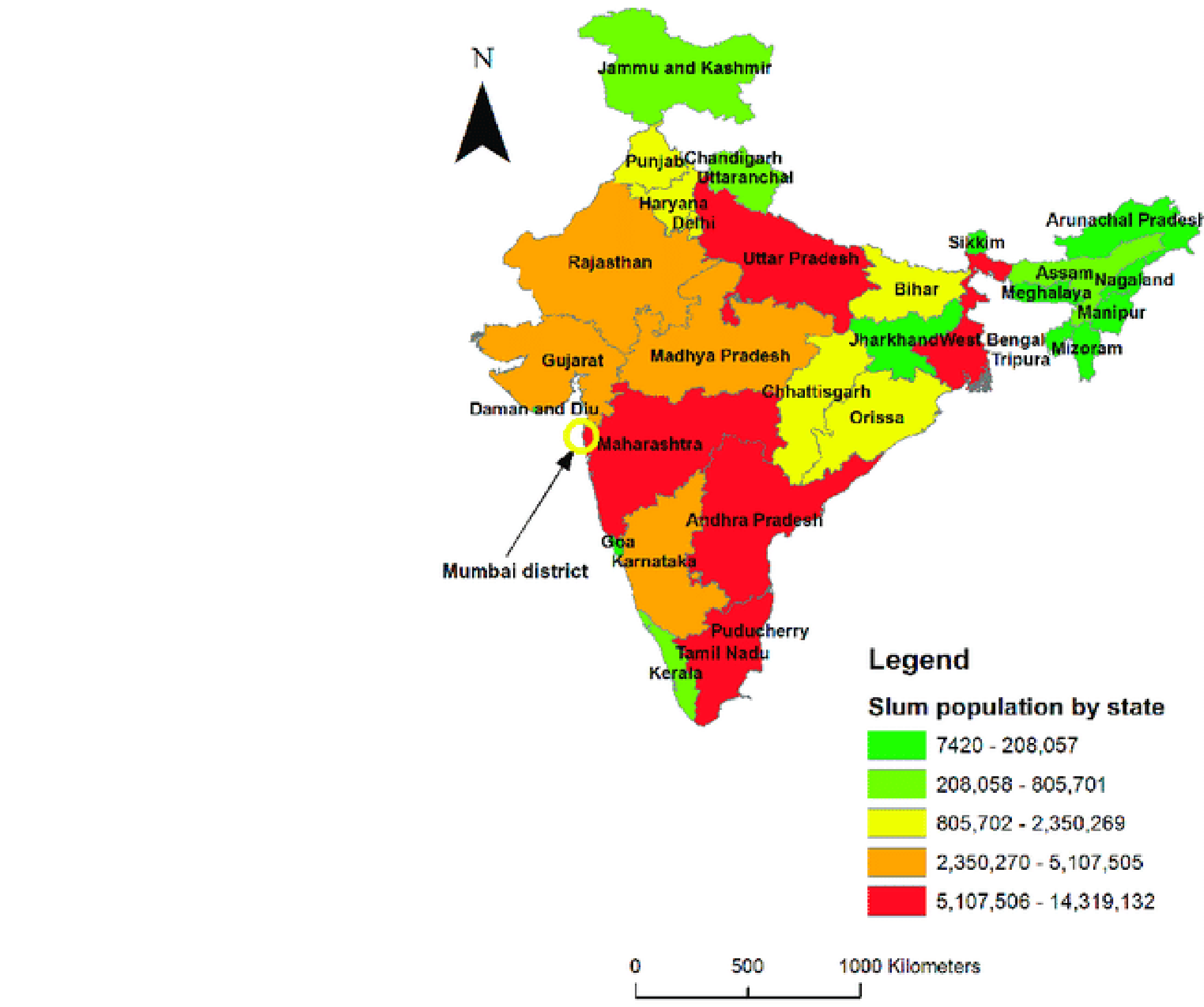


Fig 6.55. India slum distribution map
Source: Author



Fig 6.56. "Sanitation issues in Dharavi highlight inhabitable conditions (before)" by Kristian Bertel is licensed under CC by SA 4.0.
Source: https://commons.wikimedia.org/wiki/File:Slum_in_Dharavi_-_Kristian_Bertel.jpg, License: <https://creativecommons.org/licenses/by-sa/4.0/deed.en>



Fig 6.57. The New Dharavi Culture (After)
Source: Author

Appendix A: Energy Simulations for Units and Courtyards

The residents of Kumbharwada are strongly dependent on the artificial lighting since they must keep their windows closed to avoid the smoke coming out of the kilns in their current set-up. Since the thesis proposes smoke-free electric kilns, natural light can be used as much as possible to reduce the dependance on electricity, thus reducing their energy costs as well. Since the housing block dimensions were derived from the existing setup, it was tested in Climate studio to determine the amount of natural light it would receive.

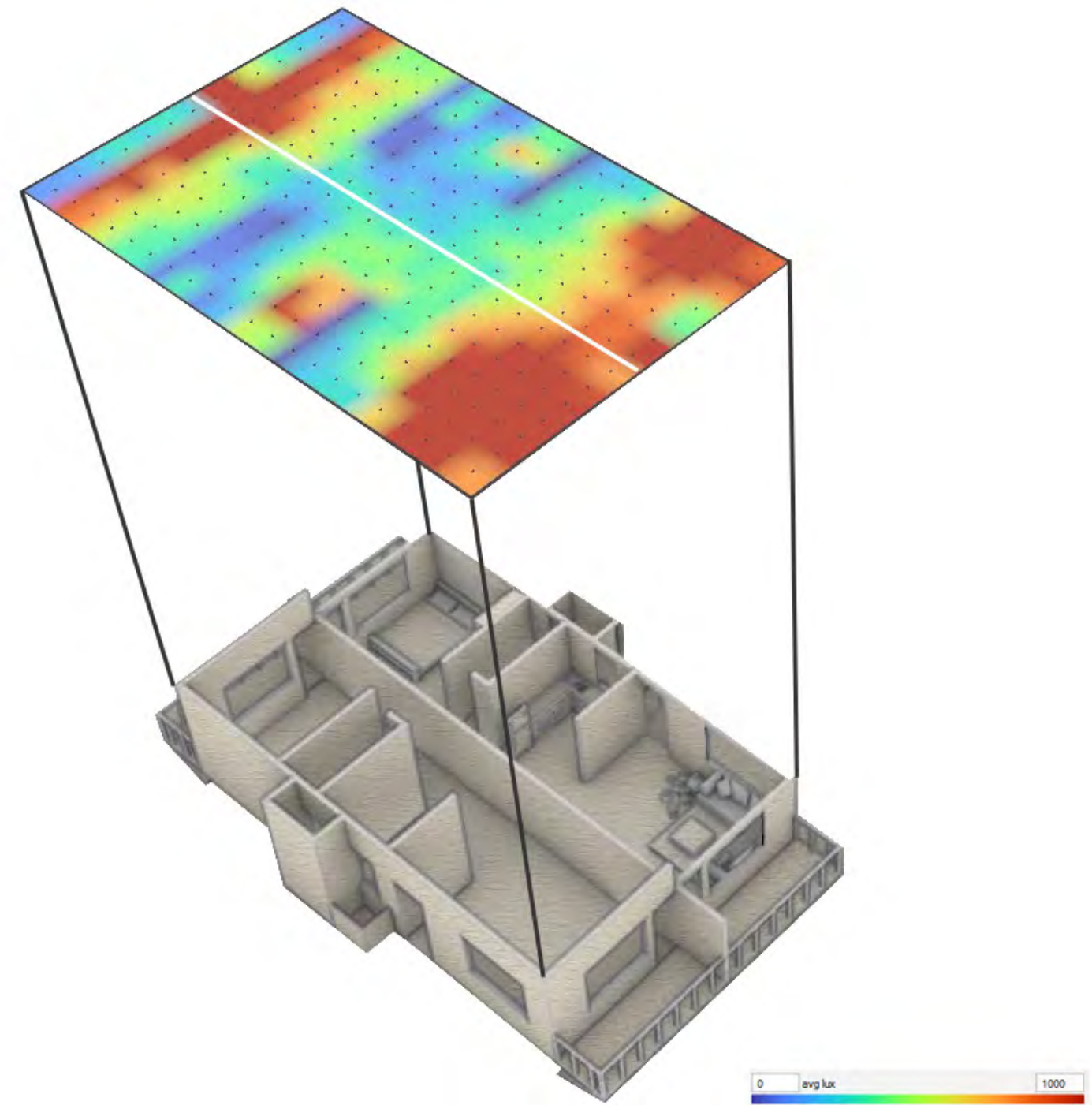


Fig A.1. Energy simulation for housing unit
Source: Author

Thus, the units overall receive an average luminance of about 743 lux on average, which is more than sufficient for carrying out all the daily activities. The unit layout and window placement also help for cross-ventilation. Along with the units, couple of courtyards were also tested for thermal comfort, in Ladybug, to understand where more trees can be planted for shading.

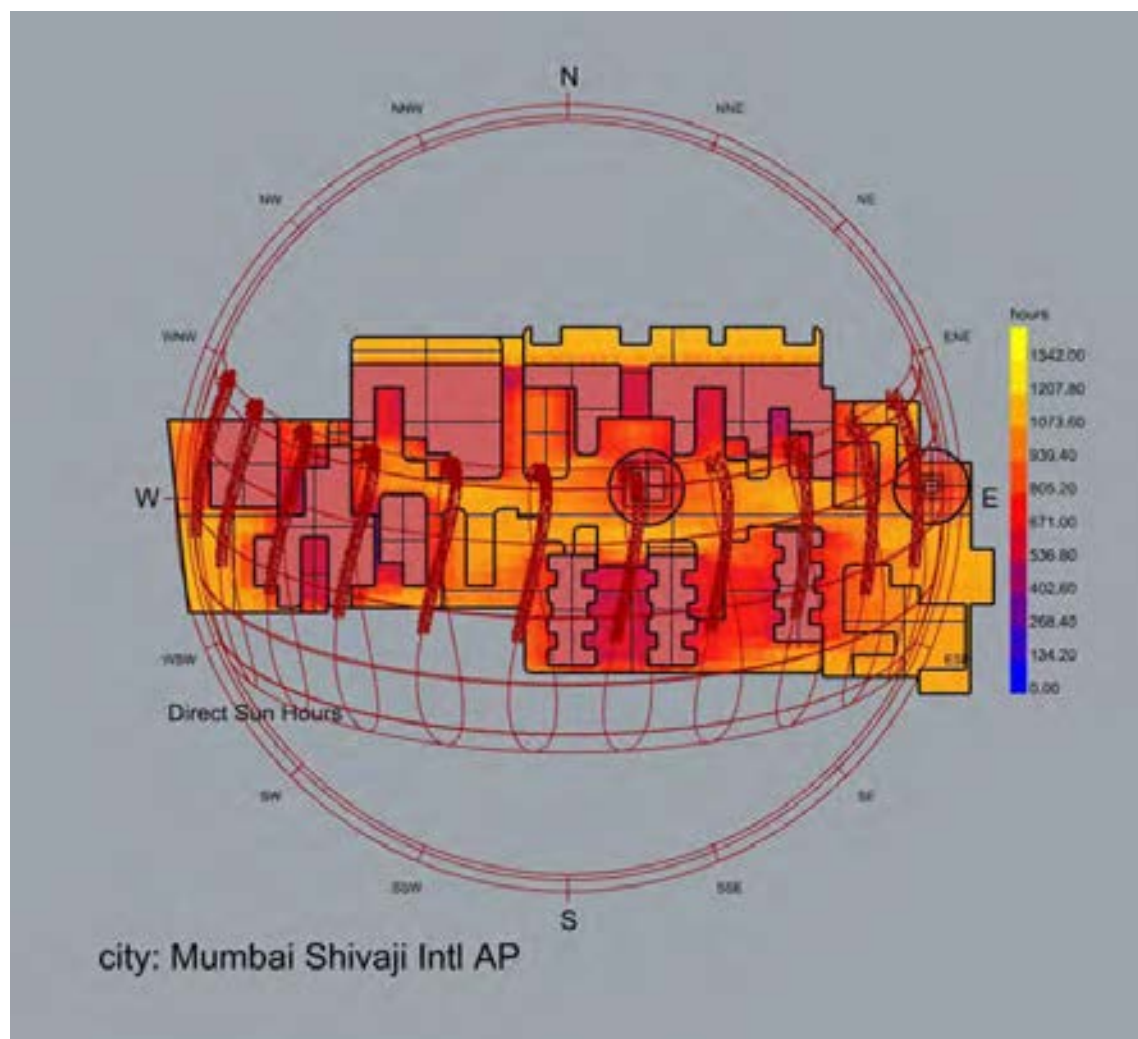


Fig A.2. Courtyard 01- Direct analysis done in ladybug 01
Source: Author

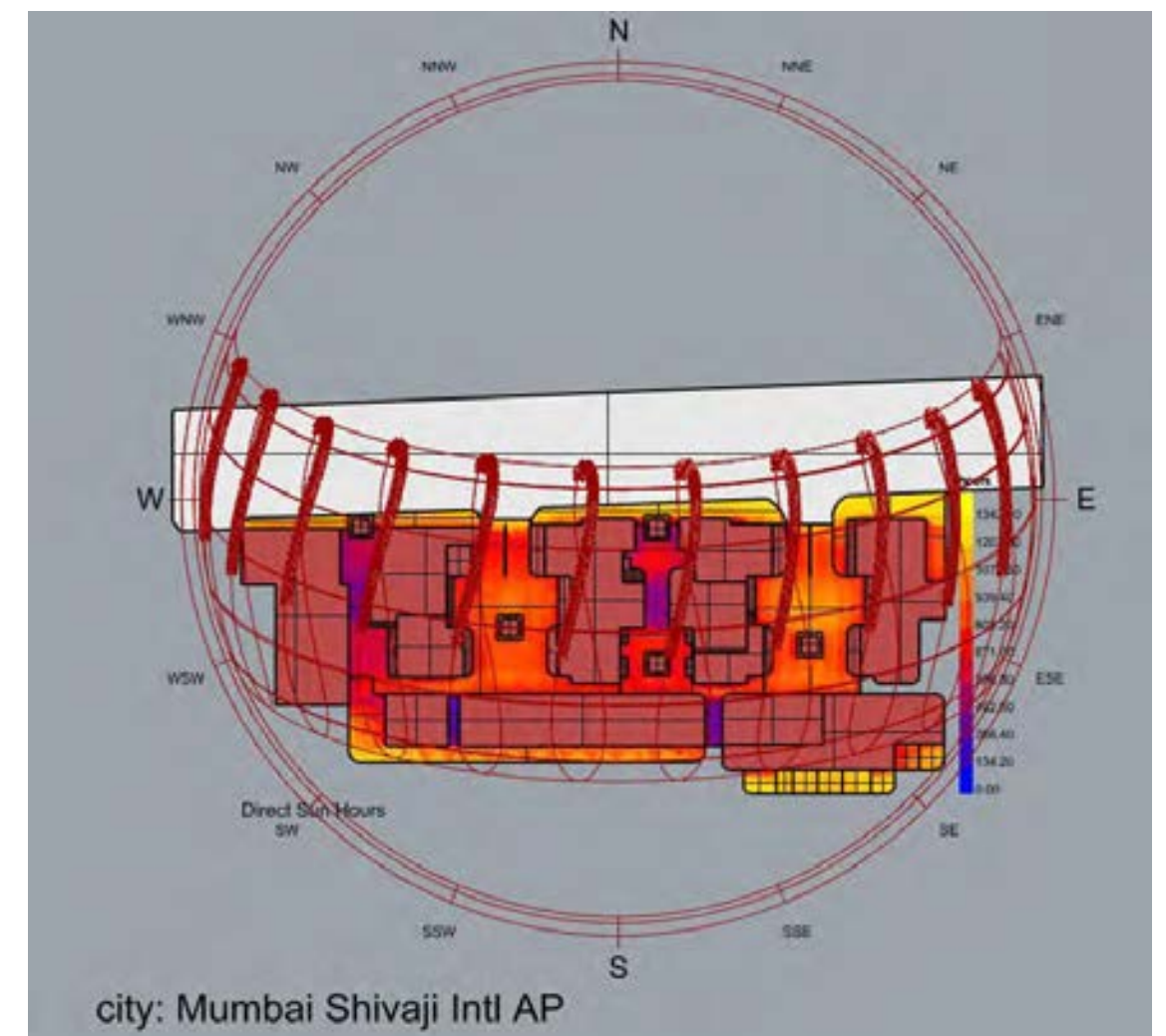


Fig A.4. Courtyard 02- Direct analysis done in ladybug 01
Source: Author

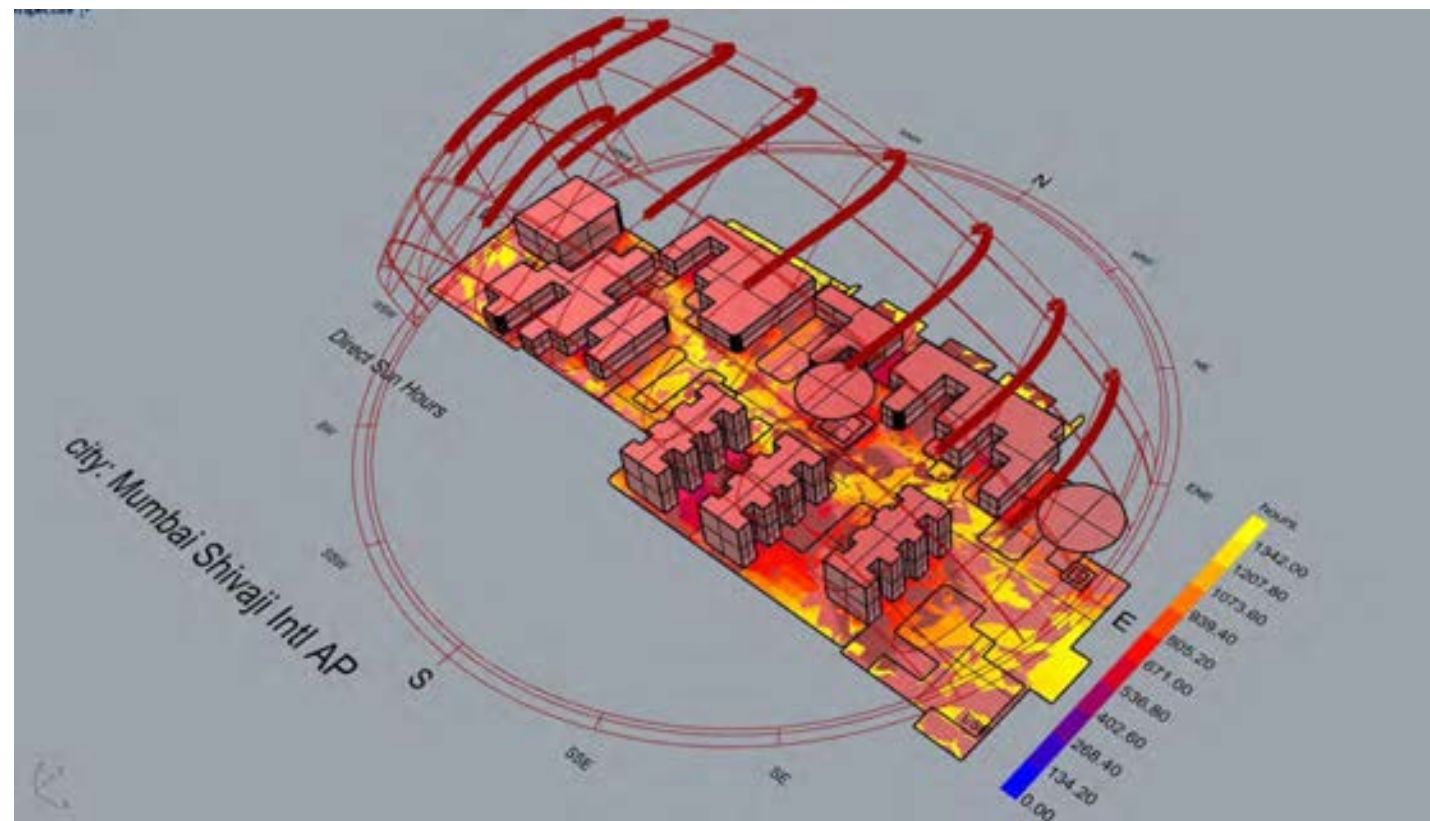


Fig A.3. Courtyard 01- Direct analysis done in ladybug 02
Source: Author

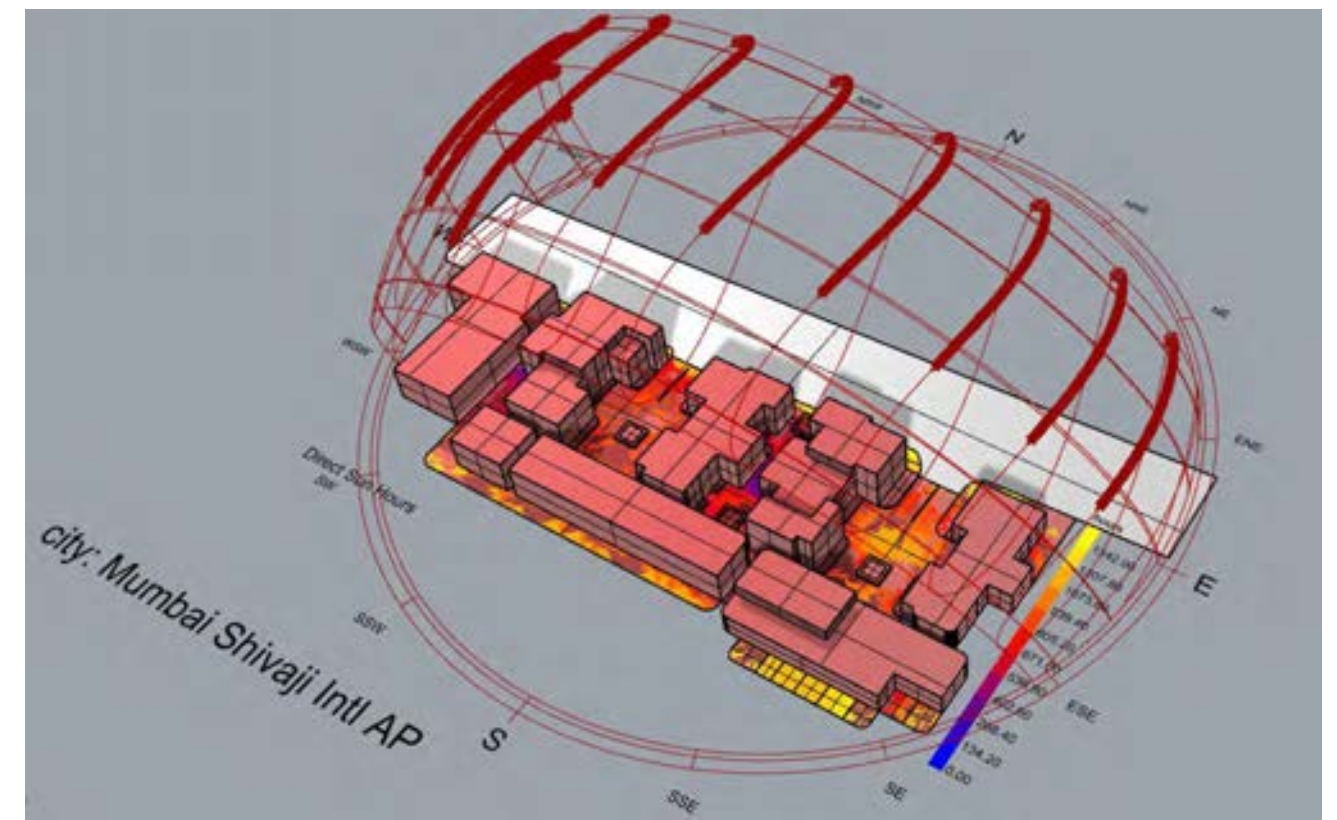


Fig A.5. Courtyard 02- Direct analysis done in ladybug 02
Source: Author

The trees are modeled as a circular surface just for study purposes. The intent is to roughly include the size of tree canopy which can help for shading in courtyards.

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